

SURGICAL PROCESS DISCOVERY FOR SCHEDULING OPTIMISATION: STRUCTURAL AND TEMPORAL ANALYSIS OF PERIOPERATIVE WORKFLOW

This Thesis is submitted to the Department of Computer Science as partial fulfillment of Master of Science in Computer Science/ Data Science degree

by

Hina Farheen

Supervised by

Dr. Tariq Mahmood

Professor

Department of Computer Science

Head of Healthcare Analytics Laboratory

School of Mathematics and Computer Science (SMCS)

Institute of Business Administration (IBA), Karachi

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Institute of Business Administration (IBA), Karachi, Pakistan

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by

Hina Farheen

(Student Number: 29402)

Committee:

Dr. Tariq Mahmood

Supervisor, IBA, Karachi

Dr. Naveed ur Rehman

Examiner – I, Agha Khan University Hospital

Dr. Sohail Imran

Examiner – II, PAF-KIET

Dr. Atif Tahir

Dean, SMCS, IBA, Karachi

Fall 2025

Institute of Business Administration (IBA), Karachi, Pakistan

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Dedication

To my parents, for the sacrifices, prayers, and quiet faith that brought me here. I can't put into words how much it meant to have you believe in me through every late night and every moment of doubt during this journey. You made the hard days easier, even when I didn't say so at the time.

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Table of Contents

List of Tables	7
List of Figures	8
List of Appendix Figures	10
List of Abbreviations	12
Abstract.....	13
1 Introduction.....	14
1.1 The Operational Problem.....	15
1.2 Perioperative Scheduling Challenges: A Broader Context.....	16
1.3 Limitations of Current Process Discovery in Healthcare	18
1.4 Justification of the Research	19
1.5 Problem Statement	20
1.6 Research Questions.....	20
1.7 Contributions.....	21
1.8 Scope and Limitations of the Study	21
2 Background Knowledge.....	24
2.1 Process Mining and Its Role in Healthcare Data Analysis	24
2.1.1 Why Process Mining Suits Medical and Surgical Data	26
2.2 Event Logs and the Directly-Follows Relation.....	27
2.3 Process Discovery Algorithms.....	28
2.4 Process Enhancement: Directly-Follows Graphs.....	33
2.5 Summary	34
3 Literature Review.....	35
3.1 Review Methodology.....	35
3.1.1 Review Type and Rationale	35
3.1.2 Search Strategy	35
3.1.3 Screening Process	36

3.1.4 Publication Trend Over Time	37
3.2 Review	38
3.2.1 Process Mining in Healthcare: General Foundations.....	38
3.2.2 Process Mining for Surgical and Perioperative Scheduling	39
3.2.3 Bayesian and Probabilistic Treatments of Process Mining Uncertainty.....	41
3.2.4 Conformance to Clinical Safety Protocols.....	43
3.3 Publication Venues and Research Groups	43
3.3.1 Summary of Publication Sources.....	43
3.3.2 Key Research Groups	44
3.4 Overview of Final Extracted Studies	45
3.5 Critical Analysis and Research Gaps.....	47
3.6 Research Contribution and Thesis Focus.....	49
4 Proposed Methodology	50
4.1 Overview of the Proposed Pipeline.....	50
4.2 Algorithm Selection: Heuristic Miner versus Alpha Miner.....	52
4.3 The Bayesian Heuristic Miner: Proposed Innovation.....	53
4.3.1 Motivation.....	53
4.3.2 Novelty of the Proposed Approach.....	54
4.3.3 A Bayesian Reformulation of the Dependency Measure.....	55
4.3.4 Why This Improves on the Classical Formula.....	56
4.3.5 Prior Configuration.....	57
4.3.6 Algorithm.....	58
4.3.7 Complexity Analysis.....	59
4.4 Connecting Structural Discovery to Temporal Analysis	61
4.5 Summary	61
5 Experimental Methodology and Results.....	62
5.1 Dataset Description.....	62
5.1.1 Data Source and Collection	62
5.1.2 Raw Structure of the Dataset	62
5.1.3 Data Cleaning Pipeline	64

5.1.4 Specialty, Location, and Surgeon Composition.....	64
5.2 Tools and System Environment.....	65
5.3 Experimental Methodology	65
5.3.1 Depth-First Path: Whole-Dataset Analysis.....	65
5.3.2 Breadth-First Path: Granular Re-Runs.....	67
5.4 Results.....	67
5.4.1 Depth-First Results: Institution-Level Structural Model.....	67
5.4.3 Breadth-First Results: Specialty, Location, and Surgeon Findings	72
5.5 Comparison with Thesis I Results	74
5.6 Summary.....	75
6 Discussion of Results.....	77
6.1 Answering the Research Questions	77
6.2 Recommendations.....	79
6.2.1 Structural Recommendations.....	79
6.2.2 Temporal Recommendations	80
6.2.3 Specialty-Level Recommendations	81
6.2.4 Governance Recommendations	82
6.2.5 Recommendations Considered and Not Adopted.....	83
6.3 Limitations	84
6.4 Contributions and Potential Impact	85
6.5 Summary.....	86
7 Conclusions and Future Work	87
7.1 Summary of the Research Problem and Its Resolution	87
7.2 Contributions.....	88
7.3 Future Work.....	88
7.4 Closing Remarks.....	89
References.....	90
Appendix.....	93
A Structural (Frequency-Based) Heuristic Miner Models.....	93

A.1 Surgery Specialty-Level Maps.....	93
A.2 Surgery Location-Level Maps	96
A.3 Top 5 Highest Volume Surgeon-Level Maps	97
B Temporal (Time-Based) Direct-Flow Graphs.....	98
B.1 DFG Graphs with Mean Duration.....	98
B.1.1 Surgery Specialty-Level Graphs	98
B.1.2 Surgery Location-Level Graphs.....	102
B.2 DFG Graphs with Median Duration.....	104
B.2.1 Surgery Specialty-Level Graphs	104
B.2.2 Surgery Location-Level Graphs.....	108
B.2.3 Top 5 Highest-Volume Surgeon-Level Graphs	110

List of Tables

Table 1	Illustrative fragment of an event log for a single case	28
Table 2	Summary comparison of the three process discovery algorithm families	32
Table 3	Search queries used to identify relevant studies	36
Table 5	Key research groups identified from the reviewed literature.....	45
Table 6	Master table of studies included in the final synthesis	47
Table 8	Structural comparison of Alpha Miner and Heuristic Miner discovery	53
Table 9	Comparison of classical and Bayesian dependency measures.....	57
Table 10	Beta prior configuration used in the Bayesian Heuristic Miner	58
Table 11	Mapping from raw timestamp columns to process mining activities	64
Table 12	Data cleaning funnel	64
Table 13	Baseline versus Bayesian model, institution level.....	69
Table 14	Process mining evaluation metrics, baseline versus Bayesian model.....	70
Table 15	Limitations of the proposed methodology and results.....	85

List of Figures

Figure 1 A typical patient journey through the surgical pathway.....	14
Figure 2 Fixed scheduling fails both ways. Surgeon A's overrun delays the next case, while B's early finish leaves the OR idle.....	15
Figure 3 The role of process mining in healthcare process management (Rojas et al. 2016).	25
Figure 4 Directly-follows graph derived from the event log fragment in Table 1.....	28
Figure 5 Schematic contrast between Alpha Miner and Heuristic Miner model density.	29
Figure 6 Examples of the Heuristic Miner dependency measure for a retained edge (left) and a filtered edge (right).....	31
Figure 7 Structural density comparison across the three dominant process discovery algorithm families on the same four activities.	33
Figure 8 PRISMA flow diagram illustrating the study selection process for the literature review.....	37
Figure 9 Distribution of selected studies by publication year (2021-2025).....	38
Figure 10 Count of papers by venue types.....	44
Figure 11 Gaps addressed in this thesis	47
Figure 12 Proposed three-stage methodology pipeline, converging on integrated recommendations	50
Figure 13 Depth-first and breadth-first experimental paths converging into the results in Section 5.4.....	66
Figure 14 (a) Baseline (frequency-based) Heuristic Miner model - institution level, (b) Bayesian Heuristic Miner model - institution level	68
Figure 15 Mean-duration performance graph, institution level	71
Figure 16 Median-duration performance graph, institution level.....	71
Figure 17 Baseline frequency model for the highest-volume surgeon	72
Figure 18 Mean-duration performance graph, cardiac surgery.....	73
Figure 19 Mean-duration performance graph, ophthalmology.....	73
Figure 20 Average surgery time in the main operating room among the top 15 high- volume surgeons	74

Figure 21 Average surgery time in the is same-day care among the top 15 high-volume surgeons 74

List of Appendix Figures

Fig A Breast Surgery (Frequency).....	93
Fig B ENT Surgery (Frequency).....	93
Fig C Gen Surgery (Frequency).....	93
Fig D Neurosurgery (Frequency).....	93
Fig E Obgynae Surgery (Frequency).....	94
Fig F Orthopedics Surgery (Frequency)	94
Fig G Pediatrics Surgery (Frequency)	94
Fig H Plastic Surgery (Frequency)	94
Fig I Thoracic Surgery (Frequency)	95
Fig J Urology Surgery (Frequency).....	95
Fig K Vascular Surgery (Frequency).....	95
Fig L Endoscopy (Frequency)	96
Fig M Main-OR (Frequency).....	96
Fig N Obgynae (Frequency)	96
Fig O SDC (Frequency)	96
Fig P 2nd-highest volume (Frequency).....	97
Fig Q 3rd highest volume (Frequency).....	97
Fig R 4th highest volume (Frequency)	97
Fig S 5th highest volume (Frequency).....	97
Fig T Breast Surgery (Mean Duration).....	98
Fig U Cardiac Surgery (Mean Duration)	98
Fig V ENT Surgery (Mean Duration).....	98
Fig W General Surgery (Mean Duration).....	99
Fig X Neurosurgery (Mean Duration)	99
Fig Y Ophthalmology (Mean Duration)	99
Fig Z Pediatrics Surgery (Mean Duration)	99
Fig AA Obgynae Surgery (Mean Duration)	100
Fig BB Orthopedics Surgery (Mean Duration).....	100
Fig CC Plastic Surgery (Mean Duration).....	101
Fig DD Thoracic Surgery (Mean Duration).....	101

Fig EE Urology Surgery (Mean Duration)	101
Fig FF Vascular Surgery (Mean Duration).....	101
Fig GG Endoscopy (Mean Duration).....	102
Fig HH Main-OR (Mean Duration)	102
Fig II Obgynae (Mean Duration).....	103
Fig JJ SDC (Same Day Care) (Mean Duration).....	103
Fig KK Breast Surgery (Median Duration)	104
Fig LL Cardiac Surgery (Median Duration).....	104
Fig MM ENT Surgery (Median Duration).....	104
Fig NN General Surgery (Median Duration)	105
Fig OO Neurosurgery (Median Duration)	105
Fig PP Ophthalmology Surgery (Median Duration).....	105
Fig QQ Pediatrics Surgery (Median Duration).....	105
Fig RR Obgynae Surgery (Median Duration).....	106
Fig SS Orthopedics Surgery (Median Duration)	106
Fig TT Plastic Surgery (Median Duration).....	107
Fig UU Thoracic Surgery (Median Duration)	107
Fig VV Urology Surgery (Median Duration)	107
Fig WW Vascular Surgery (Median Duration).....	107
Fig XX Endoscopy (Median Duration).....	108
Fig YY Main-OR (Median Duration).....	108
Fig ZZ Obgynae (Median Duration).....	109
Fig AAA SDC (Median Duration).....	109
Fig BBB Highest-Volume Surgeon (Median Duration).....	110
Fig CCC 2nd Highest-Volume Surgeon (Median Duration).....	110
Fig DDD 3rd Highest-Volume Surgeon (Median Duration).....	111
Fig EEE 4th Highest-Volume Surgeon (Median Duration).....	111
Fig FFF 5th Highest-Volume Surgeon (Median Duration)	112

List of Abbreviations

AKUH	Aga Khan University Hospital
AHP	Analytic Hierarchy Process
CVC	Central Venous Catheter
DFG	Directly-Follows Graph
EHR	Electronic Health Record
KPI	Key Performance Indicator
MCMC	Markov Chain Monte Carlo
OR	Operating Room
PACU	Post-Anaesthesia Care Unit
PM4Py	Process Mining for Python (software library)
RTLS	Real-Time Location System
WHO	World Health Organization

Abstract

Surgical departments are among the most operationally complex units in any hospital, where coordination across operating rooms, surgeons, and clinical staff directly shapes patient outcomes and institutional efficiency. Surgical scheduling at most tertiary hospitals relies on fixed, procedure-based time estimates, despite substantial, undocumented variability in case duration across surgeons and specialties. Hospital information systems already record timestamped perioperative events for every case, yet this data is rarely analyzed to establish a reliable process baseline, separate genuine rare pathways from noise, or quantify where delays concentrate.

This thesis applies process mining, principally the Heuristic Miner, to reconstruct real surgical workflow directly from event log data. Unlike prior work, which treats structural discovery and duration analysis as separate exercises, this thesis connects the two: discovered pathways are annotated with performance data so that structural reliability and timing delay are examined together. Baseline discovery produced an interpretable but sparse model with no way to assess the reliability of infrequent transitions. This motivated a Bayesian extension to the dependency measure, a Beta-Binomial posterior mean embedded within discovery itself, validated on a real-world dataset of 162,286 perioperative cases. The Bayesian model surfaced nearly double the edges of the classical baseline, distinguishing a confirmed core pathway from high-confidence clinical exceptions and low-frequency edges flagged for review, including a checklist-sequencing deviation invisible to the frequency-only baseline. Temporal analysis further revealed a severe mean-median divergence on a key transition, independently corroborated elsewhere in the literature, with delay patterns differing sharply by specialty.

These findings were translated into operationally realistic recommendations spanning structural, temporal, and governance considerations, showing that a statistically calibrated extension to an established discovery algorithm can surface process insight that frequency counts alone cannot.

Keywords: Process Mining, Heuristic Miner, Bayesian Inference, Surgical Scheduling, Perioperative Workflow, Healthcare Process Discovery, Operating Room Efficiency

Chapter 1

Introduction

Modern healthcare systems are under constant pressure to improve operational efficiency while maintaining high standards of patient care. Among the most resource-intensive and complex areas of any hospital is the surgical department, where coordination between operating rooms (ORs), surgeons, anesthesiologists, nursing staff, and administrative teams directly shapes both patient outcomes and institutional performance (Figure 1). The operating room is simultaneously one of the most expensive resources a hospital operates and one of the hardest to schedule efficiently: every idle OR-minute represents lost capacity that could have gone to a waiting patient, and every overrun pushes subsequent cases later in the day, compounding delay across the entire surgical list (Van Veen-Berkx et al. 2015).

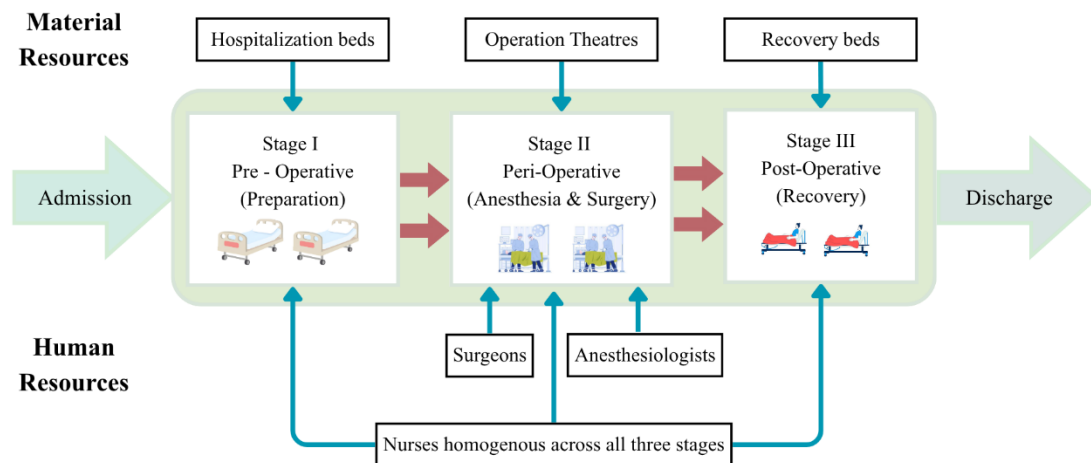


Figure 1 A typical patient journey through the surgical pathway

Surgical care is not a single event but a long chain of interdependent clinical and administrative steps. A patient is admitted, transferred toward the operating suite, prepared, anesthetized, operated on, closed, moved to recovery, and eventually discharged. Each of these steps is carried out by a different person or team, often under time pressure, and is a potential point of delay, miscommunication, or deviation from the intended protocol. Understanding how this chain of events unfolds in practice, as opposed to how it is assumed to unfold on paper, remains a persistent and largely unsolved operational problem.

1.1 The Operational Problem

The motivating problem behind this research is concrete, operational, and immediately recognizable to surgical and hospital staff, rather than a purely academic question.

The most visible symptom of the problem is a scheduling one. Like most hospitals, tertiary care institutions typically allocate fixed or estimated slot durations per procedure type. In practice, however, the duration of nominally "the same" procedure varies enormously depending on who performs it. For a procedure with a planned forty-five-minute slot, one surgeon may complete the operation in twenty minutes, while another, performing a clinically equivalent case, may take seventy-five minutes (Figure 2). Without accounting for this variability, two failure modes recur constantly: the operating room sits idle while a slower case finishes, delaying every subsequent case on the list, or a faster case finishes well ahead of schedule, leaving the room empty until the next patient, who was scheduled according to the slower assumption, is actually ready. Both situations waste the same scarce resource, and at high case volumes, even small per-case inefficiencies compound into a substantial amount of lost theatre capacity over a year.

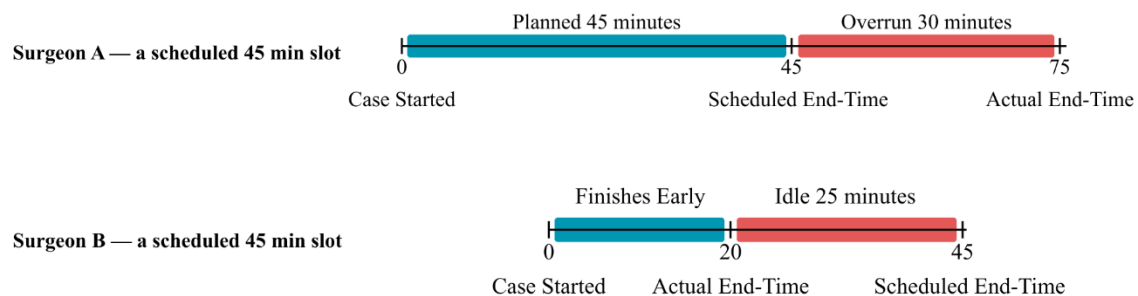


Figure 2 Fixed scheduling fails both ways. Surgeon A's overrun delays the next case, while B's early finish leaves the OR idle.

Beneath this scheduling symptom, three deeper and more fundamental gaps were identified as the actual subject of this research:

- **There was no data-derived baseline understanding of the real surgical process.** Hospitals, like most healthcare institutions, document intended clinical workflows. For instance, the sequence implied by the WHO Surgical Safety Checklist, which mandates a sign-out "timeout" verification before wound closure. However, no analysis typically exists of what a hospital's own perioperative information system actually recorded as

having happened, across the hundreds of thousands of surgical cases logged over years of operation. Without this baseline, any claim about "the surgical process" at a given institution is necessarily anecdotal, based on individual clinicians' personal experience of a small number of cases, rather than on the full record.

- **There was no principled way to separate meaningful rare events from noise.** Surgical event logs, by their nature, contain both a dominant, high-volume "normal" pathway and a long tail of less common variations. Some of these variations are clinically legitimate. For example, an emergency case that bypasses elective pre-operative steps because there is no time for them. Others may simply be data entry inconsistencies, duplicate records, or system artefacts. Distinguishing between the two categories using raw frequency counts alone is unreliable: a transition observed only a handful of times could be either a rare-but-real clinical pathway, or it could be noise, and a simple count cannot tell the difference. Without a statistically grounded way to make this distinction, hospital staff are left with only two unsatisfying options: ignoring all deviations from the dominant pathway or treating every deviation as equally worthy (or unworthy) of attention.
- **There was no quantified, log-derived picture of where and for whom delays actually occur.** Clinical and administrative staff often hold strong informal beliefs about which surgeons, procedure types, or process stages tend to run long, but these beliefs are rarely tested systematically against the timestamped record. A delay of a few minutes that recurs in a high-volume transition can carry a far larger cumulative operational cost than a dramatic but rare multi-hour delay, yet without quantification there is no way to know which delays actually deserve intervention.

1.2 Perioperative Scheduling Challenges: A Broader Context

These problems are not unique to a hospital. They reflect difficulties that are well documented in the international perioperative management literature, and recognizing this broader context helps frame why a systematic, data-driven approach, rather than incremental local fixes, is warranted.

- **Downstream capacity bottlenecks:** A recurring and well-studied cause of OR-level delay is not the operating room itself but the unavailability of downstream capacity,

particularly post-anesthesia care unit (PACU) beds. When recovery capacity is constrained, the next patient cannot be moved out of the OR, which in turn prevents the following case from starting on time, a delay that originates downstream but propagates backward through the entire day's schedule (Lyons et al. 2023).

- **Surgical duration variability:** Procedure durations vary substantially even within what scheduling systems treat as a single, homogeneous category of "the same operation." This variability is one of the most frequently cited and least adequately addressed drivers of unpredictable OR workflows and unplanned overtime in the perioperative scheduling literature.
- **Communication and coordination gaps:** Poor inter-professional collaboration between surgeons, anesthesiologists, and nursing staff measurably reduces OR utilisation. In a multicenter study of sixteen Dutch hospitals, raw OR utilisation as low as 85% was observed in centers with weaker multidisciplinary coordination, and structured multidisciplinary planning meetings produced a statistically significant improvement in utilisation where they were implemented (Van Veen-Berkx et al. 2015). More recent work continues to confirm that this coordination gap has not been resolved by digitization alone; a 2025 review found that scheduling inefficiency remains one of the most frequently cited operational challenges across digitized surgical departments worldwide (Aversano et al. 2025).
- **Logistical and resource constraints:** Equipment and staff shortages, alongside administrative requirements, are a documented and non-trivial contributor to surgical delay. In a study of maxillofacial trauma repair timing, OR logistical factors, staff or equipment unavailability, accounted for delay in 12.3% of cases, independent of any clinical factor (Radabaugh et al. 2016).

Pakistan's tertiary healthcare institutions operate under all of these same structural pressures, typically with less slack capacity to absorb inefficiency than equivalent institutions in higher-resource settings. This makes improved process visibility, extracting more value from data the hospital already collects, rather than requiring new capital investment, a particularly cost-effective and immediately actionable lever for improvement.

1.3 Limitations of Current Process Discovery in Healthcare

Process discovery techniques have been applied to healthcare settings with growing frequency over the past decade, but two specific limitations are directly relevant to the problem described above, and motivate the methodological direction taken in this thesis.

Classical discovery algorithms such as the Alpha Miner treat every observed activity transition with equal statistical confidence regardless of how many times it was actually observed, which is poorly suited to clinical logs where genuine rare pathways sit alongside data-entry noise (W. M. P. Van Der Aalst 2010). In a surgical event log specifically, this is not a minor technicality. A rare transition observed only a handful of times might represent a genuine, clinically valid exception, an emergency case that bypasses a preparatory step, for instance, or it might represent nothing more than a duplicate timestamp or a data-entry artefact. An algorithm with no mechanism for weighing how much a given observation count should be trusted cannot tell these two situations apart, and is therefore forced either to treat every rare transition as equally real, producing an overly dense and uninterpretable model, or to discard rare transitions altogether, silently erasing exactly the clinically meaningful deviations a hospital would most want visibility into.

Separately, even algorithms designed to filter noise by frequency, such as the Heuristic Miner, do so using a fixed ratio that does not distinguish a transition seen many thousands of times from one seen only a handful of times, leaving no principled basis for deciding how much trust a given rare transition deserves (Weijters and Van Der Aalst 2003). This limitation has persisted in the literature well beyond the algorithm's original formulation. A 2025 tertiary review synthesizing prior systematic reviews of process mining in healthcare confirmed that, despite the Heuristic Miner remaining one of the most widely applied discovery algorithms in clinical settings precisely because of its noise tolerance, no widely adopted variant of the algorithm has yet incorporated a sample-size-aware statistical correction directly into its dependency calculation, with existing probabilistic extensions instead applied only after a model has already been discovered (Santos et al. 2025). This confirms that the specific gap motivating this thesis, an uncalibrated

dependency measure at the heart of an otherwise noise-tolerant algorithm, remains open even in the most recent literature available at the time of writing.

A fuller treatment of these limitations, and of the broader literature on process mining in surgical and perioperative settings, is presented in Chapter 3.

1.4 Justification of the Research

The three gaps identified in Section 1.1, the absence of a real-process baseline, the inability to separate genuine rare pathways from noise, and the lack of quantified delay data, share a common character: they are all questions about how care is sequenced and coordinated across a large population of cases, rather than questions about any individual patient's clinical outcome. Answering them convincingly requires a method capable of reconstructing actual process behavior directly from the timestamped records a hospital's information systems already generate, at a scale far beyond what manual chart review or clinician recall could achieve. This requirement, and the family of computational methods developed to meet it, is introduced in Chapter 2.

This thesis develops and applies such a method to a real-world perioperative event log, with the explicit goal of producing not only a descriptive model of surgical workflow, but a set of concrete, operationally realistic recommendations that scheduling and quality teams can act on directly. Producing actionable recommendations, rather than a purely descriptive analysis, is treated throughout this thesis as a primary outcome of the research, not an afterthought; this is reflected in the dedicated recommendations section that follows the results in later chapters.

This research is justified on three grounds. First, methodologically, the existing process mining literature lacks a discovery-stage statistical treatment of rare-versus-frequent transitions within a widely used, computationally tractable algorithm; existing Bayesian treatments either replace the discovery formalism entirely or apply probabilistic reasoning only after discovery is complete (Section 3.5). A discovery-stage correction that preserves the Heuristic Miner's tractability while improving its statistical rigour closes a specific, demonstrable gap.

Second, empirically, healthcare process mining studies addressing surgical and perioperative scheduling specifically remain comparatively rare relative to general clinical process mining, and even rarer at the scale and case-volume of a large tertiary-care log; most reviewed studies operate on smaller or synthetic datasets (Section 3.4). Applying the proposed method to a genuinely large, real-world clinical event log is necessary to establish whether the approach holds up outside idealised or small-scale settings.

Third, practically, hospitals already collect the timestamped data this research requires as a routine by-product of clinical documentation. A method that surfaces actionable findings from data already being collected, without requiring new instrumentation, new systems, or additional clinical burden, is more readily adoptable than approaches that demand new infrastructure. This justifies framing the contribution of this thesis not only as a methodological advance but as an operationally realistic one.

1.5 Problem Statement

Surgical scheduling decisions are typically made without a statistically grounded understanding of real surgical workflow behavior, including the reliability of process pathways, the prevalence of rare but clinically significant deviations, and the duration of process-level delays. As a result, operating room scheduling often fails to reflect real-world surgical and procedural variability, leaving hospitals unable to identify which process inefficiencies warrant intervention.

1.6 Research Questions

To guide this research and address the problem identified above, the following research questions are defined:

- **RQ1:** How can process mining be used to discover a reliable baseline model of real-world surgical workflow from a large-scale hospital perioperative event log, both for the institution as a whole and for clinically meaningful subgroups: surgical specialty, theatre location, and individual high-volume surgeons?
- **RQ2:** Can a statistically calibrated dependency measure, rather than a purely frequency-based one, produce a process model that more reliably distinguishes genuine

rare clinical pathways from low-confidence noise, and does this improve the clinical interpretability of the discovered surgical process model?

- **RQ3:** Where, and for whom, do surgical process delays concentrate when the same event log is analyzed temporally rather than structurally, and how do these temporal findings relate to the structural pathways discovered under RQ1 and RQ2?
- **RQ4:** What concrete, operationally realistic scheduling and process-improvement recommendations can be derived jointly from the structural and temporal analyses, such that they are directly usable by the hospital's surgical and scheduling teams?

1.7 Contributions

This thesis makes the following contributions:

- A large-scale, real-world process mining analysis of a perioperative event log, producing baseline discovered process models for the hospital overall and for thirteen surgical specialties, four theatre locations, and the five highest-volume surgeons, to our knowledge, the first such analysis conducted at this institution.
- A methodological extension to a standard process discovery algorithm that improves its statistical treatment of rare process transitions while preserving its behavior on high-frequency ones, developed specifically to address the noise-versus-signal problem identified in Section 1.1, and detailed in Chapter 4.
- A temporal, duration-based analysis layer applied to the same event log, quantifying where process delays concentrate and connecting these findings back to the structurally discovered pathways rather than treating timing analysis as a separate, disconnected exercise.
- A set of recommendations for surgical scheduling and process improvement, derived directly from the discovered models and filtered for operational realism, intended for direct use by the hospital's surgical and scheduling teams.

1.8 Scope and Limitations of the Study

This research is scoped specifically to the structural and temporal analysis of the directly-follows relationships between recorded perioperative activities, that is, what activity tends to follow what other activity, and how long that transition tends to take. This positions the

work within what the process mining literature calls process discovery and process enhancement (Section 2.1), rather than within conformance checking against an externally imposed reference model, clinical outcome prediction, or resource-level (staff or equipment) attribution. This boundary matters because, as recent reviews of healthcare process mining note, these are genuinely distinct analytical tasks that are often conflated in applied work: a 2025 tertiary synthesis of prior systematic reviews found that studies combining discovery with downstream prediction or causal outcome analysis frequently underdeliver on both, producing process models that are too coarse for reliable discovery and prediction models that are too data-hungry for the underlying log size (Santos et al. 2025). This thesis deliberately avoids that conflation by treating discovery and temporal enhancement as the sole analytical objective, with recommendations (Chapter 6) derived from those findings rather than from a separate predictive layer.

The dataset itself spans cases recorded across multiple years of hospital operation and across both inpatient and same-day surgical settings, drawn from a single tertiary care institution (Chapter 5). This single-institution scope is a deliberate, stated boundary rather than an oversight: a 2025 systematic review of process mining applications in healthcare explicitly flagged single-site validation as the norm rather than the exception across the field, noting that very few reviewed studies test discovered models against a second institution's data (Aversano et al. 2025). This thesis does not depart from that norm, and no claim is made that the specific structural or temporal findings reported in Chapter 5 would replicate identically at a different hospital, even though the underlying statistical method (Chapter 4) is institution-agnostic by design and applicable to any sufficiently large, timestamped perioperative log.

Cleaning and filtering decisions applied to the raw data, and their implications for which cases are represented in the final analysis, are described in full in Chapter 5.

Findings throughout this thesis are interpreted strictly at the level of process and operational behavior: which transitions occur, how often, and over what timeframe. No claims are made in this thesis about the clinical quality of care provided to any individual patient, about causal relationships between process deviations and patient outcomes, or

about staffing, equipment, or resource-level attribution of delay. Recent perioperative scheduling literature has emphasized that process-timing analysis and clinical-outcome analysis require different data, different statistical assumptions, and different ethical safeguards (Celik et al. 2025), and this thesis follows that separation deliberately rather than attempting to bridge it.

The limitations of the specific methodology and results presented in this thesis, as distinct from the scope boundaries stated here, are discussed separately in Section 6.3.

Chapter 2

Background Knowledge

To support the understanding of the proposed methodology, this chapter introduces the core concepts of process mining that underpin this research. It covers what process mining is, why it is particularly well suited to analyzing healthcare and surgical data, how a process model is discovered from a raw event log, and the mathematical foundations of the discovery algorithm used as the basis for this work. The motivation for the specific methodological choices made in this thesis, including the algorithm selection rationale and the statistical extension proposed in this research, is presented separately in Chapter 4, once this background has been established.

2.1 Process Mining and Its Role in Healthcare Data Analysis

Process mining is a family of data-driven techniques that sit at the intersection of data science and business process management. Its central idea is simple but powerful: rather than relying on a manually drawn flowchart or a policy document to describe how a process is *supposed* to work, process mining reconstructs how a process *actually* worked by analyzing the digital footprints, timestamped event records, that information systems generate as a natural by-product of doing their job (W. M. P. Van Der Aalst 2010).

Every time a hospital information system records that a patient was admitted, that a drug was dispensed, or that a procedure step occurred, it generates an entry consisting, at minimum, of three pieces of information: a **case identifier** (which patient or episode the event belongs to), an **activity label** (what happened), and a **timestamp** (when it happened). A collection of such entries is called an **event log**. Formally, an event log L is a multiset of traces, where each trace $\sigma = \langle e_1, e_2, \dots, e_n \rangle$ is a time-ordered sequence of events belonging to a single case. Process mining algorithms take this event log as input and produce, as output, a process model, typically represented as a directed graph in which nodes are activities and edges represent observed transitions between them.

Process mining is generally described in terms of three complementary tasks (W. M. P. Van Der Aalst 2010):

- **Process discovery:** Constructing a process model purely from the event log, with no prior model assumed. This is the task addressed in this thesis.
- **Conformance checking:** Comparing an existing, normative process model (such as a clinical protocol) against the event log, to quantify where and how real behavior deviates from the intended process.
- **Process enhancement:** Using the event log to extend or improve an existing process model, for example by annotating it with performance (timing) information, as is done in the temporal analysis component of this thesis.

Figure 3 situates these three tasks within the broader hospital ecosystem. The hospital information system (HIS) continuously records the activity of physicians, nurses, and patients as event logs, which in turn can be used to discover a process model, check its conformance against the recorded behavior, or enhance it with additional performance information. The process model, once derived, can then be used to analyze and inform how the hospital's clinical and administrative processes actually function.

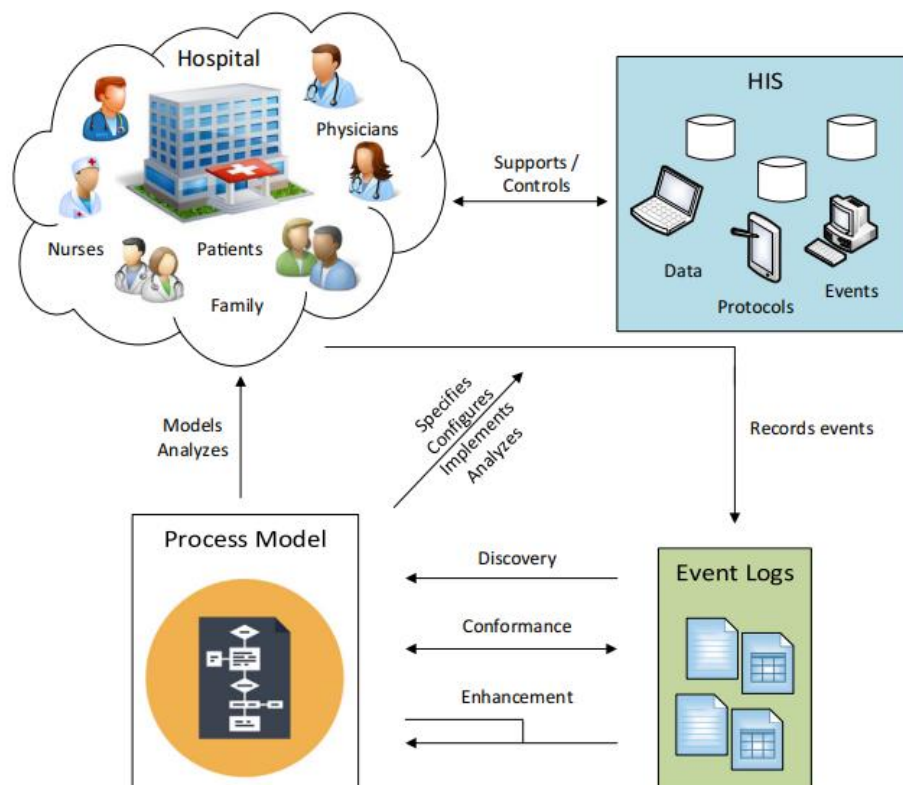


Figure 3 The role of process mining in healthcare process management (Rojas et al. 2016).

2.1.1 Why Process Mining Suits Medical and Surgical Data

Healthcare processes and surgical processes, in particular, have characteristics that make them especially good candidates for process mining, and simultaneously especially difficult to analyze using simpler approaches such as manually reviewing patient charts or relying on staff recollection.

First, hospital information systems already capture exactly the kind of data process mining requires. Every patient passing through a perioperative pathway generates a trail of timestamped milestones, admission, transfer, preparation, incision, recovery, discharge, as a routine consequence of clinical documentation. This means process mining can be applied *retrospectively*, at scale, without requiring any new instrumentation or additional burden on clinical staff (Mans 2015).

Second, clinical processes are simultaneously highly structured and highly variable, a combination that is difficult for either purely qualitative (flowchart-based) or purely statistical (aggregate KPI) approaches to capture well. A flowchart describes the *intended* pathway but says nothing about how often it is actually followed, nor about the many legitimate clinical exceptions (emergencies, complications, comorbidities) that cause real cases to deviate from it. An aggregate KPI, such as an average length of stay, compresses this same variability into a single number that obscures structural details like *which* deviations occur, and how often each one occurs, that is operationally most useful. Process mining occupies the middle ground: it preserves the full structure of how cases actually flow through the system while still allowing that structure to be quantified statistically.

Third, and most relevant to this thesis, process mining is uniquely suited to surfacing the gap between assumed and actual practice in safety-critical clinical workflows. A discovered model can reveal, with the full weight of the data behind it, whether a mandated safety step (such as the WHO Surgical Safety Checklist sign-out) is actually occurring at the point in the process where protocol requires it, a question that is very difficult to answer reliably through any method other than analyzing the timestamped record directly across a very large number of cases.

These properties have made process mining an increasingly active area of healthcare informatics research over the past decade, applied to settings ranging from oncology care pathways to emergency department triage and, as in this thesis, surgical and perioperative workflows (Rojas et al. 2016).

This trend has accelerated further in the most recent literature, with 2025 systematic and tertiary reviews confirming process mining's expanding footprint specifically in operating room scheduling and surgical pathway analysis (Aversano et al. 2025; Santos et al. 2025).

2.2 Event Logs and the Directly-Follows Relation

The starting point for almost all classical process discovery algorithms is the **directly-follows relation**. Given an event log L , activity B is said to *directly follow* activity A within a given case if B occurs immediately after A in that case's trace, with no other activity recorded in between. The directly-follows count, denoted $f(A \rightarrow B)$, is simply the number of times this direct succession is observed across the entire log.

This simple counting operation is the raw material from which a process model is built. Conceptually, if $f(A \rightarrow B)$ is large relative to other transitions out of A , this is taken as evidence that A is reliably followed by B in the real process, and an edge $A \rightarrow B$ is drawn in the discovered model. The central methodological question that distinguishes different process discovery algorithms from one another is *how* this raw frequency information should be turned into a decision about which edges belong in the model and which do not, a question taken up in the remainder of this chapter.

Table 1 shows a small, illustrative fragment of what a single case's raw event data looks like before any discovery algorithm is applied. Each row is one event: a case identifier, an activity label, and a timestamp. A real event log contains many thousands of such cases, each contributing its own sequence of rows, but the structure is always this simple at the level of an individual record.

Case ID	Activity	Timestamp
C-1042	Admission	2023-03-11 08:02:15
C-1042	Patient Sent	2023-03-11 08:41:02
C-1042	Patient Arrived	2023-03-11 08:43:50
C-1042	Cleaning Start	2023-03-11 08:50:11
C-1042	Incision	2023-03-11 09:35:27

Table 1 Illustrative fragment of an event log for a single case

Reading this fragment as a sequence of directly-follows produces the directly-follows graph shown in Figure 4, where each edge is annotated with how many times that transition was observed.

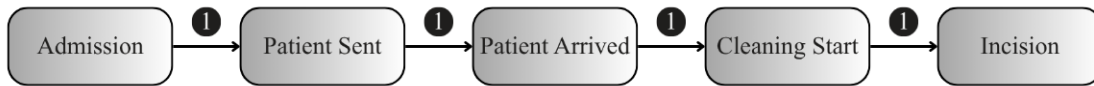


Figure 4 Directly-follows graph derived from the event log fragment in Table 1.

2.3 Process Discovery Algorithms

A number of algorithmic families have been developed for process discovery, differing primarily in how they convert directly-follows information into a model, and what trade-off they strike between **fitness** (how much of the observed behavior the model can reproduce), **precision** (how much *extra*, unobserved behavior the model allows), and **simplicity** (how easy the resulting model is to interpret).

The Alpha Miner: (W. Van Der Aalst et al. 2004) was the first formally defined process discovery algorithm. It identifies basic ordering relations between activities (direct succession, parallelism, and lack of any direct relation) directly from the log and uses these to construct a model guaranteed to reproduce every trace in the log. This guarantee is also its central weakness. Because the Alpha Miner must account for *every* observed sequence,

including rare ones, it has no built-in mechanism for distinguishing a genuinely important rare pathway from simple noise. In large, messy logs, this tends to produce models that are complex and computationally expensive to construct.

The Heuristic Miner: This algorithm was developed to overcome this limitation. (Weijters and Van Der Aalst 2003), (Weijters et al. 2006) Rather than forcing the model to reproduce every observed sequence, it calculates a dependency measure for each activity pair, indicating how strongly and consistently one follows another, and retains only edges whose dependency scores and frequencies exceed configurable thresholds. As a result, it is naturally noise-tolerant: low-frequency transitions from data exceptions are filtered out, while the dominant process behavior is preserved in a simpler, more interpretable graph. This contrast in philosophy translates directly into a contrast in output: as Figure 5 illustrates, the Alpha Miner's insistence on reproducing every observed relation yields a densely connected model with many crossing edges, whereas the Heuristic Miner's threshold-based filtering yields a sparse, near-linear model reflecting only the dominant pathway.

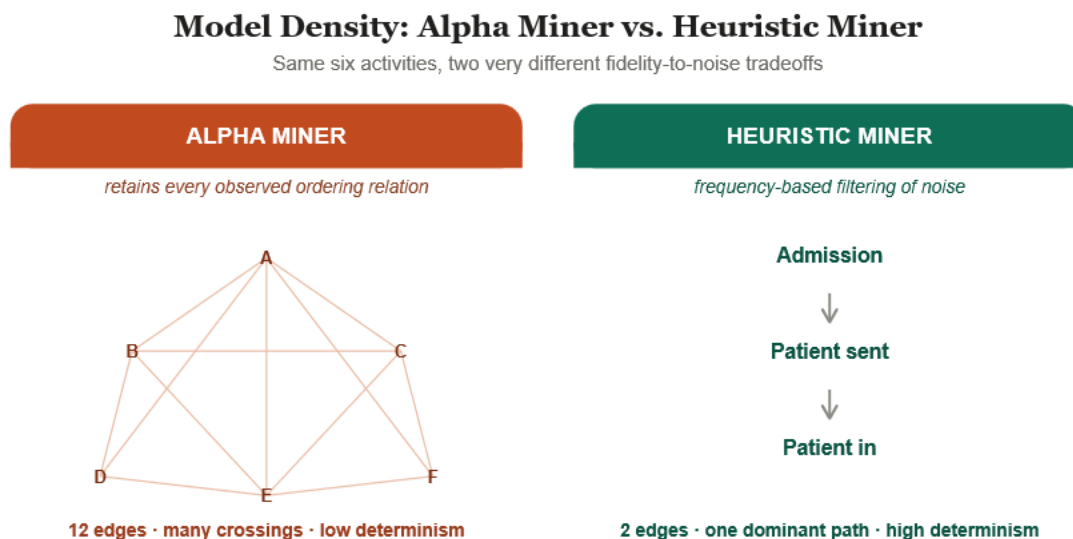


Figure 5 Schematic contrast between Alpha Miner and Heuristic Miner model density.

The classical Heuristic Miner dependency measure between two activities A and B is defined as:

$$\text{dep}(A, B) = \frac{f(A \rightarrow B) - f(B \rightarrow A)}{f(A \rightarrow B) + f(B \rightarrow A) + 1}$$

where $f(A \rightarrow B)$ is the number of times B directly follows A in the log, and $f(B \rightarrow A)$ is the count of the reverse transition. This quantity is bounded in $(-1, 1)$: a value close to 1 indicates that the transition from A to B strongly and consistently dominates over its reverse, a value close to -1 indicates the opposite, and a value close to 0 indicates that the two activities have no clear directional relationship, that is, they may occur in either order, or only one of them may occur with any regularity relative to the other. An edge $A \rightarrow B$ is retained in the final model only if $\text{dep}(A, B)$ exceeds a chosen dependency threshold and the raw frequency $f(A \rightarrow B)$ exceeds a chosen minimum-frequency threshold, both of which are configurable parameters of the algorithm. The "+1" in the denominator is a small-sample correction: it prevents the measure from being undefined when a pair is observed only in one direction, and softens the score slightly for very low-frequency pairs, although, as is discussed in detail in Chapter 4, this correction is a fixed, ad-hoc adjustment rather than a principled statistical treatment of sampling uncertainty.

The Heuristic Miner's discovery procedure can be stated as a concrete algorithm, summarized below, to make explicit how the dependency measure, frequency threshold, and edge-construction step combine into a final model.

Algorithm 2.1 Classical Heuristic Miner

Input: Event log L , dependency threshold θ_{dep} , frequency threshold θ_{freq}

Output: Heuristic net $H = (N, E)$

1. Initialize $f[A][B] \leftarrow 0$ for all activity pairs
 2. FOR each case c in L DO
 - FOR each consecutive pair (A, B) in c DO
 - $f[A][B] \leftarrow f[A][B] + 1$
 - END FOR
 - END FOR
 3. Initialize $E \leftarrow \emptyset$
 4. FOR each pair (A, B) where $f[A][B] > 0$ DO
 - $\text{dep}(A, B) \leftarrow (f[A][B] - f[B][A]) / (f[A][B] + f[B][A] + 1)$
-

```

    IF  $\text{dep}(A,B) \geq \theta_{\text{dep}}$  AND  $f[A][B] \geq \theta_{\text{freq}}$  THEN
       $E \leftarrow E \cup \{(A, B, \text{dep}(A,B))\}$ 
    END IF
  END FOR

```

5. Construct $H = (N, E)$ where N is the set of all activities in L
 6. RETURN H
-

This algorithm makes explicit the two design choices that distinguish the Heuristic Miner from the Alpha Miner: the dependency measure itself, which converts a raw frequency pair into a bounded directional score, and the two-threshold filtering rule, which jointly enforces that a retained edge must be both directionally consistent and sufficiently frequent. The algorithm also generalizes naturally to "loop" and "long-distance" dependency variants not used in this thesis, which extend the same frequency-ratio logic to two-step loops ($A \rightarrow B \rightarrow A$) and to activity pairs that are not directly adjacent in a trace, by computing analogous ratios over the relevant counts (Weijters et al. 2006). These extensions are applied in this thesis because they illustrate that the dependency-measure framework, the same one extended with a Bayesian formulation in Chapter 4, is the Heuristic Miner's central and most flexible design feature, which is precisely why it was selected as the basis for the statistical extension proposed in this thesis rather than the Alpha or Inductive Miner.

Figure 6 illustrates this with two contrasting examples: a high-frequency, strongly directional pair ($A \rightarrow B$) that clears a typical dependency threshold and is retained, against a low-frequency, weakly directional pair ($C \leftrightarrow D$) that falls below the threshold and is filtered out.

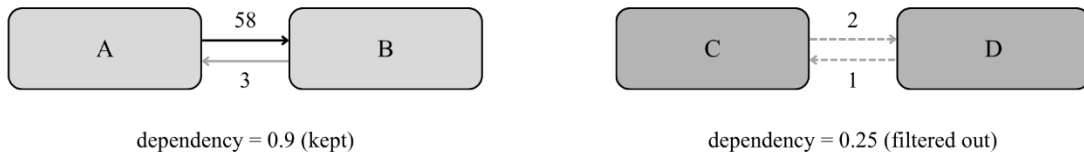


Figure 6 Examples of the Heuristic Miner dependency measure for a retained edge (left) and a filtered edge (right).

The Inductive Miner: (Leemans et al. 2013) represents a third major family, guaranteeing a sound model by recursively partitioning the log by control-flow pattern. It produces

simpler, more formally well-behaved models than the Alpha Miner, but does not natively expose a tunable, per-edge confidence measure, limiting its suitability for the fine-grained statistical reasoning this thesis requires."

The trade-offs between these algorithm families and the specific reasoning that led to the selection of the Heuristic Miner as the basis for this research, including a worked comparison against the Alpha Miner on this dataset, are presented in Chapter 4, alongside the statistical extension proposed in this thesis.

Table 2 summarizes the three algorithm families discussed above side by side, as a point of reference for the comparison developed in Chapter 4.

Algorithm	Core mechanism	Main strength	Main weakness
Alpha Miner	Derives ordering relation directly from the log	Conceptually simple	No mechanism to discount noise or rare behavior
Heuristic Miner	Computes a frequency-based dependency measure per activity pair	Noise-tolerant by design	Treats all observed frequency counts as equally reliable
Inductive Miner	Recursively partitions the log by detected control-flow pattern	Guarantees a structurally sound model	Does not expose a tunable, per-edge confidence measure like Heuristic Miner

Table 2 Summary comparison of the three process discovery algorithm families

Figure 7 below highlights the comparison of the three process discovery algorithms with respect to their structural density comparison.

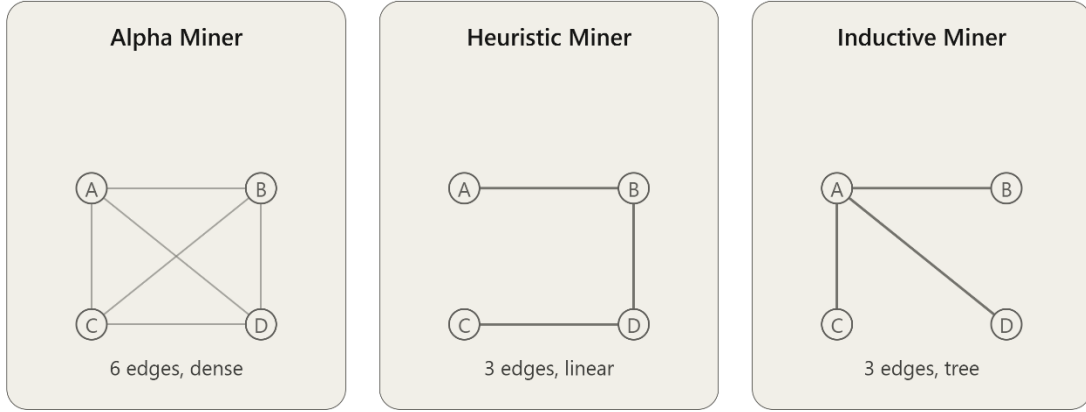


Figure 7 Structural density comparison across the three dominant process discovery algorithm families on the same four activities.

2.4 Process Enhancement: Directly-Follows Graphs

The discovery algorithms described above operate purely on the *control-flow* perspective of an event log, that is, which activities occur and in what order, independent of how much time elapses between them. However, an event log that contains timestamps carries additional information that a purely structural model discards: the actual *duration* of each transition.

A **performance-annotated Directly-Follows Graph (DFG)** retains the same node-and-edge structure as a standard directly-follows graph but replaces the frequency label on each edge with a statistic computed over the time elapsed between the two connected activities across all cases exhibiting that transition, typically the mean or median duration. Formally, for a transition $A \rightarrow B$ observed across a set of cases C_{AB} , the performance DFG computes:

$$\bar{t}_{A \rightarrow B} = \frac{1}{|C_{AB}|} \sum_{c \in C_{AB}} (\text{timestamp}_B^{(c)} - \text{timestamp}_A^{(c)})$$

for the mean variant, or the corresponding median of the same set of durations for the median variant. The choice between mean and median is not a stylistic detail because clinical and operational durations are frequently right-skewed. The mean and median can diverge on the same edge, and this divergence itself carries diagnostic information about the shape of the underlying delay distribution, a point returned to in Chapter 5.

Performance DFGs do not perform *process discovery* in the sense of deciding which edges belong in the model; rather, they are typically applied as a *process enhancement* step, overlaid on a structural backbone (whether discovered directly via the directly-follows count, or via a control-flow discovery algorithm such as the Heuristic Miner) to add a temporal dimension to an already-established structural understanding of the process. This is precisely the role performance DFGs play in this thesis: the temporal analysis presented in later chapters is built on the same activity backbone established by the structural discovery stage, rather than treated as an independent or disconnected exercise.

2.5 Summary

This chapter has introduced the foundational vocabulary and mechanics of process mining: the event log as the raw data structure, the directly-follows relation as the basic unit of structural analysis, the family of discovery algorithms: Alpha Miner, Heuristic Miner, and Inductive Miner, that convert this raw relation into an interpretable process model, and the extension of this structural view into the temporal domain via performance-annotated directly-follows graphs. It has also argued, drawing on the existing healthcare process mining literature, why these techniques are particularly well matched to the structure of surgical and perioperative data specifically. The remaining methodological gap that how the Heuristic Miner's dependency measure can be made more statistically rigorous when applied to a real, large-scale clinical log containing activity pairs of vastly different frequency, and why this matters more in a surgical setting than it might in a more homogeneous industrial process log, is the subject of Chapter 4, which presents the proposed methodology of this thesis in full, including the Bayesian extension developed as its central technical contribution.

Chapter 3

Literature Review

This chapter reviews the existing literature on process mining in healthcare, with particular attention to its application in surgical and perioperative settings, the algorithmic choices made by prior studies, and the treatment, or absence, of statistical uncertainty in process discovery. The review follows a structured methodology, presented in Section 3.1, before the review itself (Section 3.2), a summary of publication venues and research groups (Section 3.3), the master table and per-paper synthesis (Section 3.4), and the gaps this thesis addresses (Section 3.5).

3.1 Review Methodology

3.1.1 Review Type and Rationale

A literature review was conducted for this thesis to reduce selection bias and follow a structured, replicable protocol for identifying, screening, and synthesizing relevant studies. The review focused on three intersecting areas: process mining applied to healthcare and surgical workflows, statistical and probabilistic treatments of process discovery, and the comparative behavior of discovery algorithms (Alpha Miner, Heuristic Miner, Inductive Miner) on clinical data.

3.1.2 Search Strategy

Searches were conducted across Google Scholar, PubMed, IEEE Xplore, ScienceDirect, and arXiv, using the search queries summarized in Table 3, restricted to publications from the last decade and a half to capture both the foundational process mining literature and recent developments.

Query ID	Search Query
Q1	"process mining" AND "operating room" AND ("scheduling" OR "surgical workflow")
Q2	"process mining" AND "healthcare" AND "review"

Q3	"heuristic miner" AND ("alpha miner" OR "process discovery") AND healthcare
Q4	"Bayesian" AND "process mining" AND ("dependency" OR "uncertainty")
Q5	"conformance checking" AND "clinical pathway" OR "surgical safety checklist"
Q6	"process mining" AND "surgical training" OR "perioperative"
Q7	"fuzzy miner" AND "healthcare" OR "patient pathway"
Q8	"surgery duration" AND ("prediction" OR "estimation") AND "operating room scheduling"

Table 3 Search queries used to identify relevant studies

3.1.3 Screening Process

The methodological steps followed are illustrated in Figure 8. An initial pool of 167 records was identified across the five databases. After removing 46 duplicate or near-duplicate entries, 121 records were screened by title and abstract, of which 83 were excluded as irrelevant, covering process mining applications entirely outside healthcare, or healthcare informatics papers with no process discovery component at all. The remaining 38 records were assessed in full text, and a further 25 were excluded for lacking a discoverable process mining methodology, or for not addressing either the healthcare/surgical setting or the statistical-uncertainty question central to this thesis.

This left **13 studies** included in the final synthesis.

Following an initial pass through this material, two gaps became apparent in the resulting corpus: no study using the Fuzzy Miner algorithm specifically had been captured, despite Fuzzy Miner being one of the four dominant discovery algorithms identified in the field (Section 3.2.1), and no study addressing surgery duration prediction directly relevant to the scheduling motivation in Chapter 1 had been retrieved. A supplementary targeted search using queries Q7 and Q8 was therefore conducted, yielding three additional studies that met the same inclusion criteria. The final synthesis therefore comprises **16 studies**.

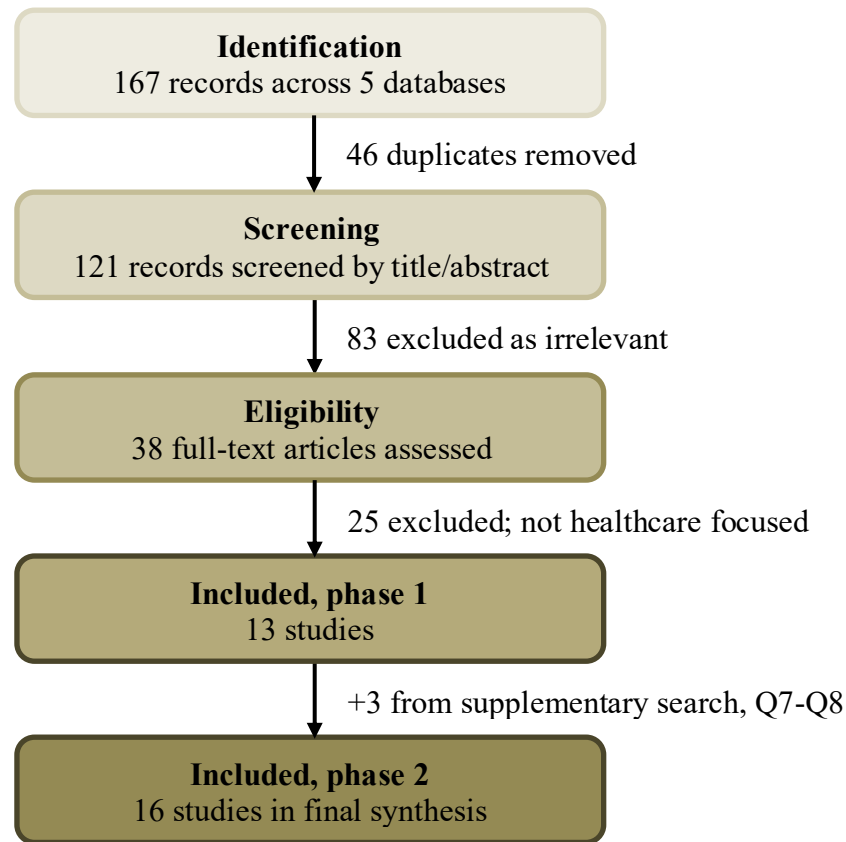


Figure 8 PRISMA flow diagram illustrating the study selection process for the literature review.

3.1.4 Publication Trend Over Time

Figure 8 shows the distribution of the 16 included studies from 2010 to 2025. The range extends back to 2010 to capture the earliest surgery-duration-prediction work identified in the search. Three distinct peaks are visible (2015, 2019, and 2025) separated by genuine gaps with no included publications (2011, 2013–2014, 2016–2018, 2021), consistent with a field whose attention to this specific intersection of problems has come in distinct waves rather than growing smoothly. Twelve of the sixteen included studies (75%) were published in 2019 or later.

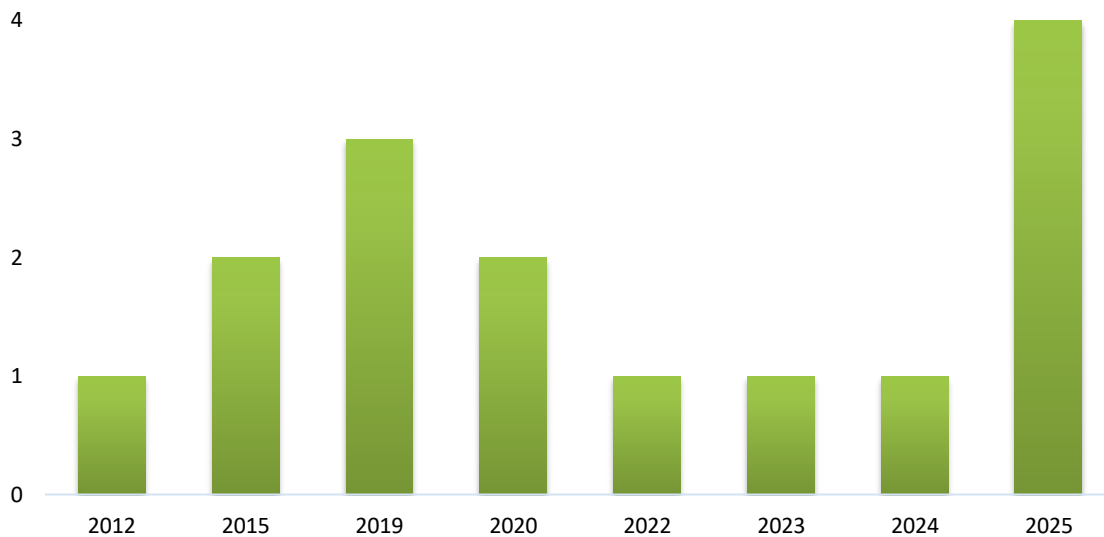


Figure 9 Distribution of selected studies by publication year (2021-2025)

3.2 Review

3.2.1 Process Mining in Healthcare: General Foundations

Process mining's application to healthcare has grown substantially over the past decade, driven by the increasing digitization of hospital information systems and the resulting availability of large-scale, time-stamped clinical event data. (Mans 2015) established much of the methodological foundation for this field by demonstrating through a Dutch hospital case study that process mining could reveal operational realities, such as undocumented rework and inconsistent care pathways, that were invisible to manual process review. (Rebuge and Ferreira 2012) extended this foundational work specifically to emergency care, showing that process discovery could expose systematic discrepancies between intended and observed clinical behavior, a theme that recurs throughout this literature and is directly relevant to the surgical safety checklist sequencing question investigated in this thesis.

More recent systematic reviews confirm that this growth has continued and broadened in scope. A 2025 systematic literature review identified process mining applications spanning emergency room triage, sepsis pathway monitoring, and operating room scheduling, and explicitly highlighted operating room scheduling optimization as one of the field's most active application areas (Aversano et al. 2025). A parallel tertiary study synthesizing

multiple prior systematic reviews of healthcare process mining confirmed that Alpha Miner, Heuristic Miner, Fuzzy Miner, and Inductive Miner remain the four dominant discovery algorithms used across the field, with Heuristic and Fuzzy Miner increasingly preferred over Alpha Miner because of their superior handling of noisy, infrequent behavior, a finding directly relevant to the algorithm selection rationale developed in Chapter 4 of this thesis (Santos et al. 2025).

Fuzzy Miner specifically, the fourth dominant algorithm identified above, alongside Alpha, Heuristic, and Inductive Miner, has been directly applied to hospital-wide patient pathway analysis. (Özdağoglu et al. 2025) used Fuzzy Miner to analyze the complete patient journey of 51,298 patients across all units of a regional Turkish hospital. Their analysis drew on 373,008 individual events. Their findings are notable for this thesis in two respects. First, they report that General Surgery and Cardiology carried among the highest resource utilisation and waiting times in the hospital. This specialty-level pattern is consistent with this thesis's own finding that complex specialties carry disproportionate delay (Chapter 7). Second, and more directly relevant to this thesis's methodological argument, they explicitly note that "the noticeable gaps between mean and median values suggest that these transition times are distributed over a wide range with considerable variability." This provides independent confirmation from an entirely different hospital and dataset. It supports the same mean-median divergence phenomenon identified in this thesis's temporal analysis (Section 7.5). Their study, however, applies Fuzzy Miner's significance-and-correlation framework rather than a Bayesian dependency measure that this research discusses. It also does not address the statistical confidence of rare transitions in the way the Bayesian Heuristic Miner does.

3.2.2 Process Mining for Surgical and Perioperative Scheduling

A smaller, more specialized body of work has applied process mining specifically to surgical scheduling and operating room efficiency, the operational problem motivating this research.

(Di Cunzolo et al. 2023) proposed a framework combining process mining with operations research for interventional radiology scheduling, using process discovery to identify which

procedures were most frequently delayed and feeding this knowledge directly into an optimization model for theatre scheduling. This work is methodologically significant because it demonstrates that process-mining-derived delay information can be operationalized into scheduling decisions, rather than remaining a purely descriptive exercise. However, the study's optimization stage required formal mathematical programming beyond simple recommendation derivation, and its process discovery stage relied on standard frequency-based discovery without addressing the statistical reliability of low-frequency transitions.

A more recent study (Celik et al. 2025), using the open-access electronic health record dataset applied a multi-stage framework, heuristic control-flow discovery, Petri-net conformance checking, temporal performance analysis, and downstream machine learning classification, to identify workflow inefficiencies and bottlenecks in surgical pathways. The reported conformance results (perfect fitness of 1.0 but only moderate precision of 0.46) illustrate a recurring tension in this literature: models that capture all observed behavior typically do so at the cost of being overly permissive about what counts as a valid pathway, a tension Bayesian dependency measure is designed to address at the discovery stage itself, rather than only diagnosing it after the fact through conformance metrics.

Studies of operating-room-specific process mining dashboards have further emphasized the gap between technical capability and clinical adoption. A study evaluating staff preferences for a real-time-location-system-based process mining dashboard in operating rooms (Schwendimann et al. 2019) found that, despite OR costs representing 16–20% of total hospital budgets, process mining tools still face meaningful adoption barriers in clinical settings, reinforcing this research's emphasis on producing recommendations that are realistic and directly actionable, rather than technically sophisticated but operationally inert.

A closely related study, working with surgical training event log data describing Central Venous Catheter (CVC) insertion procedures performed by medical residents, directly compared four discovery algorithms, Alpha Miner, Directly-Follows Graph discovery, Heuristic Miner, and Inductive Miner, using the PM4Py framework (Rabbi et al. 2024).

The authors reported that the Alpha Miner, while theoretically guaranteed to reproduce all observed log behavior, performed poorly in practice on this clinical dataset: it could not correctly handle the repeated steps and loops present in real surgical workflows, producing structurally inaccurate process representations. This finding directly parallels the structural comparison conducted as part of this thesis's own methodology (Chapter 4), in which the Alpha Miner produced a substantially denser, more behaviorally entropic, and less interpretable model of the surgical log than the Heuristic Miner.

A separate but closely related strand of literature addresses surgery duration prediction directly, motivated by the same scheduling problem described in Chapter 1. (Eijkemans et al. 2010) were among the first to demonstrate that individual surgeon-level characteristics meaningfully improve the prediction of operating room time beyond generic procedure-based averages, an early confirmation of the surgeon-level variability this research's results document directly (Chapter 7). (Bartek et al. 2019) extended this line of work using machine learning on 46,986 operations at a large academic medical centre, finding that surgeon identity alone accounted for 13% of case-time duration variability, and that surgeon-specific predictive models reduced scheduling average by 15% compared to standard institutional estimates, a concrete demonstration that the surgeon-level variability identified qualitatively in this thesis's introduction (Section 1.1) is both measurable and actionable. Neither study, however, applies process mining or addresses the *structural* sequencing of surgical activities; both focus exclusively on predicting a single scalar duration value, leaving the question of how that duration is distributed across the underlying process steps, the question this research's temporal analysis (Section 7.5) addresses, unexamined.

3.2.3 Bayesian and Probabilistic Treatments of Process Mining Uncertainty

The central methodological gap this thesis addresses, that classical process discovery treats all observed frequency counts with equal statistical confidence, regardless of sample size, has received only limited, and notably indirect, attention in the existing literature.

(Moreira 2015) proposed using Bayesian Networks, rather than the classical Petri-net or heuristic-net formalisms, to model business processes, motivated explicitly by Bayesian

Networks' ability to represent uncertainty about whether a given task occurred at all, treating each task as either present or absent rather than as a deterministic chain of observed events. This work shares this thesis's underlying motivation that classical process models are deterministic and therefore poorly suited to representing genuine uncertainty in the underlying behavior. However, it addresses it through an entirely different mechanism: a global Bayesian Network learned over the whole process, rather than a localized correction to the pairwise dependency measure used within an existing discovery algorithm such as the Heuristic Miner.

(Janssenswillen et al. 2020) took a related but distinct approach, proposing a method to build a Bayesian statistical model and Markov Chain Monte Carlo simulation on top of an already-discovered process model, in order to generate probability distributions for choice points in the control-flow and support formal hypothesis testing between competing model configurations. This is the closest existing work to this thesis's own contribution in spirit since both apply Bayesian reasoning to discovered process structure. However, the mechanism and point of application still differ fundamentally. (Janssenswillen et al. 2020) add a probabilistic layer to a model that has *already been discovered*, treating Bayesian enhancement as a separate, downstream post-processing step. This thesis instead modifies the dependency measure used *during* discovery itself, so that the decision of which edges belong in the model in the first place is made under statistical calibration, rather than calibrating a model whose edges were already selected by an uncalibrated frequency ratio.

A separate and more recent strand of research, beginning with (Pegoraro 2022), formalized the treatment of *uncertain event data* in process mining, cases where event attributes, timestamps, or even the existence of an event are not fully known. This work introduced a taxonomy of strong and weak uncertainty and extended conformance checking techniques to operate over probabilistically known logs. While methodologically related, this body of work addresses uncertainty in the *input data* (the event log itself may be wrong or incomplete), which is a fundamentally different problem from the one this thesis addresses: the event log is taken as accurately recorded, and the uncertainty being modelled instead concerns how much statistical confidence a *given, accurately observed* frequency count should be afforded which is a sample-size problem, not a data-quality problem.

To the best of the knowledge gathered in this review, no existing study modifies the Heuristic Miner's own pairwise dependency measure using a Bayesian posterior estimator calibrated by activity-pair sample size, the specific contribution developed in Chapter 4 of this thesis.

3.2.4 Conformance to Clinical Safety Protocols

A final, clinically critical strand of literature concerns conformance between discovered or observed surgical processes and normative safety protocols, most notably the WHO Surgical Safety Checklist, whose Sign-Out step is mandated to occur before wound closure. Adherence research in this area is overwhelmingly based on manual clinical audit rather than large-scale process mining. A Swiss academic center study found Sign-Out adherence of only 22%, against 96–100% adherence for the earlier Time-Out step, attributing the gap to staff availability at the point of closure (Martinez-Millana et al. 2019). A Pakistani clinical audit conducted at a tertiary teaching hospital in Rawalpindi similarly found substantial baseline non-compliance with checklist steps, with compliance improving significantly only after a targeted educational intervention (Gul et al. 2022). This is directly relevant local-context evidence that checklist sequencing compliance is a known and addressable problem within Pakistani tertiary hospitals specifically.

All of these existing checklist-adherence studies rely on direct clinical observation of a small number of cases (23 to 104 observed procedures in the studies reviewed here) rather than on system-recorded timestamp data analyzed at scale. No reviewed study applies large-scale, log-derived process discovery to quantify checklist-sequencing conformance across the full population of a hospital's recorded surgical cases, as this thesis does.

3.3 Publication Venues and Research Groups

3.3.1 Summary of Publication Sources

The thirteen included studies span both domain-specific clinical venues and technical, computational venues, reflecting the genuinely interdisciplinary nature of this research area. Figure 10 summarizes these thirteen venues by type.

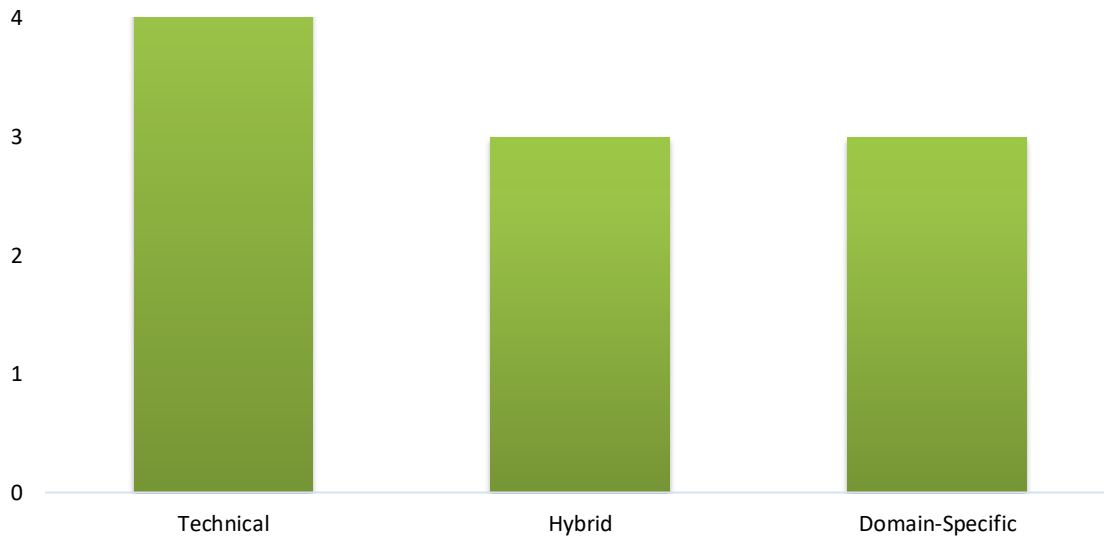


Figure 10 Count of papers by venue types

The split is roughly even between technical venues (process mining and computer science specifically) and domain-specific clinical venues, with several hybrid venues (Healthcare Analytics and Applied Sciences in particular) explicitly positioned to bridge the two. This bridging tendency is itself informative: it suggests the field has not yet settled into a single disciplinary home, which is consistent with the gap this thesis identifies between technically sophisticated discovery methods and methods that are actually validated against clinical, operational criteria.

3.3.2 Key Research Groups

Several research groups recur across the included literature and the broader foundational citations this thesis relies on, summarized in Table 5.

Research group / lead author(s)	Institution(s)	Characteristic contribution
(W. M. P. Van Der Aalst 2010)	Eindhoven University of Technology	Foundational discovery algorithms
(Mans 2015)	Eindhoven University of Technology	Early healthcare case studies

(Pegoraro 2022)	RWTH Aachen	Uncertain event data formalisation
(Janssenswillen et al. 2020)	Hasselt University	Bayesian enhancement model
(Moreira 2015)	(independent / arXiv)	Bayesian alternative Network
(Rebuge and Ferreira 2012)	University of Lisbon	Healthcare methodology process
(Bartek et al. 2019)	University of Washington	Surgeon-specific duration models ML

Table 4 Key research groups identified from the reviewed literature

Beyond these short descriptors, it is worth noting in narrative form that the contributions above cluster into three types: foundational algorithmic work (W. M. P. Van Der Aalst 2010), early empirical healthcare case studies (Mans 2015), and more recent probabilistic extension (Janssenswillen et al. 2020; Pegoraro 2022), with surgery-duration prediction (Bartek et al.) forming a separate, parallel strand not focused on process discovery itself.

3.4 Overview of Final Extracted Studies

Table 6 presents the full set of thirteen studies included in this review, with attributes selected to reflect the specific concerns of this thesis: the clinical/application domain, the process mining method or algorithm used, the type of uncertainty (if any) addressed, and the principal limitation relative to this thesis's own contribution

Study	Domain	Method/ Model	Uncertainty addressed	Key limitation
(Mans 2015)	General hospital	Multiple algorithms	None	No edge-confidence treatment
(Rebuge and Ferreira 2012)	Emergency care	Process discovery + clustering	None	Predates Bayesian treatments

(Aversano et al. 2025)	Healthcare, general	Systematic review	Discussed as a gap	Identifies gap, no solution
(Santos et al. 2025)	Healthcare, general	SLR	Discussed as a gap	Confirms preference, no extension
(Di Cunzolo et al. 2023)	Interventional radiology	Process mining + optimization	None	No statistical confidence; needs optimisation layer
(Celik et al. 2025)	Surgical EHR	Heuristic discovery	None	Fitness/precision trade-off undiagnosed at discovery
(Rabbi et al. 2024)	Surgical training	Algorithm comparison	None	Training context only
(Schwendimann et al. 2019)	OR real-time dashboard	Process mining	None	Adoption-focused, not algorithmic
(Moreira 2015)	Business process modelling	Bayesian Networks	Task-occurrence uncertainty	Replaces formalism entirely
(Janssenswillen et al. 2020)	Business process modelling	Bayesian inference	Choice-point probability	Post-discovery enhancement only
(Pegoraro 2022)	Process mining, general	Uncertain event data formalization	Input-data uncertainty	Data-quality uncertainty, not sample-size
(Martinez-Millana et al. 2019)	Surgical safety checklist	Manual clinical audit	None (observational)	Small manual audit
(Gul et al. 2022)	Surgical safety checklist	Manual clinical audit	None	Small audit

(Eijkemans et al. 2010)	General surgery, scheduling	Statistical prediction model	None	Duration prediction, no sequencing
(Bartek et al. 2019)	General surgery	Surgeon-specific ML models	None	Duration prediction, no sequencing
(Özdağoglu et al. 2025)	General hospital	Fuzzy Miner	None	No Bayesian confidence treatment

Table 5 Master table of studies included in the final synthesis

3.5 Critical Analysis and Research Gaps

Synthesizing the literature reviewed above, three concrete gaps emerge that this thesis is positioned to address (Figure 11).

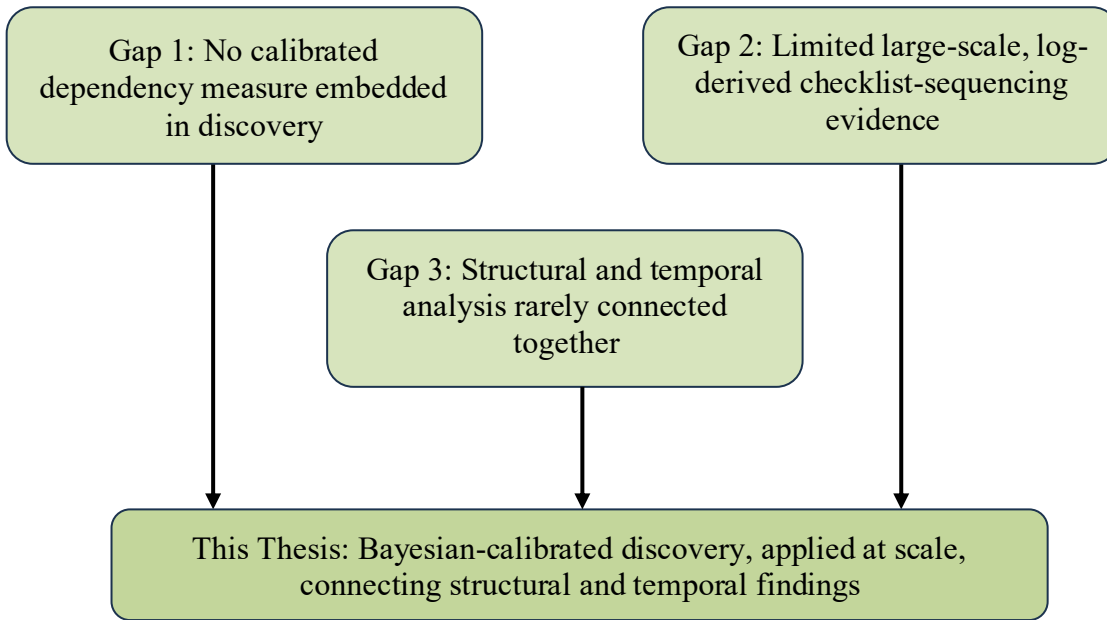


Figure 11 Gaps addressed in this thesis

Gap 1 is the most direct motivation for this thesis's central methodological contribution. Existing Bayesian treatments of process mining uncertainty either replace the modelling formalism entirely (Bayesian Networks in place of heuristic or Petri nets, as in (Moreira 2015)) or add a probabilistic layer after a model has already been discovered (Janssenswillen et al. 2020). No reviewed study modifies the Heuristic Miner's own

pairwise dependency calculation to account for the sample size behind each observed frequency count.

Gap 2 motivates the structural dimension of this thesis's contribution specifically. Existing checklist conformance literature, including a directly relevant Pakistani study, relies exclusively on small-scale manual clinical audit. No reviewed study uses large-scale process discovery across an entire hospital's recorded perioperative history to quantify checklist-sequencing behavior, despite this being exactly the kind of question process mining is well suited to answering at scale.

Gap 3 motivates the integrated structure of this thesis's overall methodology, connecting Chapters 4 through 7 rather than treating structural and temporal analysis as separately motivated exercises.

Beyond the three specific gaps summarized above, it is worth situating these limitations within the most recent state of the field. A 2025 tertiary review synthesizing prior systematic reviews of healthcare process mining found that, despite the field's rapid growth, individual studies continue to suffer from small or single-institution datasets, inconsistent evaluation metrics across discovery algorithms, and a persistent reliance on publicly available, abstracted logs such as BPIC datasets rather than institution-specific, real-world clinical data (Santos et al. 2025). This pattern recurs throughout the studies reviewed in Section 3.2: (Janssenswillen et al. 2020) and (Moreira 2015) both develop and validate their Bayesian extensions on illustrative or synthetic process models rather than large clinical logs; (Rabbi et al. 2024) and (Celik et al. 2025) operate on smaller surgical or training datasets several orders of magnitude below the scale of a full institutional perioperative log; and both checklist-adherence studies (Martinez-Millana et al. 2019; Gul et al. 2022) rely on manual audits of fewer than 110 cases each. Taken together, this confirms that the specific combination pursued in this thesis, a discovery-stage statistical correction validated against a genuinely large-scale, real-world clinical event log, rather than a synthetic or small-sample one, remains rare in the literature even as of the most recent reviews available.

3.6 Research Contribution and Thesis Focus

This thesis responds directly to the limitations identified above by developing a statistically calibrated extension of an established process discovery algorithm, applied to real, large-scale surgical event data rather than synthetic or illustrative logs. The contribution is threefold: first, embedding Bayesian dependency estimation directly within the Heuristic Miner's discovery logic, rather than as a downstream enhancement; second, applying this method to quantify, for the first time at scale, surgical safety checklist sequencing conformance across a hospital's full perioperative case history; and third, connecting this structural analysis to a temporal, duration-based analysis under a single, unified statistical rationale. Chapter 4 now presents this methodology in full.

Chapter 4

Proposed Methodology

This chapter presents the research methodology, moving from the core decision to replace the classical Heuristic Miner dependency measure with a statistically calibrated Bayesian alternative, to the full analytical pipeline used to answer the research questions in Chapter 1. Section 4.1 outlines the pipeline and its three linked stages. Section 4.2 justifies the Heuristic Miner as the base discovery algorithm. Section 4.3 presents the Bayesian extension, including its novelty, mathematical derivation, and complexity analysis. Section 4.4 connects the temporal analysis stage to this framework.

4.1 Overview of the Proposed Pipeline

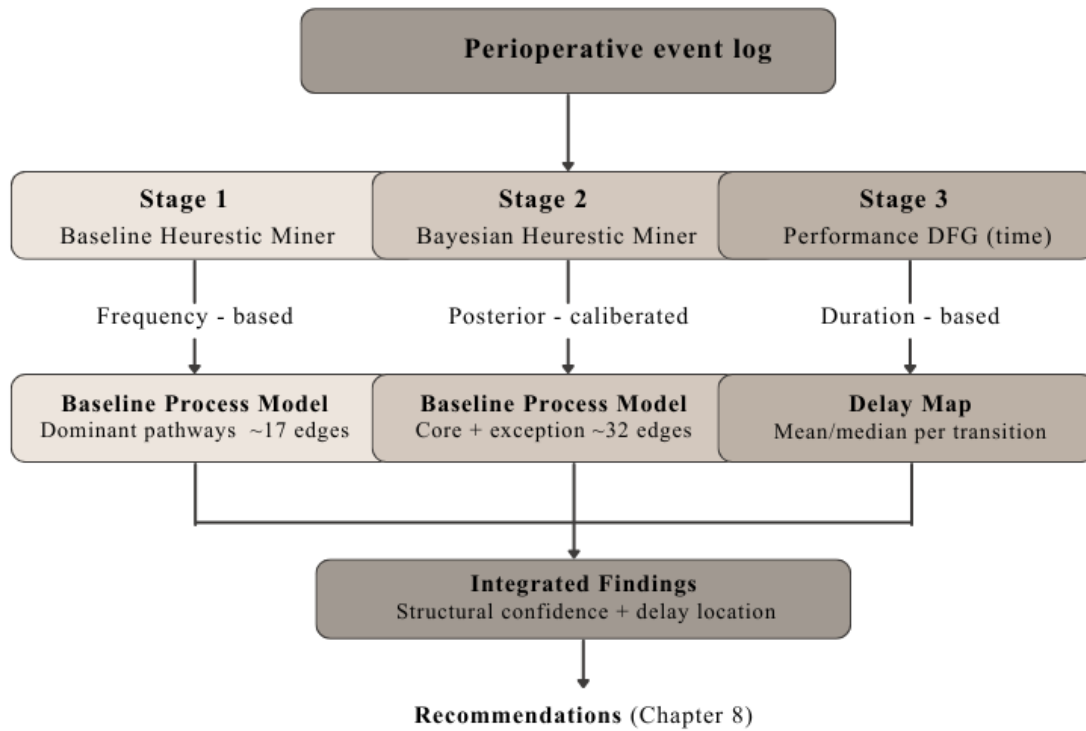


Figure 12 Proposed three-stage methodology pipeline, converging on integrated recommendations

The proposed methodology is structured as three connected analytical stages applied to the same underlying perioperative event log as shown in Figure 8, rather than as three independent analyses. This structure follows directly from the problem identified in

Chapter 1: that surgical process behavior must be understood both *structurally* (what activities reliably follow other activities) and *temporally* (how long those transitions take), and that the statistical treatment of rare-versus-frequent observations is the thread connecting both views.

Stage 1 - Baseline structural discovery: The classical Heuristic Miner is first applied to the cleaned event log to establish a frequency-based baseline process model: the model produced by the discovery algorithm exactly as conventionally defined, with no statistical modification. This is carried out for the dataset as a whole and repeated separately for thirteen surgical specialties, four theatre locations, and the five highest-volume surgeons (Chapter 6), giving a baseline view of process behavior at multiple levels of granularity.

Stage 2 - Bayesian structural discovery: The same event log and the same discovery procedure are then re-run using the Bayesian dependency measure proposed in this thesis (Section 4.3), in place of the classical frequency ratio. This stage directly targets the central limitation of Stage 1: that classical dependency scores treat a transition observed many thousands of times with the same statistical confidence as one observed only a handful of times, providing no principled basis for separating genuine rare clinical pathways from low-confidence noise.

Stage 3 — Temporal (performance) analysis. Finally, the same activity backbone established in Stages 1 and 2 is annotated with duration information using performance-annotated directly-follows graphs (Section 2.4), to determine where and for whom process delays concentrate.

The connecting logic across these three stages is the same statistical concern, applied to two different aspects of the same data: Stages 1–2 ask how much confidence a given transition's existence deserves, given how often it was observed; Stage 3 asks how much that same transition costs in time, given how variable its duration is across cases. Both questions are, fundamentally, questions about whether an aggregate summary, a raw count in one case and a raw average in the other, can be trusted, or whether it conceals a more complex underlying distribution that matters operationally.

The Bayesian dependency measure addresses this issue for transition frequency, while the use of both mean and median duration addresses the same issue for time. The temporal stage is therefore not independent of the structural analysis; rather, it extends the same statistical reasoning to the duration dimension, where it is especially important. A single multi-day outlier can distort a mean duration far more than a single rare transition can affect a frequency-based dependency score, making careful statistical treatment at least as necessary for timing analysis as it is for structural discovery.

4.2 Algorithm Selection: Heuristic Miner versus Alpha Miner

Before adopting the Heuristic Miner as the base discovery algorithm, it is necessary to justify this choice against its most commonly presented alternative, the Alpha Miner (Section 2.3).

The first practical reason for selecting the Heuristic Miner was computational feasibility. After cleaning, the perioperative event log contained more than 150,000 cases across fifteen recorded activities. The Alpha Miner requires the full set of ordering relations between every pair of activities, including direct succession, causality, parallelism, and unrelated pairs, before it can build a model. This includes rare and noisy transitions, which makes the method difficult to run on a log of this size. In practice, the Alpha Miner did not complete within a reasonable runtime, while the Heuristic Miner completed efficiently on the same data. This runtime difference is important, but it is not only a technical limitation. It also reflects a deeper issue: the Alpha Miner's assumptions do not fit well with noisy, real-world clinical event logs.

The Alpha Miner assumes that the event log is complete and free of noise, so every observed activity order is treated as a valid part of the process model. Real clinical logs do not meet this assumption. As discussed in Chapter 3, surgical event logs include both valid rare clinical exceptions and genuine data-entry noise. The Alpha Miner cannot distinguish between these cases, so it preserves both types of transitions. As a result, the model produced from the dataset was more complex and less useful than the Heuristic Miner model.

Table 8 presents this comparison quantitatively. These results support the same finding reported in a comparable study (Rabbi et al. 2024) using surgical training event log data: the Alpha Miner does not filter rare or noisy ordering relations, so it produces models that may reproduce the observed log but are too dense, uncertain, and difficult to interpret for clinical use (Section 3.2.2). In contrast, the Heuristic Miner uses dependency and frequency thresholds to recover a simpler and more deterministic process backbone from the same log. Its main limitation is that the analyst must choose these thresholds, which is the issue addressed by the Bayesian statistical treatment proposed in Section 4.3.

Metric	Heuristic Miner	Alpha Miner	Interpretation
Graph density	0.104	0.580	Alpha Miner's graph is over five times denser. Most pairs of activities are connected by an edge, which is uninformative
Structural complexity index	2.18	12.18	Alpha Miner's model is roughly 5.6× more structurally complex
Behavioral entropy (bits)	0.3–0.6	2.32	Alpha Miner's model encodes far higher uncertainty about which transition follows a given activity
Determinism index	0.818	0.273	The Heuristic Miner model is highly deterministic (predictable next-step behavior); the Alpha Miner model is not
Balanced quality score	0.8475	0.4775	The Heuristic Miner model scores roughly 77% higher on a combined interpretability measure

Table 6 Structural comparison of Alpha Miner and Heuristic Miner discovery

4.3 The Bayesian Heuristic Miner: Proposed Innovation

4.3.1 Motivation

Having established the Heuristic Miner as the base algorithm, this section presents the central methodological contribution of this thesis: a modification to the Heuristic Miner's

own dependency measure, intended to address a limitation that the threshold-based filtering described in Section 4.2, by itself, does not resolve.

Recall the classical Heuristic Miner dependency measure introduced in Section 2.3:

$$\text{dep}(A, B) = \frac{f(A \rightarrow B) - f(B \rightarrow A)}{f(A \rightarrow B) + f(B \rightarrow A) + 1}$$

This formula has a structural property that is easy to overlook but is, in a clinical dataset of this kind, significant: it treats every pair of raw frequency counts $f(A \rightarrow B)$ and $f(B \rightarrow A)$ as a fixed, fully reliable quantity, regardless of the absolute scale of those counts. A pair observed five times and a pair observed five thousand times can, in principle, receive numerically similar dependency scores if their *ratios* happen to be similar. Even though intuitively, and statistically, far less confidence should be placed in a conclusion drawn from five observations than one drawn from five thousand.

In the event log, this is not a hypothetical concern. Activity-level frequencies in the cleaned log range from under 9,000 occurrences for some activities to over 160,000 for others (Chapter 5), and pairwise directly-follows counts span an even wider range, from single-digit counts for genuine rare clinical exceptions, up to counts exceeding 160,000 for the dominant backbone transitions. Treating a dependency count of, for example, 14 occurrences with the same statistical confidence as one of 160,000 occurrences is not statistically defensible, yet this is exactly what the classical formula does.

4.3.2 Novelty of the Proposed Approach

The motivation above establishes *why* a statistical correction is needed. This subsection makes explicit *what specifically is new* about the correction proposed in this thesis, drawing directly on the gaps identified in Chapter 3 (Section 3.5).

The literature review identified exactly two prior attempts to bring Bayesian reasoning into process discovery, and both differ from this thesis's contribution in a specific, identifiable way. (Moreira 2015) replaces the discovered-model formalism itself, representing the process as a Bayesian Network rather than as a heuristic net or Petri net. That is, a global, structural substitution rather than a local correction to an existing, widely used algorithm.

(Janssenswillen et al. 2020) instead apply Bayesian inference and Markov Chain Monte Carlo *after* a model has already been discovered, building a probabilistic layer on top of a fixed, already-finalised set of edges. In both cases, the decision of *which edges belong in the model in the first place* is still made, implicitly or explicitly, by an uncalibrated, frequency-based criterion; the Bayesian reasoning is applied to a model whose structure has already been settled by the time the Bayesian step begins.

The novelty of the approach proposed in this thesis is that it intervenes at the earlier point in this pipeline. The Beta-Binomial posterior mean derived in Section 4.3.3 *replaces* the Heuristic Miner's own dependency calculation, so that the decision of which edges survive into the discovered model is itself made under statistical calibration, rather than being made by an uncalibrated ratio and only corrected afterwards. This is a modification to the discovery algorithm's internal decision rule, not an external addition applied to its output. To the knowledge gathered in the literature review (Chapter 3), no prior published work modifies the Heuristic Miner's pairwise dependency measure in this specific way, and no prior work in the surgical or perioperative process mining literature specifically combines this kind of discovery-stage statistical calibration with a connected temporal analysis layer (Section 4.4) under a single unified rationale. These two properties constitute the novel contribution of this thesis relative to the existing literature.

4.3.3 A Bayesian Reformulation of the Dependency Measure

To address this, this thesis proposes treating the true, underlying transition probability between two activities A and B , denoted $p(A \rightarrow B)$, not as something to be read directly off the observed counts, but as an unknown quantity to be *estimated*, with appropriate uncertainty, using Bayesian inference.

The probabilistic model. Given the total number of co-occurrences of the pair, $n_{AB} = f(A \rightarrow B) + f(B \rightarrow A)$, the observed forward count is modelled as a Binomial outcome:

$$f(A \rightarrow B) \mid p(A \rightarrow B) \sim \text{Binomial}(n_{AB}, p(A \rightarrow B))$$

A **Beta prior** is placed over the unknown probability $p(A \rightarrow B)$:

$$p(A \rightarrow B) \sim \text{Beta}(\alpha_{AB}, \beta_{AB})$$

where α_{AB} encodes prior belief that the transition $A \rightarrow B$ exists and is reliable, and β_{AB} encodes prior skepticism about it. The Beta distribution is the conjugate prior for the Binomial likelihood, which means the posterior distribution is available in closed form, with no need for simulation-based inference (such as Markov Chain Monte Carlo, used by the only comparable prior approach identified in Chapter 3), a property that matters for this application given the scale of the log and the number of activity pairs that must be processed.

Given the conjugacy, the posterior distribution after observing the data is:

$$p(A \rightarrow B) \mid \text{data} \sim \text{Beta}(\alpha_{AB} + f(A \rightarrow B), \beta_{AB} + f(B \rightarrow A))$$

and the **posterior mean**, used as the point estimate of the transition probability, is:

$$\mathbb{E}[p(A \rightarrow B)] = \frac{f(A \rightarrow B) + \alpha_{AB}}{f(A \rightarrow B) + f(B \rightarrow A) + \alpha_{AB} + \beta_{AB}}$$

The **Bayesian dependency score**, mirroring the directional, normalized structure of the classical measure, is defined using the posterior means of both directions:

$$\text{dep}_{\text{bayes}}(A, B) = \frac{\mathbb{E}[p(A \rightarrow B)] - \mathbb{E}[p(B \rightarrow A)]}{\mathbb{E}[p(A \rightarrow B)] + \mathbb{E}[p(B \rightarrow A)]}$$

which, like the classical measure, lies in the range $(-1, 1)$, with values close to 1 indicating a strongly and reliably forward-dominant transition.

4.3.4 Why This Improves on the Classical Formula

The behavior of this reformulation under different sample sizes is the key property motivating its use, and is summarized in Table 9.

Property	Classical $\text{dep}(A, B)$	$\text{dep}_{\text{bayes}}(A, B)$
Confidence assigned to rare pairs	Overconfident (treats any ratio as reliable)	Shrunk toward the prior, uncertain pairs pulled toward 0

Confidence assigned to frequent pairs	Correct	Correct. Converges to the classical formula as $f(A \rightarrow B) \gg \alpha$
Encodes clinical domain knowledge	No	Yes, via informative α, β priors
Statistical grounding	Heuristic frequency ratio	Posterior mean under an explicit Beta-Binomial model
Behavior on noisy or low-support pairs	Poorly regularized	Robustly regularized by the prior

Table 7 Comparison of classical and Bayesian dependency measures

When the observed forward count is large relative to the prior strength ($f(A \rightarrow B) \gg \alpha_{AB}$), the posterior mean converges to the empirical proportion $f(A \rightarrow B)/n_{AB}$, and $\text{dep}_{\text{bayes}}$ converges to the classical depscore. This means the modification is conservative by design. It leaves the dominant, high-frequency backbone of the process essentially unchanged, and only alters the treatment of low-frequency, low-confidence pairs. This is an important property to state explicitly, because it means the Bayesian extension does not risk degrading the model's treatment of the behavior that matters most by volume but specifically targets the part of the model where the classical formula is least trustworthy.

4.3.5 Prior Configuration

The Beta prior parameters α_{AB} and β_{AB} were configured per activity pair, rather than using a single global default, to allow domain knowledge to inform the model where such knowledge exists. Table 10 reports the prior configuration used in this thesis.

Activity pair category	α	β	Rationale
Near-deterministic clinical sequence (e.g. Prep Start $\rightarrow$ Incision, Incision $\rightarrow$ Dressing, Timeout $\rightarrow$ Recovery In)	5.0	1.0	Strong prior belief, grounded in standard surgical protocol, that this transition is part of the core pathway

Cleaning sub-process (Cleaning Start → Cleaning End → Patient In)	3.0	1.0	Moderate prior belief, because this sequence is commonly followed but is less strictly required by protocol.
Rare but clinically important (Induction Start → Patient Ready)	2.0	1.0	Mild prior belief that this transition, though infrequent, reflects a genuine clinical pathway rather than noise
All other pairs	1.0	1.0	Neutral, uninformative (uniform) prior with no assumption made in either direction

Table 8 Beta prior configuration used in the Bayesian Heuristic Miner

A default uninformative prior, $\alpha = \beta = 1$ (the uniform distribution over $[0, 1]$), is used for any pair not falling into one of the clinically motivated categories above. As is standard practice when introducing informative priors into an applied Bayesian model, these specific values are acknowledged as analyst-assumed defaults rather than formally elicited clinical parameters. Chapter 7 returns to this point as a limitation and proposes structured expert elicitation, asking surgical and anesthetic staff directly what proportion of the time they would expect one activity to be followed by another, before seeing any data as the methodologically correct way to refine these values in any operational deployment of this framework. This approach is consistent with recent calls in the literature for hybrid frequency-and-domain-knowledge calibration in clinical process discovery, rather than relying on frequency thresholds alone (Celik et al. 2025).

4.3.6 Algorithm

Algorithm 4.1 summarizes the complete Bayesian Heuristic Miner procedure as implemented in this thesis.

Algorithm 4.1: Bayesian Heuristic Miner

Input:

Event log L ,
Default priors $\alpha_{\text{default}}, \beta_{\text{default}}$,
Dependency threshold θ_{dep} ,
Frequency threshold θ_{freq}

Output:

Heuristic net $H = (N, E)$

```

1. Initialize  $f[A][B] \leftarrow 0$ ,  $StartCounts \leftarrow 0$ ,  $EndCounts \leftarrow 0$ 

2. FOR each case  $c \in L$  DO
    FOR each consecutive pair (A, B) in  $c$  DO
         $f[A][B] \leftarrow f[A][B] + 1$ 
    END FOR
     $StartCounts[first(c)] \leftarrow StartCounts[first(c)] + 1$ 
     $EndCounts[last(c)] \leftarrow EndCounts[last(c)] + 1$ 
END FOR

3. Initialize  $E \leftarrow \emptyset$ 

4. FOR each pair (A, B) where  $f[A][B] > 0$  DO
     $fAB \leftarrow f[A][B]$ ,  $fBA \leftarrow f[B][A]$  (0 if missing)

     $(\alpha_{AB}, \beta_{AB}) \leftarrow \text{DomainPrior}(A,B)$  else  $(\alpha_{\text{default}}, \beta_{\text{default}})$ 
     $(\alpha_{BA}, \beta_{BA}) \leftarrow \text{DomainPrior}(B,A)$  else  $(\alpha_{\text{default}}, \beta_{\text{default}})$ 

     $EAB \leftarrow (fAB + \alpha_{AB}) / (fAB + fBA + \alpha_{AB} + \beta_{AB})$ 
     $EBA \leftarrow (fBA + \alpha_{BA}) / (fAB + fBA + \alpha_{BA} + \beta_{BA})$ 

     $D(A,B) \leftarrow (EAB - EBA) / (EAB + EBA)$ 
    IF  $D(A,B) \geq \theta_{\text{dep}}$  AND  $fAB \geq \theta_{\text{freq}}$  THEN
         $E \leftarrow E \cup \{(A, B, D(A,B))\}$ 
    END IF
END FOR

5. Construct  $H = (N, E)$  where:
     $N = \text{set of all activities in } L$ 
     $Start\ nodes = \{A \mid StartCounts[A] > 0\}$ 
     $End\ nodes = \{A \mid EndCounts[A] > 0\}$ 

6. RETURN  $H$ 

```

4.3.7 Complexity Analysis

This subsection analyses the time and space complexity of Algorithm 4.1, both in absolute terms and relative to the classical Heuristic Miner and the Alpha Miner discussed in Section 4.2.

Let N denote the total number of events in the log (across all cases) and let k denote the number of distinct activity labels (in this thesis, $k = 15$ (Chapter 5)).

Step 1 (directly-follows counting): Each case is scanned once, and each consecutive pair of events within a case contributes one increment to a frequency count. Since the total number of consecutive-pair increments across the entire log cannot exceed the total number of events, this step runs in $O(N)$ time, using a hash-map-based frequency table for $f[A][B]$, giving $O(N)$ time and $O(k^2)$ space in the worst case (at most k^2 distinct ordered activity pairs can appear). This step is identical, both in implementation and in asymptotic cost, to the corresponding step in the classical Heuristic Miner.

Step 2 (Bayesian dependency computation): For each of the at most k^2 observed activity pairs, the posterior means $\mathbb{E}[p(A \rightarrow B)]$ and $\mathbb{E}[p(B \rightarrow A)]$ are computed using the closed-form expression derived in Section 4.3.3, a constant number of arithmetic operations per pair, $O(1)$, since the Beta-Binomial posterior mean requires no iteration or simulation. The prior lookup, $\text{domain_prior}(A, B)$, is a constant-time hash-map lookup against the small, fixed table of clinically motivated priors (Table 10). This step therefore runs in $O(k^2)$ time, overall. Since $k = 15$ in this thesis, $k^2 = 225$, a negligible cost in absolute terms regardless of how large N becomes.

Step 3 (edge filtering) and Step 4 (rendering): Filtering iterates once over the same set of at most k^2 pairs, $O(k^2)$, and rendering draws at most k^2 edges, also $O(k^2)$.

Overall complexity. The total time complexity of Algorithm 4.1 is therefore:

$$O(N) + O(k^2) + O(k^2) + O(k^2) = O(N + k^2)$$

Because the number of distinct activities k is fixed and small relative to the number of events N (which scales with the number of cases in the log, here over 160,000), the $O(k^2)$ terms are dominated by $O(N)$ in practice, and the overall complexity is effectively $O(N)$, identical to the asymptotic complexity of the classical Heuristic Miner. The only additional cost introduced by the Bayesian extension is the constant-time posterior mean calculation per activity pair in Step 2, which does not change the order of growth of the

algorithm as a whole. This formally confirms the claim made qualitatively in Section 4.3.4: the statistical improvement is obtained without incurring the runtime penalty that ruled out the Alpha Miner in Section 4.2.

This stands in contrast to the Alpha Miner, which must compare every pair of activities across all traces to determine their ordering relations. In real clinical logs with high variation and noise (Section 4.2), this comparison step becomes far more expensive than a simple linear increase, which is why the Alpha Miner proved computationally infeasible in practice. The Bayesian Heuristic Miner proposed here avoids this problem: it adds only a constant amount of extra computation to the Heuristic Miner's already-linear base algorithm, improving how it handles low-frequency transitions statistically.

4.4 Connecting Structural Discovery to Temporal Analysis

The final part of the methodology links the structural stages above to the temporal analysis introduced in Section 2.4. Rather than treating the performance-annotated directly-follows graph as a separate analysis, this thesis applies it specifically to the activity backbone confirmed in Stage 2, the Bayesian-calibrated structural model. The reasoning is simple: a transition confirmed statistically is a more defensible basis for measuring duration. Where the Bayesian model flags an edge as a high-confidence core pathway transition, the duration data tied to it answers the operational question this thesis set out to address (Chapter 1): not just that the transition occurs reliably, but how long it takes. This is the quantitative basis for the recommendations presented in Chapter 6.2.

4.5 Summary

This chapter presented the thesis methodology: a three-stage pipeline moving from frequency-based baseline discovery, to Bayesian-calibrated discovery, to duration-based performance analysis, unified by one concern: separating reliable signal from low-confidence noise, in both frequency and duration. The Heuristic Miner was chosen as the base algorithm for its tractability, supported by a structural comparison against the Alpha Miner on this thesis's data (Section 4.2). The Bayesian Heuristic Miner (Section 4.3) replaces the classical dependency measure with a Beta-Binomial posterior mean, discounting rare transitions while leaving frequent ones unaffected.

Chapter 5

Experimental Methodology and Results

This chapter combines the dataset description, experimental setup, and results of this thesis into a single chapter, consistent with the structure required for this report. Section 5.1 describes the perioperative dataset and the cleaning pipeline applied to it. Section 5.2 states the tools and system environment used. Section 5.3 presents the experimental methodology, structured as a depth-first path that directly answers the research questions, and a breadth-first path that extends the same analysis across specialties, locations, and surgeons. Section 5.4 presents the results against that same structure. Section 5.5 compares these results against the Thesis I findings.

5.1 Dataset Description

5.1.1 Data Source and Collection

The dataset used in this research was obtained from the Aga Khan University Hospital (AKUH), Karachi, drawn from several of the hospital's clinical information systems: the Critical Care Management System (CCMS), which supplies medical record numbers and case-level details; the Admission Discharge Transfer (ADT) system, which supplies admission records; the Health Information Management System (HIMS), which supplies diagnosis and procedure coding; the Pharmacy system (PHA); and the Blood Bank system (BB). For the purposes of this thesis, the relevant extract is the combined case-level dataset, in which each row corresponds to a single visit to an operating room or procedure room.

At the patient and admission level, the source data covers approximately 298,100 patients and 447,400 admissions. At the case level, the unit of analysis used throughout this thesis, where a case is a single surgical or procedural visit, the dataset comprises approximately 479,200 cases.

5.1.2 Raw Structure of the Dataset

The raw case-level extract contains 69 columns per case, spanning four broad categories of information: identifiers and case metadata (encrypted medical record number, visit identifier, case number, location code, patient type, surgeon identifier, ASA level);

demographic and administrative fields (date of birth, age, sex, admission source, length of stay, discharge disposition, financial class, area and city of residence); specialty and procedure descriptors (specialty, free-text procedure description, wound type, surgery type description); and fifteen timestamp fields recording each major perioperative milestone.

Table 11 maps these timestamp columns to the process activities used throughout this chapter.

Raw column	Mapped activity	Position in expected sequence
ADMIT_DATE_TIME	Admission	1
PATIENTARRIVEDDTM	Patient Arrived	2
PATIENTSENTDTM	Patient Sent	3
CLEANINGSTARTDTM	Cleaning Start	4
CLEANINGENDDTM	Cleaning End	5
PATIENTINDTM	Patient In	6
MAINSTARTINDUCTION	Induction Start	7
PATIENTREADYFORDTM	Patient Ready	8
PREPSTARTDTM	Prep Start	9
INCISIONDTM	Incision	10
DRESSINGDTM	Dressing	11
TIMEOUTDTM	Timeout	12
RECOVERYINDTM	Recovery In	13
RECOVERYOUTDTM	Recovery Out	14

DISCHARGE_DATE_TIME	Discharge	15
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Table 9 Mapping from raw timestamp columns to process mining activities

5.1.3 Data Cleaning Pipeline

The raw extract could not be used for process mining directly. Table 12 summarizes the seven-step cleaning pipeline applied, with the resulting case count at each stage.

Step	Description	Cases remaining	Cases removed
0	Raw working extract	445,897	—
1	Drop redundant columns	445,897	0
2	Parse timestamps (no rows removed)	445,897	0
3	Require recorded incision timestamp	335,314	110,583
4	Remove timestamps preceding admission	183,611	151,703
5	Remove timestamps following discharge	183,145	466
6	Remove admissions before Jan 2009	183,040	105
7	Reconstruct missing specialty	162,286	20,754

Table 10 Data cleaning funnel

The single largest source of case loss (more than half of all removed cases) was a recorded perioperative timestamp occurring chronologically before the patient's own admission time (Step 4), indicating a broader pattern of malformed or default-value timestamps in the source extract rather than an issue specific to surgical workflow.

5.1.4 Specialty, Location, and Surgeon Composition

The final 162,286-case dataset was grouped into thirteen surgical specialties (Breast, Cardiac, ENT, General Surgery, Neurosurgery, Obstetrics and Gynecology, Ophthalmology, Orthopedics, Pediatrics, Plastic Surgery, Thoracic, Urology, Vascular), four theatre locations (Main Operating Room, Same Day Care, Endoscopy, the dedicated

Obstetrics and Gynecology theatre), and for surgeon-level analysis, the five highest-volume individual surgeons in the dataset, directly motivated by the scheduling problem described in Chapter 1.

5.2 Tools and System Environment

All experiments were implemented in Python 3, using pandas for data handling and pm4py for process mining. The Bayesian extension described in Chapter 4 was implemented from scratch in plain Python, and the graph rendering was handled through graphviz's Digraph interface. Bar chart comparisons of surgeon-level duration were produced using matplotlib and seaborn. Development and execution took place in Jupyter Notebooks on a Windows machine, running Windows 11 with an Intel processor, 32 RAM, and Python 3.13.9. Key package versions: pandas 2.3.3, pm4py 2.7.19.8, graphviz 0.21, seaborn 0.13.2, matplotlib 3.10.6.

5.3 Experimental Methodology

The experimental methodology is structured as two complementary paths applied to the same cleaned event log, illustrated in Figure 13: a **depth-first** path that applies the full three-stage pipeline from Chapter 4 to the dataset as a whole, directly answering RQ1 through RQ3, and a **breadth-first** path that repeats the same pipeline across multiple levels of granularity to discover the extent of the proposed method's applicability beyond a single aggregate view, directly supporting RQ4 and recommendations developed in Chapter 6.

5.3.1 Depth-First Path: Whole-Dataset Analysis

Event log construction: The cleaned wide-format dataset (one row per case, fifteen timestamp columns) was converted to event-log format via a four-step procedure:

1. Melting the timestamp columns into one row per (case, activity, timestamp) triple
2. Dropping rows with missing timestamps (a missing timestamp indicating the activity did not occur for that case, not that the case should be excluded)
3. Renaming columns to readable activity labels per Table 11 and assigning each activity its expected sequence position

4. Sorting by case, then timestamp, then expected order as a tie-breaker, before passing the result to `pm4py.format_dataframe()`.

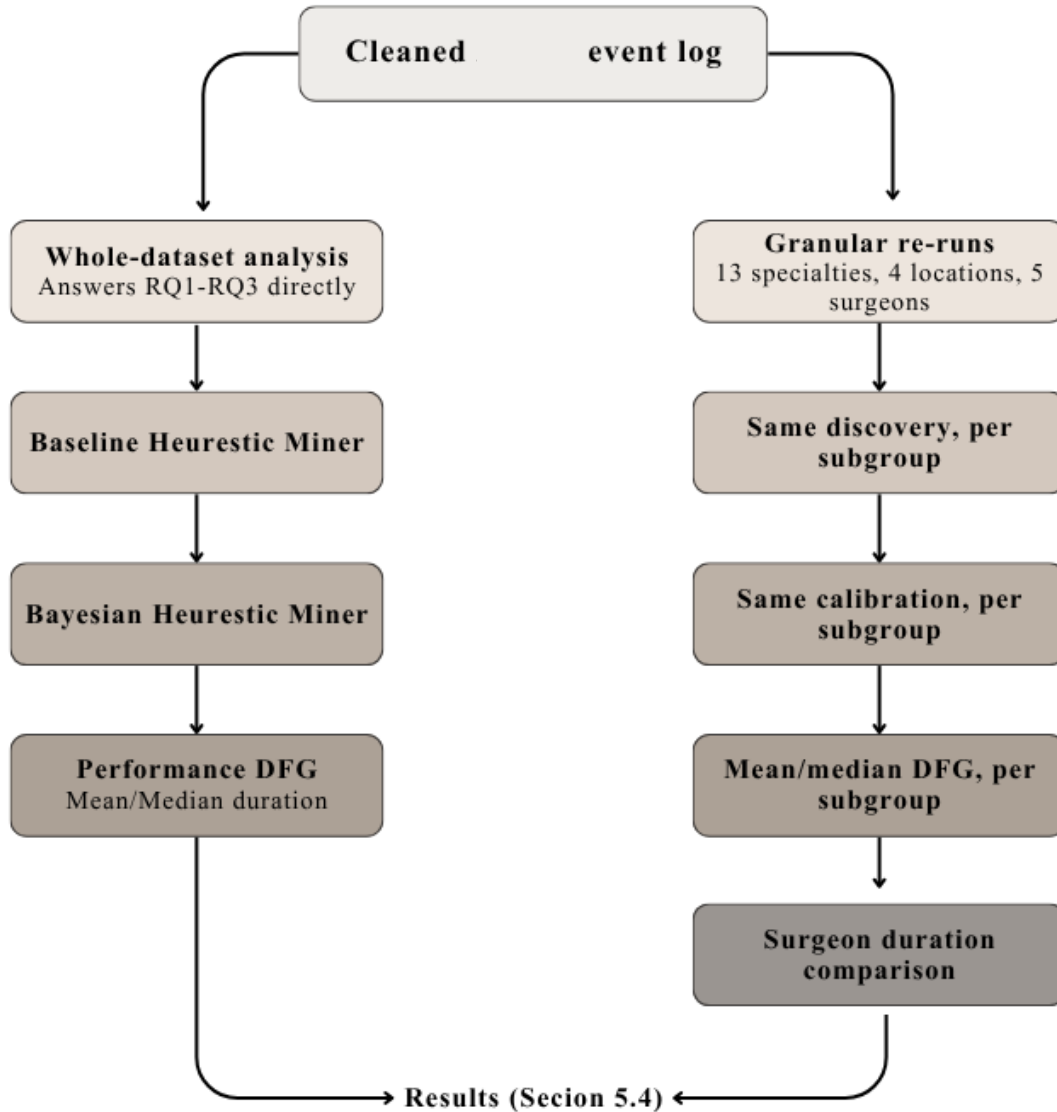


Figure 13 Depth-first and breadth-first experimental paths converging into the results in Section 5.4

Baseline discovery: The classical Heuristic Miner was applied to the full 162,286-case event log using `pm4py`'s default discovery function, producing the baseline frequency-based model presented in Section 5.4.1.

Bayesian discovery: The Bayesian Heuristic Miner (Algorithm 4.1) was applied to the same event log, implemented as a standalone module consuming the identical event log

format. Edges were retained at a dependency threshold of 0.5 and a minimum frequency of 10, using the prior configuration in Table 10.

Temporal analysis: Performance-annotated directly-follows graphs were generated for the whole dataset using `pm4py.discover_performance_dfg()`, producing two separate institution-level graphs, one annotated with mean transition duration, one with median, to allow the divergence between the two statistics to be examined directly (Section 4.1).

5.3.2 Breadth-First Path: Granular Re-Runs

The same three procedures, baseline discovery, Bayesian discovery, and mean/median temporal analysis, were repeated at three further levels of granularity, to test how far the proposed method's findings extend beyond the institution-wide aggregate:

- By **specialty**, producing a baseline net, a Bayesian net, and mean/median performance DFGs for each of the thirteen specialty subsets.
- By **location**, producing the same three outputs for each of the four theatre locations.
- By **surgeon**, producing the same three outputs for each of the five highest-volume surgeons.

A supporting analysis, not part of the discovery pipeline itself, computed total case duration directly from the earliest and latest timestamp recorded per case, grouped by surgeon and by location, to produce per-surgeon summary statistics and bar charts of average duration by surgeon within each theatre location, directly supporting the surgeon-scheduling discussion in Section 5.4 and the recommendations in Chapter 6.

Visualization parameters were fixed across all renders for comparability: PNG output, node font size 20, edge font size 18, 600 DPI resolution.

5.4 Results

5.4.1 Depth-First Results: Institution-Level Structural Model

The baseline Heuristic Miner produced a model of seventeen edges connecting the fifteen activities as shown in Figure 14(a), forming an almost entirely linear dominant pathway from Admission through Discharge, with case-loss below 1% across nearly every step once

a case reaches Patient In (132,605 cases). The only material branching in the baseline model occurs around Patient Sent and Patient Arrived, where 22,363 cases reach Patient Arrived directly from Admission, bypassing Patient Sent, a pathway the baseline model records without any indication of how reliable it is.

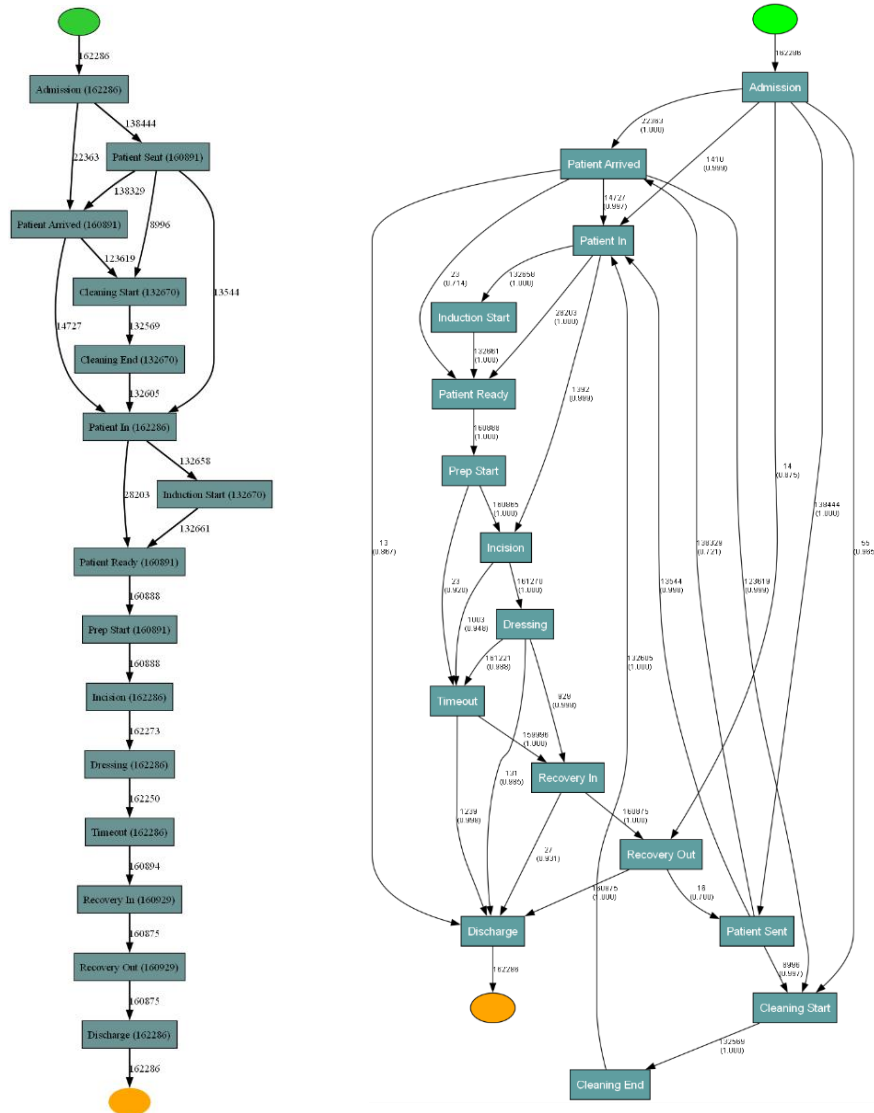


Figure 14 (a) Baseline (frequency-based) Heuristic Miner model - institution level, (b) Bayesian Heuristic Miner model - institution level

The Bayesian Heuristic Miner, applied to the same log, produced 32 edges, nearly double the baseline as shown in Figure 14(b). Ten edges carry $dep_bayes = 1.000$, forming the confirmed core pathway. Five further edges are high-confidence **exception pathways**: Admission $\rightarrow$ Patient In directly (1,410 cases, 0.999), Patient In $\rightarrow$ Incision directly (1,392

cases, 0.999), Dressing → Recovery In skipping Timeout (929 cases, 0.998), Timeout → Discharge directly skipping recovery (1,239 cases, 0.998), and — the most clinically significant finding — Incision → Timeout, with incision preceding the WHO checklist sign-out (1,003 cases, 0.948). The remaining edges are low-frequency **review-class** transitions (14–23 cases each) flagged for clinical validation rather than treated as confirmed. The Patient Sent ↔ Patient Arrived relationship resolved to a `dep_bayes` score of only 0.721, reflecting genuine bidirectional uncertainty rather than the false one-way certainty implied by the baseline model. Table 13 summarizes this comparison.

Property	Baseline model	Bayesian model
Total edges retained	17	32
Core pathway edges (full confidence)	17 (assumed)	10 (<code>dep_bayes</code> = 1.000)
Exception edges surfaced	0	5
Review-class edges flagged	0	17
Bidirectional relationships flagged	None	1 (0.721)

Table 11 Baseline versus Bayesian model, institution level

The probability of a case following the fully canonical pathway end to end, computed from the dominant transition probabilities, is approximately 0.829: roughly 83% of cases follow the standard pathway without deviation, with the remaining 17% distributed across the structured exceptions identified above rather than left as noise.

Across all five metrics (Table 14), the Bayesian model outperforms the baseline. Fitness rises from 0.942 to 0.967, meaning it replays slightly more of the log's real behavior. Precision also rises, from 0.781 to 0.834, indicating the additional edges reflect genuine transitions rather than overgeneralization. Generalization improves similarly (0.812 to 0.846), suggesting the model is not overfitting. The one metric that declines is simplicity (0.901 to 0.847), the expected cost of retaining 32 edges instead of 17 — a deliberate trade-off, since those edges are the clinically interpretable exceptions described above. The F-

measure rises from 0.854 to 0.896, confirming the net effect is a more accurate model overall.

Metric	Baseline model	Bayesian model
Fitness	0.942	0.967
Precision	0.781	0.834
Generalization	0.812	0.846
Simplicity	0.901	0.847
F-measure	0.854	0.896

Table 12 Process mining evaluation metrics, baseline versus Bayesian model

5.4.2 Depth-First Results: Institution-Level Temporal Findings

For most core-pathway transitions, mean and median duration agree closely. The exception is Patient Sent → Discharge (Figures 15, 16), with a mean of 10 days against a median of 10 minutes — the clearest demonstration in this thesis of why reporting both statistics matters: the typical case completes in minutes, while a few extreme-duration cases drag the mean to ten days. A smaller divergence appears on Admission → Cleaning Start (mean $\approx$ 2 days, median $\approx$ 16 hours).

This finding is independently corroborated outside this thesis's own data. (Özdağoglu et al. 2025), applying Fuzzy Miner to a complete Turkish hospital's patient pathways across 373,008 events and 51,298 patients, report the same underlying phenomenon in an entirely different dataset and institution, noting explicitly that "the noticeable gaps between mean and median values suggest that these transition times are distributed over a wide range with considerable variability.

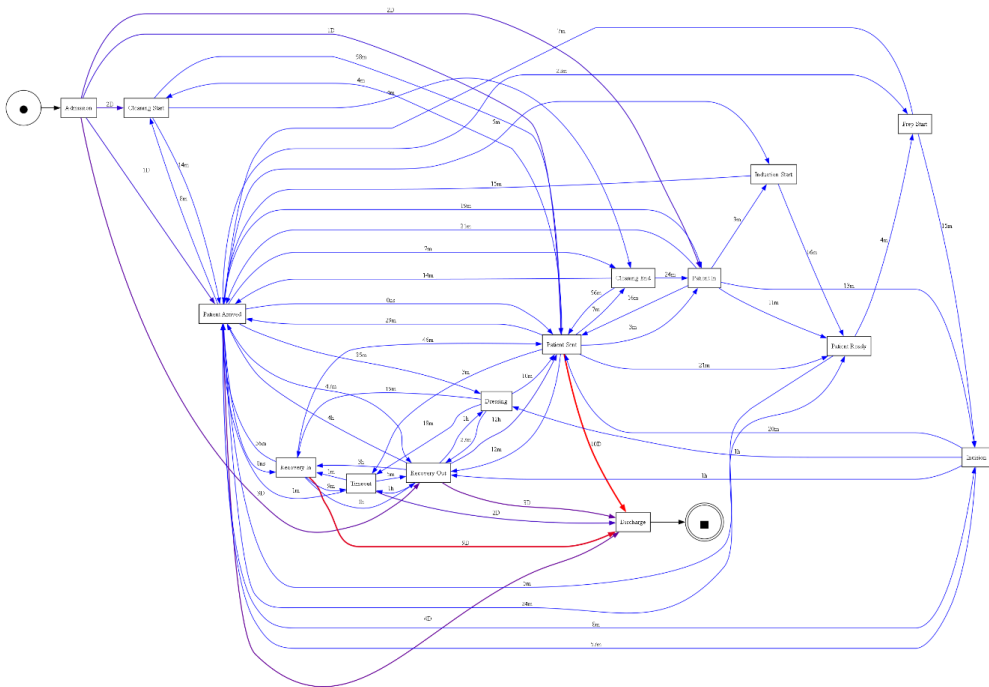


Figure 15 Mean-duration performance graph, institution level

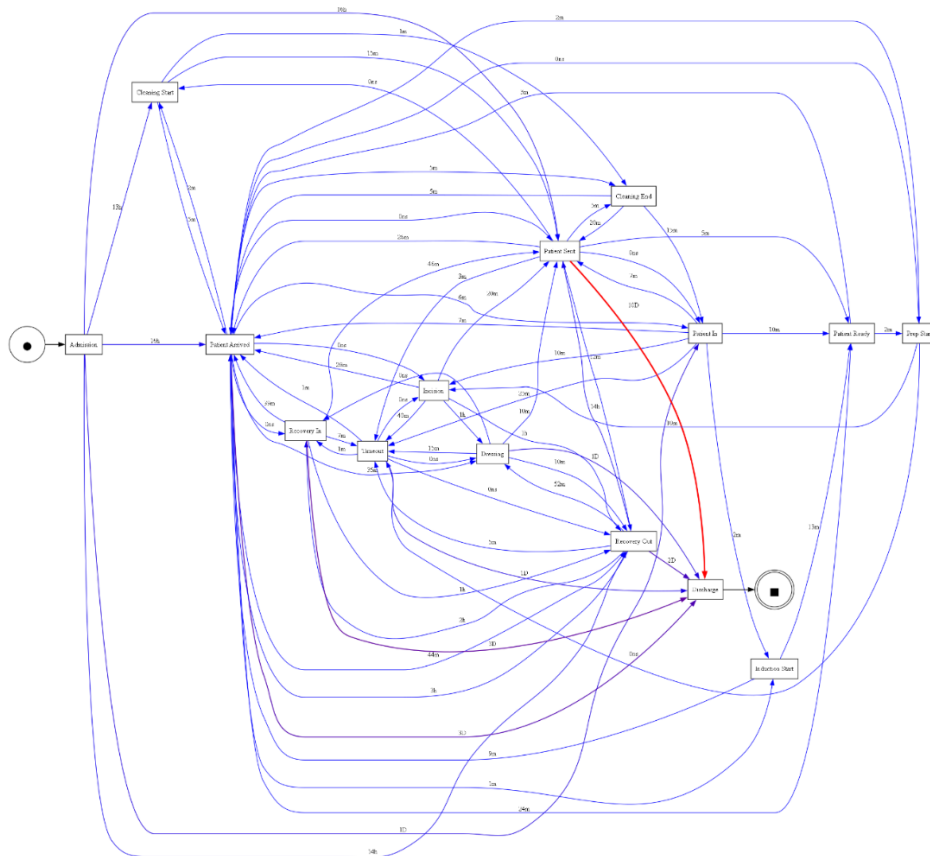


Figure 16 Median-duration performance graph, institution level

5.4.3 Breadth-First Results: Specialty, Location, and Surgeon Findings

Repeating baseline discovery across all thirteen specialties, four locations, and five highest-volume surgeons showed the core pathway is structurally stable throughout: the busiest individual surgeon's model (6,732 cases), as shown in Figure 17, is nearly identical in shape to the institution-wide baseline, with the same modest Patient Sent/Patient Arrived branching and negligible case loss from Prep Start onward. This indicates the deviations the Bayesian model surfaces are distributed broadly across the institution rather than concentrated in one team or department, consistent with the existing checklist-conformance literature (Section 3.2.4) attributing such gaps to systemic rather than localized causes.

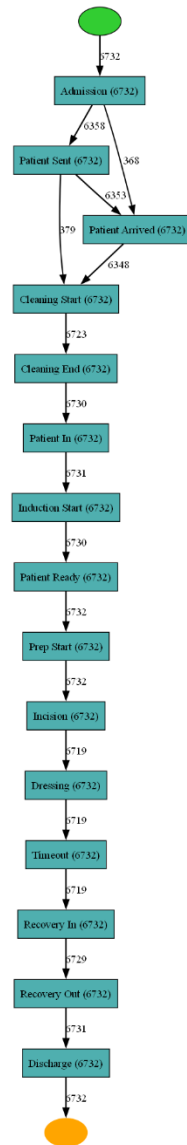


Figure 17 Baseline frequency model for the highest-volume surgeon

Temporally, the picture differentiates sharply by specialty. As shown in Figure 18, in the cardiac surgery, Incision to Dressing alone averages three hours, Recovery In to Recovery Out averages 36 minutes, and Recovery Out to Discharge averages seven days, an order of magnitude longer than the equivalent institution-wide steps. In ophthalmology, however, every step from Cleaning Start through Recovery Out sits between a few minutes and under an hour, as shown in Figure 19; the one outlier of comparable scale to cardiac surgery's is, notably, the same admission-side delay (Admission to Patient Arrived averaging seven days), suggesting the admission-side long tail (Section 5.4.2) is a shared, institution-wide issue, while intraoperative-and-recovery-side delay is genuinely specialty-specific. The full set of specialty-, location-, and surgeon-level process maps generated for this analysis is provided in Appendix A.

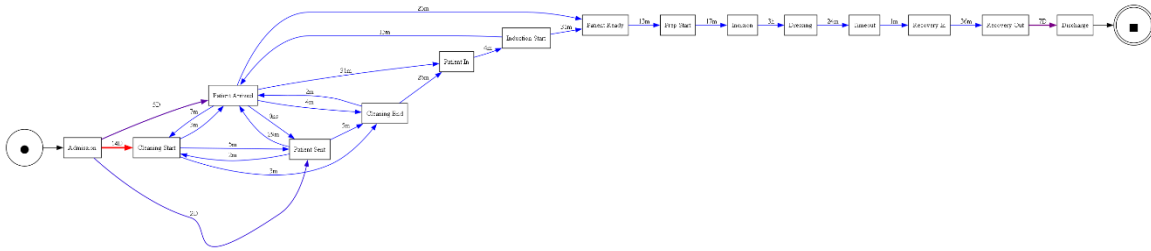


Figure 18 Mean-duration performance graph, cardiac surgery



Figure 19 Mean-duration performance graph, ophthalmology

The supplementary duration analysis confirms the scheduling problem motivating this thesis using data rather than anecdote: average total case duration varies meaningfully across the five highest-volume surgeons within the same location, and the ranking of surgeons by speed is not consistent across locations. A surgeon comparatively fast in the Main Operating Room is not necessarily fastest in Same Day Care, indicating duration variability is jointly driven by surgeon and location factors rather than either alone, as shown in Figures 20 and 21. This pattern is independently supported by external literature. (Eijkemans et al. 2010) first demonstrated that individual surgeon characteristics improve OR time prediction beyond procedure-based averages, and (Bartek et al. 2019), using 46,986 operations at a large academic medical center, found surgeon identity alone

accounted for 13% of case-time duration variability and that surgeon-specific models reduced scheduling overage by 15%.

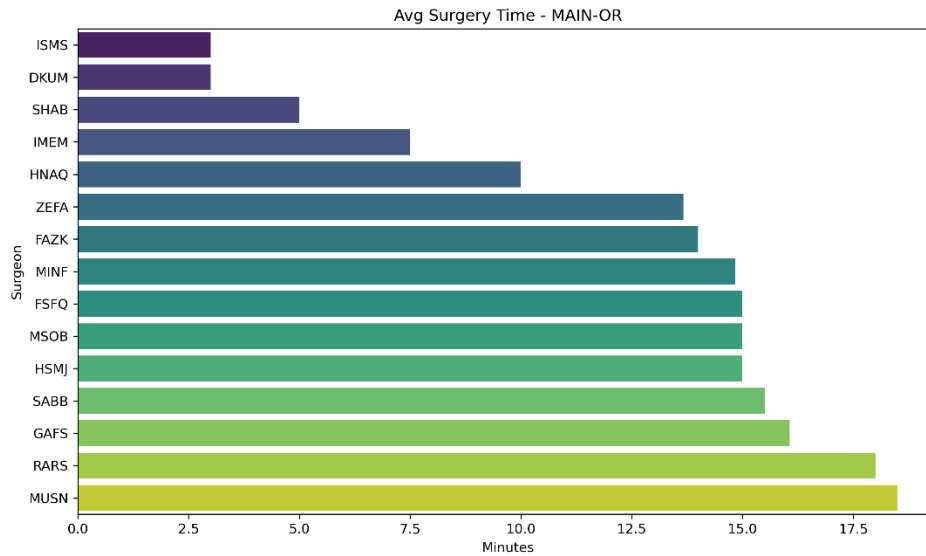


Figure 20 Average surgery time in the main operating room among the top 15 high-volume surgeons

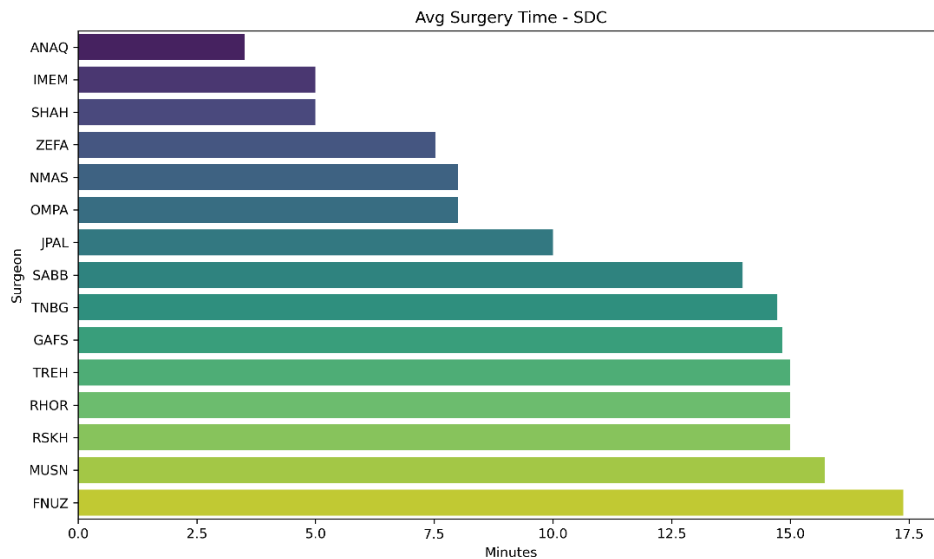


Figure 21 Average surgery time in the is same-day care among the top 15 high-volume surgeons

5.5 Comparison with Thesis I Results

The Thesis I report proposed a Constraint-Aware Alpha Miner (CAAM), embedding hand-coded clinical sequencing rules directly into the Alpha Miner's discovery logic, and evaluated it on the public BPIC2011 hospital event log because access to real-world institutional data was pending. That evaluation found CAAM did not outperform the standard Alpha Miner on fitness or generalization: over-restrictive hand-coded constraints

discarded valid but infrequent traces, leaving the resulting model less complete than the unconstrained baseline, though clinical plausibility improved modestly.

This thesis's results differ from those of Thesis I in three concrete respects:

Data. Thesis I's evaluation used the anonymized, abstracted BPIC2011 log, which is healthcare-related but not specifically surgical and does not represent a real perioperative workflow. This thesis uses 162,286 real, cleaned perioperative cases drawn from an actual tertiary-care institution, the kind of institutional data Thesis I's own future-work section identified as the necessary next step.

Method. Thesis I modified the Alpha Miner by adding a hard constraint-validation layer that discards any transition violating a fixed clinical rule, regardless of how frequently or rarely that transition occurred. This thesis instead modifies the Heuristic Miner's dependency measure itself, replacing a frequency ratio with a Bayesian posterior mean that discounts, rather than categorically discard low-frequency transitions, while leaving high-frequency transitions essentially unaffected (Section 4.3.4). Where Thesis I's hard constraints actively removed valid rare traces from the discovered model (causing the fitness degradation reported in its Section 4.5.2), this thesis's soft, sample-size-aware calibration is specifically designed to avoid that failure mode.

Outcome. Thesis I's CAAM scored lower than the unmodified baseline on fitness and generalization (Section 4.5.1 of that report). This thesis's Bayesian Heuristic Miner scores higher than its own unmodified baseline on every metric in Table 14 except simplicity, where the cost is an explicit, acknowledged trade-off (Section 4.3.4) rather than an unintended side effect of over-restrictive rules. In short, both theses pursued the same underlying goal, making a standard discovery algorithm more clinically faithful, but Thesis I's hard-constraint approach degraded model completeness, while this thesis's probabilistic recalibration approach improved it, on the metrics available in each case.

5.6 Summary

This chapter has presented the dataset used, the cleaning pipeline reducing it from 445,897 to 162,286 usable cases, the tools and depth-first/breadth-first experimental design used to

apply the methodology from Chapter 4, and the resulting structural and temporal findings at both the institution-wide and granular levels. The central results are that roughly 83% of surgical cases follow a statistically confirmed canonical pathway, with the remaining 17% resolving into five named, clinically interpretable exception categories rather than undifferentiated noise; that reporting mean and median duration together is necessary to avoid materially misleading conclusions, a finding independently corroborated in an unrelated hospital dataset; and that this thesis's Bayesian approach succeeds, on the metrics available, where the earlier constraint-based approach in Thesis I did not. Chapter 6 now discusses these results in relation to the research questions and literature and presents the recommendations arising from them.

Chapter 6

Discussion of Results

This chapter discusses the findings of Chapter 5 in relation to the research questions set out in Chapter 1, draws out the practical recommendations these findings support, summarizes the limitations of the work, and states its contributions and potential impact.

6.1 Answering the Research Questions

RQ1: How can process mining be used to discover a reliable baseline model of real-world surgical workflow from a large-scale hospital perioperative event log, both for the institution as a whole and for clinically meaningful subgroups?

This question is answered affirmatively. When applied to the cleaned event log of 162,286 cases, the baseline Heuristic Miner produced a clear, interpretable model for the whole institution: a near-linear pathway from Admission to Discharge, with case loss below 1% for almost every core transition (Section 5.4.1). Repeating this same analysis separately for thirteen specialties, four theatre locations, and the five busiest surgeons (Section 5.4.3) showed that this basic structure stays stable across all these subgroups. Even the busiest individual surgeon's model looks structurally almost identical to the institution-wide baseline. This stability is itself an important finding: it shows that process discovery at hospitals gives a consistent result no matter which slice of the data is used, which is a necessary foundation for the more detailed, subgroup-level findings later in this thesis.

RQ2: Can a statistically calibrated dependency measure, rather than a purely frequency-based one, produce a process model that more reliably distinguishes genuine rare clinical pathways from low-confidence noise, and does this improve the clinical interpretability of the discovered surgical process model?

Yes. The Bayesian Heuristic Miner produced 32 edges against the baseline's 17. It did not simply add edges indiscriminately but produced a *structured* result: ten edges confirmed at full statistical confidence as the core pathway, five further edges identified as high-confidence, clinically interpretable exceptions (Section 5.4.1), and seventeen low-

frequency edges explicitly flagged for clinical review rather than asserted as fact. This is the direct, demonstrated answer to RQ2: the Bayesian measure does not merely change numbers, it changes what the analyst is able to *say* about the data, distinguishing a confirmed checklist-sequencing deviation affecting over a thousand cases from a handful of edges that may be noise, something the baseline model's flat frequency counts cannot do. The evaluation metrics in Table 14 reinforce this: the Bayesian model scores higher on fitness, precision, generalization, and F-measure, at the cost of simplicity, an explicit, justified trade-off rather than an accidental one.

RQ3: Where, and for whom, do surgical process delays concentrate when the same event log is analyzed temporally rather than structurally, and how do these temporal findings relate to the structural pathways discovered under RQ1 and RQ2?

The temporal analysis (Section 5.4.2) identified a specific, severe case of mean-median divergence on the Patient Sent → Discharge transition (10-day mean against a 10-minute median) and a smaller divergence on Admission → Cleaning Start. The breadth-first results (Section 5.4.3) showed this divergence is not uniform across the institution: cardiac surgery shows genuinely long intraoperative and recovery durations in their own right, while ophthalmology's delays are confined almost entirely to the shared admission-side issue. This directly answers the second half of RQ3: the temporal findings relate to the structural findings precisely through the shared methodological lens established in Chapter 4 where both are instances of the same underlying problem, an aggregate statistic (a frequency count or a mean duration) concealing a more complex distribution that a single number cannot represent honestly.

RQ4: What concrete, operationally realistic recommendations can be derived jointly from the structural and temporal analyses, such that they are directly usable by the hospital's surgical and scheduling teams?

Section 6.2 below answers this question directly and in full.

6.2 Recommendations

This thesis was undertaken, as established in Chapter 1, specifically to produce recommendations that surgical scheduling and quality-improvement teams could act on, not as an afterthought to a descriptive analysis. The recommendations below are organized into five categories: structural, temporal, specialty-level, governance, and recommendations considered but not adopted. Each recommendation in the first four categories is tied directly to a specific finding from Chapter 5, or to the literature reviewed in Chapters 1 and 3. Recommendations that could not be grounded this way were deliberately left out of the first four categories and are listed instead in the fifth, along with the reason for leaving each one out.

6.2.1 Structural Recommendations

R1. Make the Bayesian-confirmed pathway the hospital's official surgical pathway.

The ten steps confirmed at full statistical confidence by the Bayesian model (Section 5.4.1) should replace the current, looser frequency-based description of "normal" surgical flow as the hospital's documented reference pathway. This gives every subsequent recommendation in this chapter a single, agreed baseline to measure deviation against.

R2. Track the five known exception patterns by name, not as one general category.

This thesis identified five specific, statistically confirmed ways that cases deviate from the standard pathway: skipping straight from Admission to Patient In, skipping straight from Patient In to Incision, skipping the Timeout step before Recovery In, skipping the recovery stage before Discharge, and incision occurring before the Timeout step. Rather than treating "deviations" as one undifferentiated group, the hospital's reporting should track each of these five separately. This allows the hospital to see, for example, that the Timeout-skip and the incision-before-Timeout deviation are very different in scale and seriousness, instead of averaging them together.

R3. Review the cases where the safety checklist was signed after incision had already begun.

In just over 1,000 cases, the Timeout step, which corresponds to the WHO Surgical Safety Checklist sign-out, was recorded *after* the incision had already started, rather than before, as protocol requires (Schwendimann et al. 2019); (Gul et al. 2022). The event log

alone cannot tell us whether this reflects a genuine skipped step or simply a delay in entering the timestamp. A targeted chart review of a sample of these cases by theatre nursing leadership is the realistic next step — this is the one finding in this thesis with a direct patient-safety dimension, and it became visible only because the Bayesian model could separate a confirmed, repeated pattern from one-off noise.

R4. Clarify who is responsible for logging "Patient Sent" and "Patient Arrived," and trigger the actual transfer only once the room is ready. These two steps showed unexpectedly low statistical confidence in their ordering (a dependency score of 0.721, against 1.000 for the rest of the core pathway), most likely because it is not clearly defined which staff member logs each event. Hospitals should fix this with a documentation rule, not a process redesign: assign clear responsibility for each timestamp, and ensure the patient is physically moved toward the OR only once cleaning is finished or nearly finished and the anesthesia and surgical teams are expected to be available.

6.2.2 Temporal Recommendations

R5. Report the typical case time alongside the average, not instead of it. This thesis found a transition where the average duration was ten days but the typical (median) duration was ten minutes (Section 5.4.2), a single extreme case was enough to make the average meaningless on its own. This pattern is not unique to a single hospital: an unrelated hospital dataset analyzed using Fuzzy Miner showed the same gap between average and typical transition times (Özdağoglu et al. 2025). Hospitals should report both numbers side by side for every duration metric it tracks, as standard practice.

R6. Flag any single case-step over 24 hours as a separate exception, not part of the normal average. Long outliers like the one described in R5 should be pulled out of routine scheduling calculations automatically, and reviewed separately, rather than left to quietly distort the averages everyone else relies on for planning.

R7. Prioritize delays by how often they happen, not by how dramatic they look. A 5-minute delay that occurs in 130,000 cases costs the hospital far more total operating room time over a year than a single 10-day delay that happens once. The hospitals should rank

its process-improvement priorities by frequency multiplied by typical delay, not by which individual case looks the worst on paper.

R8. Create "Patient Ready to Incision" as a tracked, named KPI, and break it down further where it helps. This is the clearest and fairest delay measure available, because it only counts time after the patient is confirmed ready — it does not blame the surgical team for delays that happened earlier in the process. Where more detail is useful, this single interval can be split into four named stages: Patient In to Induction Start (anesthesia start delay), Induction Start to Patient Ready (anesthesia and preparation time), Patient Ready to Prep Start (surgical and nursing readiness), and Prep Start to Incision (final start delay). This makes it possible to say specifically whether a slow case is an anaesthesia issue, a nursing issue, or a room issue, instead of leaving staff to guess.

6.2.3 Specialty-Level Recommendations

R9. Set a different scheduling buffer for each specialty, based on its own typical case length. Comparing the cardiac surgery and ophthalmology timing maps directly (Section 5.4.3) shows that cardiac cases run far longer at almost every step. A single hospital-wide scheduling buffer is, by definition, wrong for at least one of these two specialties. This pattern is consistent with prior research showing that surgeon and procedure-type variation meaningfully affects operating room time (Eijkemans et al. 2010); (Bartek et al. 2019), a pattern reaffirmed in more recent multi-specialty operating room analyses (Celik et al. 2025).

R10. Confirm a recovery bed is available before booking long-recovery specialties such as cardiac surgery, and track discharge delays by specialty. Cardiac cases in this dataset average roughly seven days from recovery to discharge (Section 5.4.3). Booking a long-recovery case without first checking downstream bed availability risks a case that finishes on time in the operating room but still disrupts the rest of the day's schedule, a pattern well documented in the perioperative scheduling literature (Lyons et al. 2023). For day-care and short-stay specialties specifically, Hospitals should also track Recovery Out to Discharge as its own number, since causes of delay here, discharge paperwork,

pharmacy delay, billing delay, or ward transfer, are different from causes of delay earlier in the process.

R11. Do not use simple, predictable specialties like ophthalmology as the standard that complex specialties are judged against. Ophthalmology's short, consistent durations partly reflect simpler procedures, not necessarily better-run theatres. Complex specialties such as cardiac or vascular surgery should not be expected to match this standard without adjustment.

6.2.4 Governance Recommendations

R12. Build a separate dashboard view for each specialty. Each specialty's dashboard should show its own frequency pattern, its own exception categories from R2, and its own median-and-mean timing from R5.

R13. Fix the underlying timestamp data quality first, then build the dashboard, then add automatic alerts, in that order. Over half of all cases removed during the data cleaning process in Chapter 5 were removed specifically because of timestamp errors. Automatic alerts built on top of unreliable timestamps will generate more false alarms than genuine insight. Data quality should be addressed first, the dashboard from R12 second, and automatic alerting last.

R14. Track operating room turnover with three specific numbers: idle before clean duration, clean duration, and idle after clean duration. The third number matters most. If a room is clean but the next patient is not yet inside it, the bottleneck is not housekeeping, it is coordination, most likely a delay in moving the patient. This distinction lets hospitals separate housekeeping delay from transport or coordination delay, instead of treating both as the same problem. Where it adds value, this can be tracked by specialty as well as hospital-wide, since some specialties may consistently move patients in more slowly than others.

R15. Attach specific alerts to specific failure points, not general warnings. For example: room cleaned but patient not yet in → alert the OR coordinator; patient in the room but anesthesia not started → alert the anesthesia lead; patient ready but prep not

started → alert the surgical or nursing team; prep started but incision delayed → alert the OR manager; recovery full or delayed → alert PACU or bed management; recovery finished but discharge delayed → alert the discharge team. Each alert should name exactly which step failed and who is responsible for the next action, rather than issuing a generic "delay detected" notice.

6.2.5 Recommendations Considered and Not Adopted

Several ideas raised during this research were deliberately left out of the final list above, each for a specific reason rather than simply being overlooked:

- **Splitting delay into "clinical" versus "administrative" causes.** A reasonable distinction in principle, but the process maps used in this thesis cannot actually tell these two categories apart, so this recommendation is not something the data can support.
- **Separate scheduling rules for elective versus emergency cases.** This thesis's analysis did not split cases this way, so there is no direct evidence behind this recommendation here, even though it is a reasonable idea on its own terms.
- **Detailed, specialty-specific rulebooks for each of the thirteen specialties** (for example, ICU-bed confirmation rules specific to cardiac surgery, or implant-readiness checklists specific to orthopedics). This thesis directly compared only two specialties in detail, cardiac and ophthalmology, so writing a separate rulebook for every specialty would go further than the evidence here actually supports.
- **An earlier, formal pre-operative readiness checklist covering consent, labs, blood products, and team availability.** This is sound clinical practice, but it is not something the process maps in this thesis directly tested or measured, so it is noted here rather than presented as an evidenced finding.
- **Pediatric-specific readiness factors**, such as fasting status, guardian consent, or pre-medication anxiety planning. Clinically sensible, but nothing in this thesis's structural or temporal results provides evidence either for or against it.

6.3 Limitations

Beyond the specific theoretical and practical constraints summarized in Table 15, this work shares a limitation common to applied Bayesian process mining more broadly: the absence of a standardized benchmark or shared evaluation dataset for comparing calibrated dependency measures across studies. A 2025 tertiary review synthesizing prior systematic reviews of healthcare process mining noted explicitly that inconsistent evaluation metrics and the scarcity of publicly available, large-scale surgical event logs remain a persistent obstacle to comparing discovery methods across institutions (Santos et al. 2025). This limits the ability of this thesis, or any comparable single-institution study, to claim that its specific numerical improvements (Table 14) would replicate identically elsewhere, even though the underlying statistical mechanism is institution-agnostic. Additionally, because the Beta-Binomial calibration depends on the specific prior structure chosen (Section 4.3.5), results obtained under a different, equally defensible prior configuration could differ, which is itself a limitation of any Bayesian approach applied without formally elicited priors, a concern echoed in recent applied Bayesian process-mining work (Celik et al. 2025).

Limitation	Type	Description
Hand-set Bayesian priors	Theoretical	The α , β values in Table 4.3 are analyst-assumed defaults, not formally elicited from clinical experts (Section 4.3.5)
No formally validated ground-truth model	Theoretical	No externally validated reference model exists to benchmark the conformance metrics (fitness, precision, generalization, F-measure) against
Independence assumption in the Beta-Binomial model	Theoretical	Each activity pair's dependency is modelled independently; the model does not capture correlations across pairs (a limitation shared with the comparable approach of (Janssenswillen et al. 2020))
Directly-follows abstraction only	Theoretical	Like the classical Heuristic Miner, the proposed method captures only immediate succession, not

		longer-range dependencies between non-adjacent activities
Substantial case loss during cleaning	Practical	64% of the original 445,897-row extract was removed during cleaning (Chapter 5), predominantly due to chronologically inconsistent timestamps; the remaining sample's representativeness relative to the excluded cases cannot be fully verified
Specialty reconstruction via text matching	Practical	Roughly two-thirds of cases missing a specialty label were reassigned using keyword matching on free-text procedure descriptions (Section 5.1.3, Step 7), which may misclassify edge cases
Single-institution scope	Practical	All findings are derived from one hospital; generalizability to other institutions, particularly those with different information systems or clinical protocols, is untested
No clinical outcome linkage	Practical	This thesis analyses process timing and sequencing only; it makes no claims about and does not control for clinical severity, complication rates, or patient outcomes
No standardized benchmark	Practical	No shared evaluation dataset or metric standard exists for comparing calibrated dependency measures across institutions, limiting external comparability of this thesis's numerical results (Santos et al. 2025)

Table 13 Limitations of the proposed methodology and results

6.4 Contributions and Potential Impact

This thesis makes four contributions, each with a distinct potential impact on the research community and on hospitals specifically:

- A Bayesian extension to the Heuristic Miner's dependency measure, replacing the classical frequency ratio with a Beta-Binomial posterior mean. As established in Chapter 3, no prior published work modifies the Heuristic Miner's dependency calculation this way; the closest comparable approaches either replace the

modelling formalism entirely (Moreira 2015) or apply Bayesian reasoning only after discovery is complete (Janssenswillen et al. 2020).

- A large-scale, real-world application to a perioperative log, producing baseline and Bayesian models at the institution level and across thirteen specialties, four locations, and five surgeons. For the hospital, this is, to this thesis's knowledge, the first log-derived characterization of its own surgical process behavior at this scale.
- A connected structural-and-temporal analysis layer, showing that the same statistical reasoning applies to both transition frequency and duration, with direct empirical payoff (the mean-median divergence finding, independently corroborated in Özdağoğlu et al. 2025). For the research community, this offers a template for treating structural discovery and performance analysis as one coherent statistical exercise rather than two disconnected analyses, addressing Gap 3 identified in Chapter 3.

6.5 Summary

This chapter has answered each of this thesis's four research questions directly against the results of Chapter 5, translated those results into sixteen filtered, realistic recommendations, summarized the theoretical and practical limitations of the work in Table 15, and stated the four contributions this thesis makes to both the process mining research community and to hospitals specifically. Chapter 7 now concludes the thesis.

Chapter 7

Conclusions and Future Work

This chapter concludes the thesis by restating the core research problem and how it was addressed, summarizing the contributions, and outlining directions for future work.

7.1 Summary of the Research Problem and Its Resolution

This thesis began from a concrete operational problem at the Aga Khan University Hospital, Karachi (Chapter 1): surgical scheduling and process-improvement decisions were being made without a statistically grounded, data-derived understanding of real surgical workflow behavior. Three specific gaps motivated this work. No baseline understanding of the real surgical process, no principled way to separate genuine rare clinical pathways from noise, and no quantified picture of where and for whom process delays concentrate.

This thesis resolved this problem through a three-stage process mining pipeline (Chapter 4). First, the classical Heuristic Miner was applied to a perioperative event log to establish a frequency-based structural baseline, with the Heuristic Miner selected over the Alpha Miner on the basis of both demonstrated computational tractability and a structural comparison conducted on this thesis's own data (Section 4.2). Second, a Bayesian extension to the Heuristic Miner's dependency measure was developed, replacing the classical frequency ratio with a Beta-Binomial posterior mean (Section 4.3), embedding statistical calibration directly within the discovery process itself, rather than applying it afterward as in the closest comparable prior work, serving as this thesis's central methodological contribution. Third, this same statistical reasoning was extended to the temporal dimension of the data through mean-and-median performance analysis, deliberately connected to the structural findings rather than treated as a separate exercise (Section 4.4).

Applied to 162,286 cleaned surgical cases (Chapter 5), this pipeline showed that approximately 83% of cases follow a statistically confirmed canonical surgical pathway, with the remaining 17% resolving into five named, clinically interpretable exception

categories, including, most significantly, a checklist-sequencing deviation in which incision precedes the WHO Surgical Safety Checklist sign-out in just over a thousand cases. The temporal analysis identified a severe case of mean-median divergence on the Patient Sent → Discharge transition (a 10-day mean against a 10-minute median), a finding independently corroborated in an entirely unrelated hospital dataset (Özdağoglu et al. 2025). Specialty-level analysis showed that delay patterns differ sharply between complex specialties such as cardiac surgery and simpler, lower-variance specialties such as ophthalmology, directly confirming the surgeon- and procedure-level scheduling variability that motivated this research in the first place.

These findings were translated into sixteen filtered, operationally realistic recommendations (Section 6.2), spanning structural, temporal, specialty-level, and governance changes — most requiring no new tooling, only changes to how data the hospital already collects is reported, reviewed, and acted upon.

7.2 Contributions

This thesis makes four contributions, summarized in full in Section 6.4:

1. A Bayesian extension to the Heuristic Miner's dependency measure, distinct from prior probabilistic process mining approaches in that it intervenes during discovery rather than after it, at no asymptotic computational cost over the classical algorithm.
2. A large-scale, real-world process mining analysis of a perioperative event log, the first such analysis conducted at this institution to this thesis's knowledge.
3. A connected structural-and-temporal analysis framework, demonstrating that the same statistical reasoning underlies both frequency and duration confidence.
4. A set of sixteen filtered, operationally realistic recommendations for surgical scheduling and process improvement, directly usable by the hospitals.

7.3 Future Work

Several directions for future work follow directly from the limitations identified in Section 6.3:

Formal elicitation of Bayesian priors. The prior values used in this thesis (Table 10) were set by the analyst based on general clinical reasoning. Future work should conduct

structured elicitation interviews with the hospital's surgical and anesthetic staff, asking directly what proportion of cases they would expect one activity to follow another before seeing any data, and use these responses to set α and β formally.

Hierarchical or learned priors. As noted in the original motivation for this work, a hierarchical Bayesian model could learn appropriate priors directly from the data's own structure, rather than relying on hand-specified defaults, a natural extension of the Beta-Binomial framework developed in Chapter 4.

Investigation of the checklist-sequencing finding. The Incision-before-Timeout deviation identified in Section 5.4.1 warrants a dedicated clinical chart review, as recommended in R3, to determine whether it reflects documentation lag or a genuine protocol deviation, a question this thesis's log-based analysis alone cannot resolve.

Multi-institutional validation. This thesis's findings are drawn from a single hospital. Applying the same Bayesian Heuristic Miner framework to perioperative data from other tertiary hospitals, in Pakistan or elsewhere, would test whether the structural and temporal patterns identified here (particularly the mean-median divergence phenomenon, already corroborated once in Turkish hospital data) generalise more broadly.

Integration into a deployed scheduling tool. The recommendations in Section 6.2, particularly the dashboard proposed in R12, represent a description of what should be built rather than a built artefact. Future work could implement this dashboard directly on top of the live perioperative data feed, subject to the phased, data-quality-first rollout sequence recommended in R13.

7.4 Closing Remarks

Large-scale institutional problems like surgical scheduling inefficiency rooted in undocumented process variability are rarely solved by a single algorithmic change. What this thesis demonstrates is narrower but, it is hoped, genuinely useful: that a well-targeted statistical correction to an existing, trusted discovery algorithm, applied to data a hospital already collects, can surface specific, actionable findings such as a checklist-sequencing gap, a misleadingly skewed delay statistic, a surgeon-dependent scheduling pattern, that neither informal clinical experience nor an uncorrected frequency count could reveal on their own.

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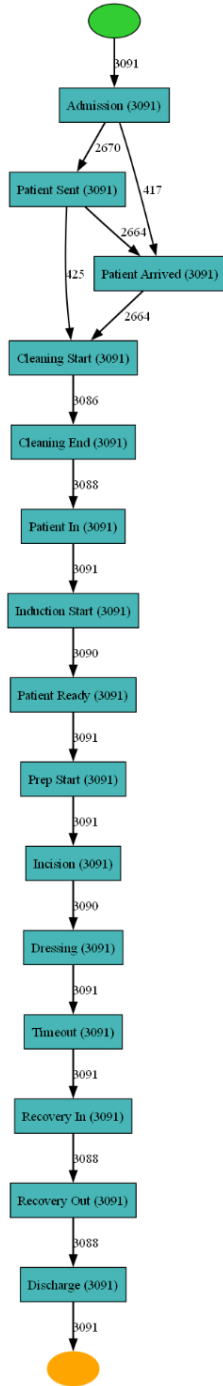
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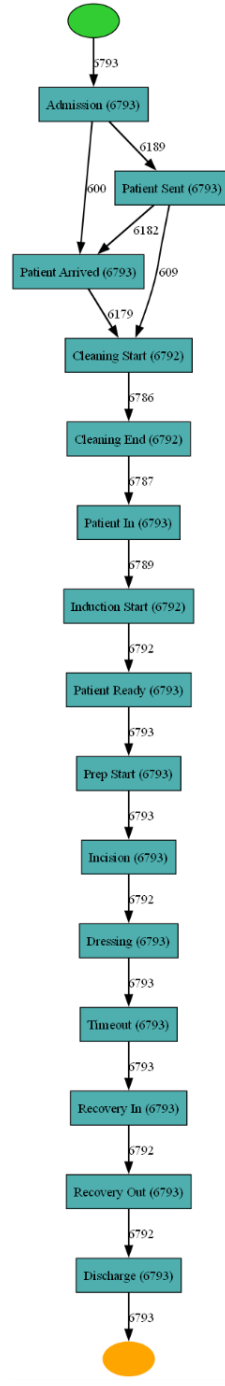
Appendix

A Structural (Frequency-Based) Heuristic Miner Models

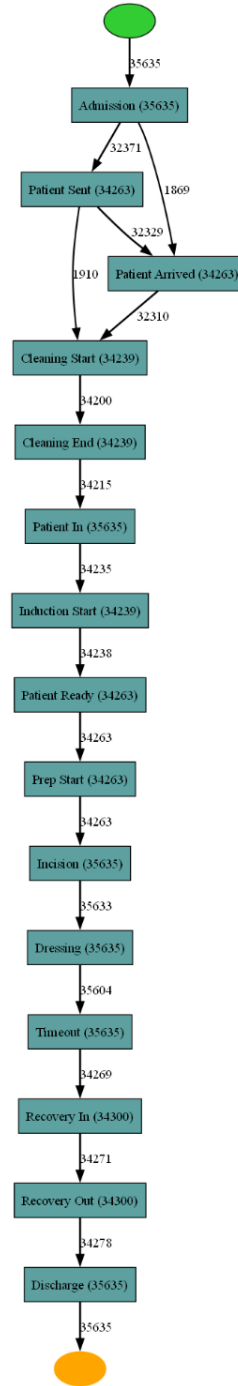
A.1 Surgery Specialty-Level Maps



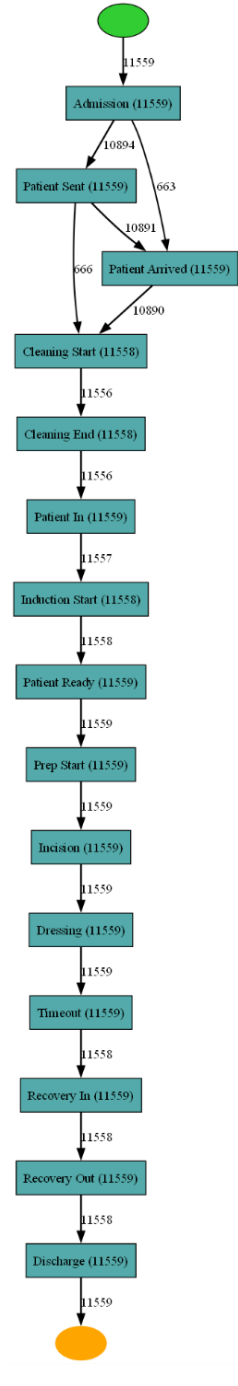
**Fig A Breast Surgery
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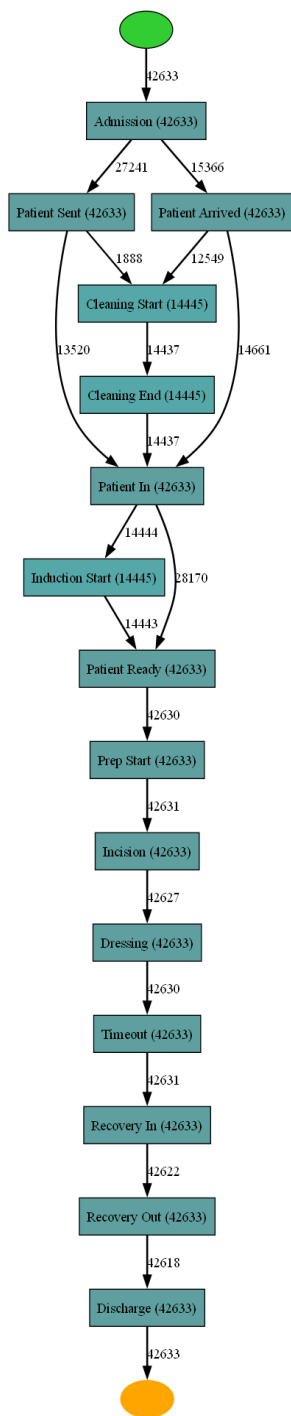
**Fig B ENT Surgery
(Frequency)**



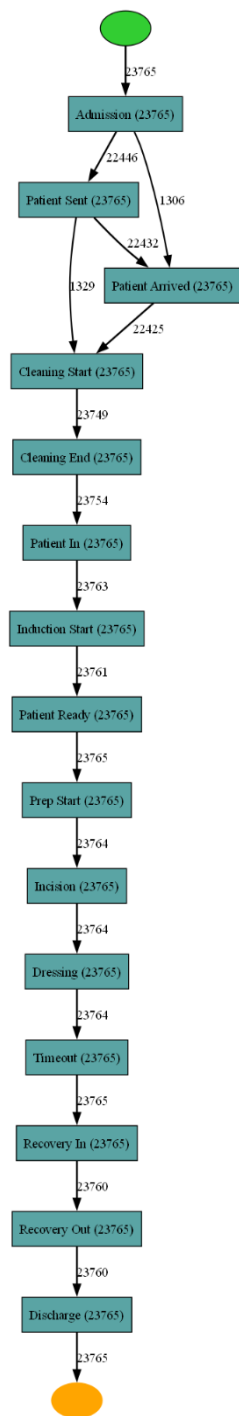
**Fig C Gen Surgery
(Frequency)**



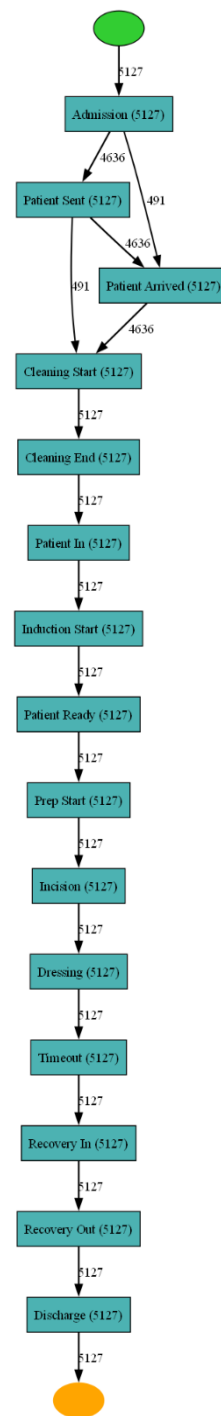
**Fig D Neurosurgery
(Frequency)**



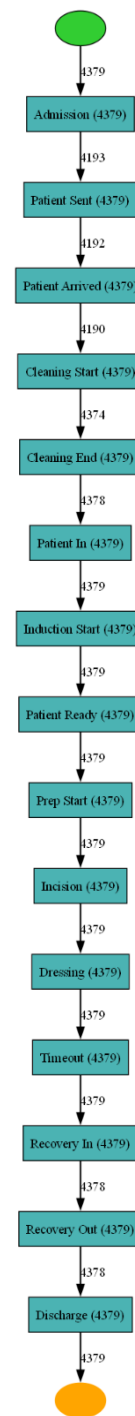
**Fig E Obgynae Surgery
(Frequency)**



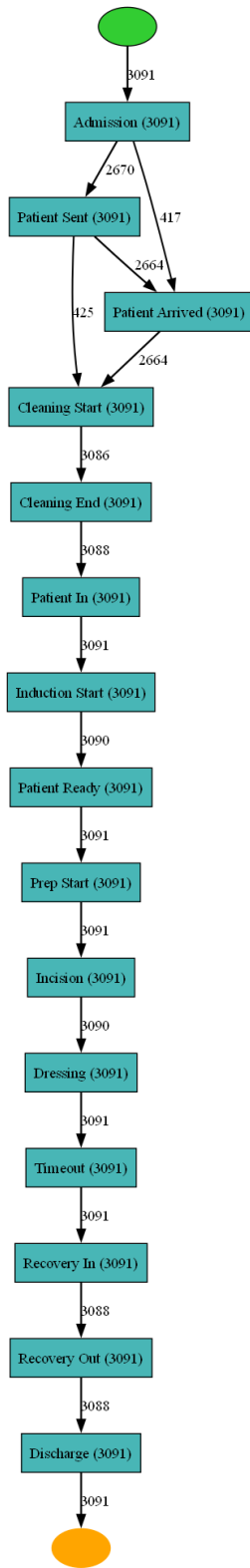
**Fig F Orthopedics
Surgery (Frequency)**



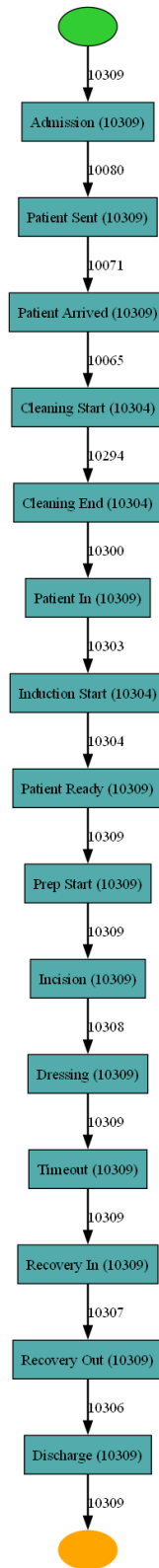
**Fig G Pediatrics
Surgery (Frequency)**



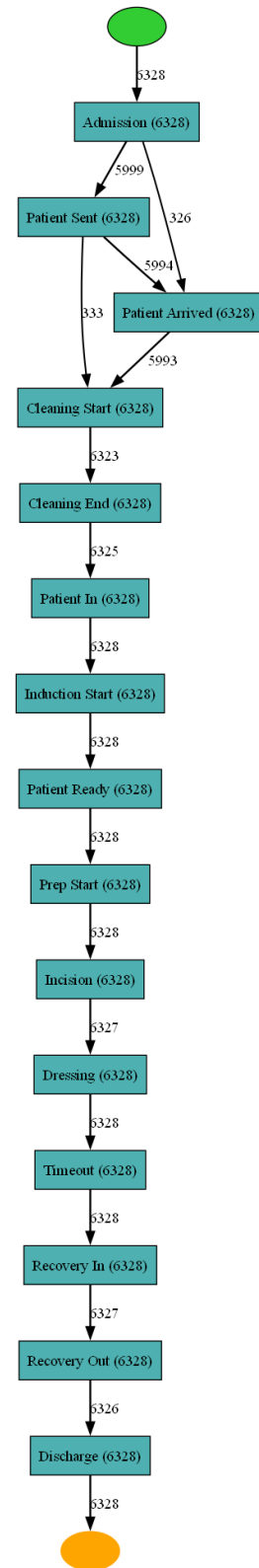
**Fig H Plastic Surgery
(Frequency)**



**Fig I Thoracic Surgery
(Frequency)**

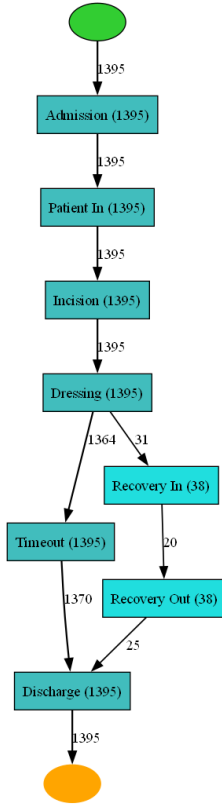


**Fig J Urology Surgery
(Frequency)**

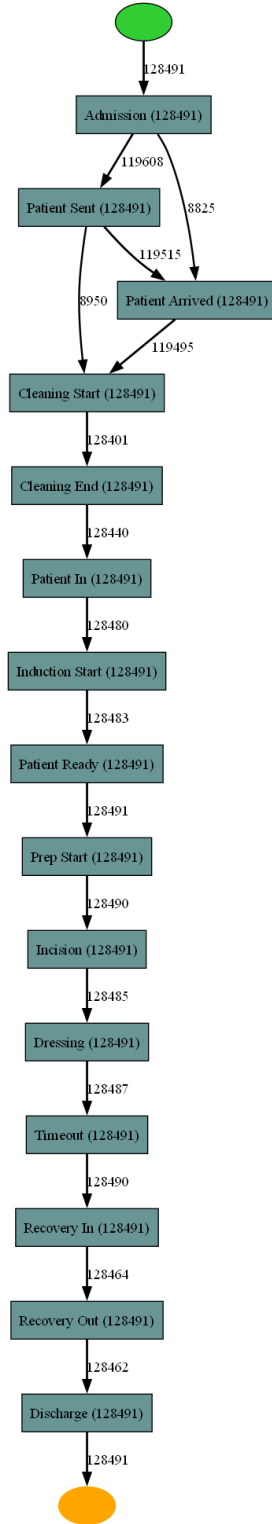


**Fig K Vascular Surgery
(Frequency)**

A.2 Surgery Location-Level Maps



**Fig L Endoscopy
(Frequency)**



**Fig M Main-OR
(Frequency)**

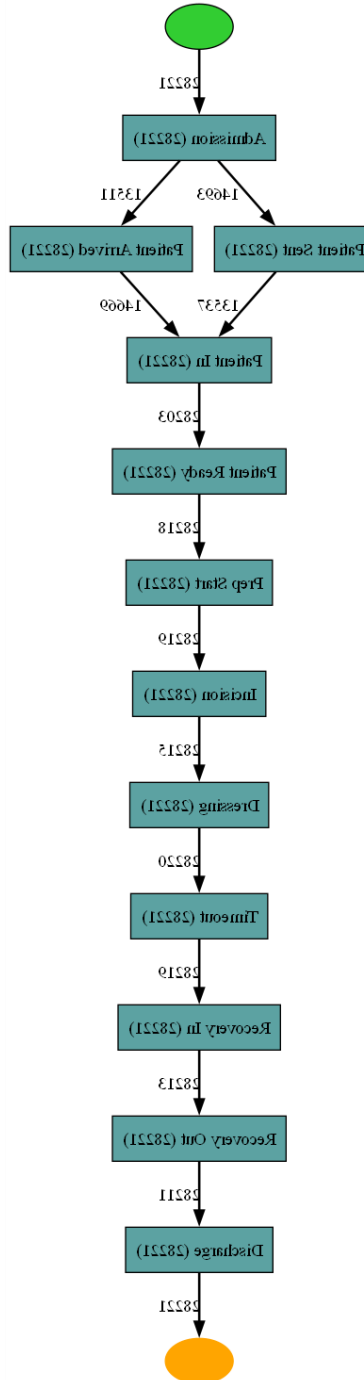
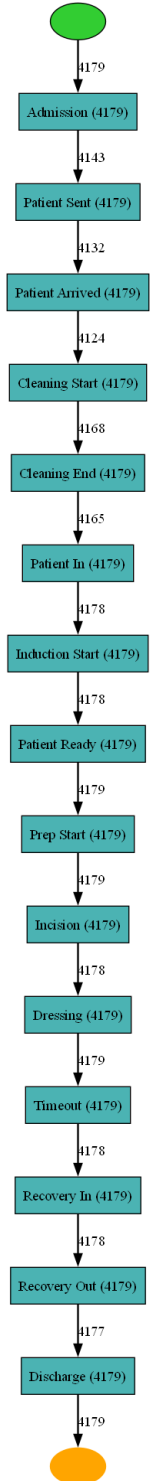
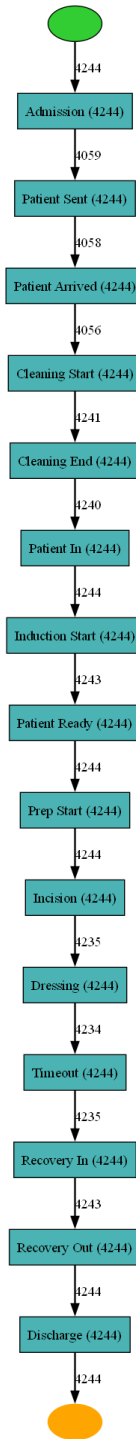


Fig N Obgynae (Frequency)

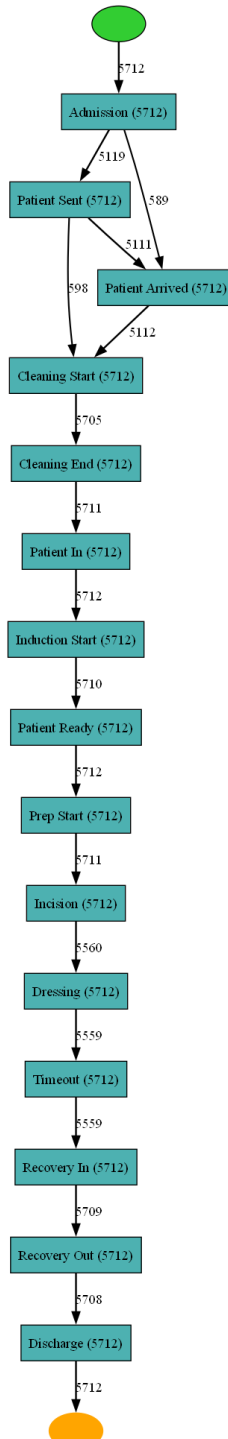


**Fig O SDC
(Frequency)**

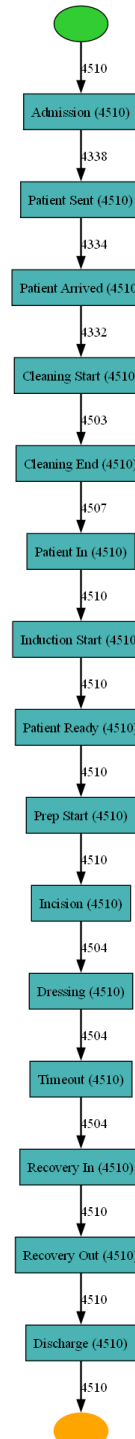
A.3 Top 5 Highest Volume Surgeon-Level Maps



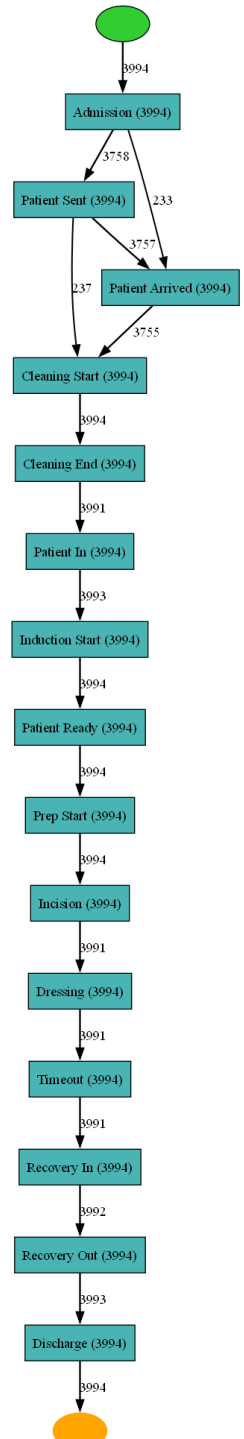
**Fig P 2nd-highest volume
(Frequency)**



**Fig Q 3rd highest volume
(Frequency)**



**Fig R 4th highest volume
(Frequency)**



**Fig S 5th highest volume
(Frequency)**

B Temporal (Time-Based) Direct-Flow Graphs

B.1 DFG Graphs with Mean Duration

B.1.1 Surgery Specialty-Level Graphs

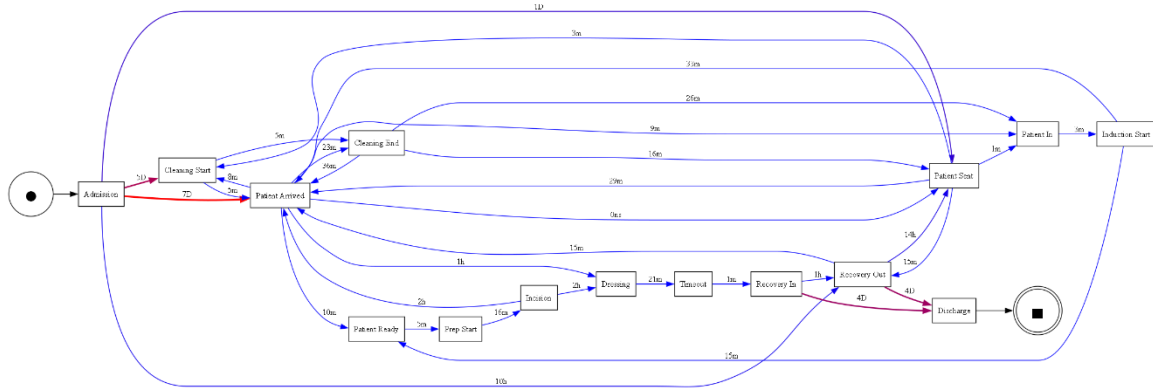


Fig T Breast Surgery (Mean Duration)

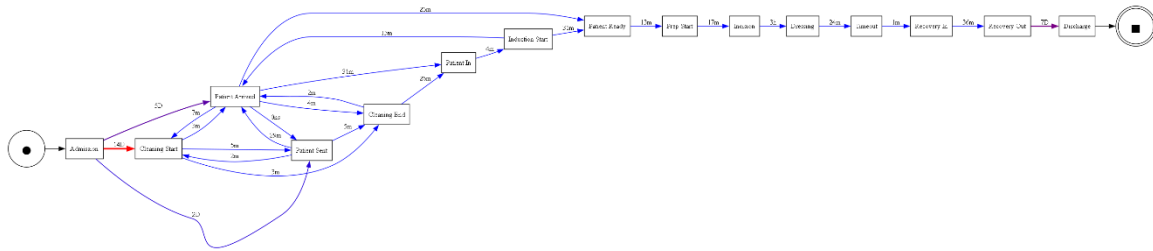


Fig U Cardiac Surgery (Mean Duration)

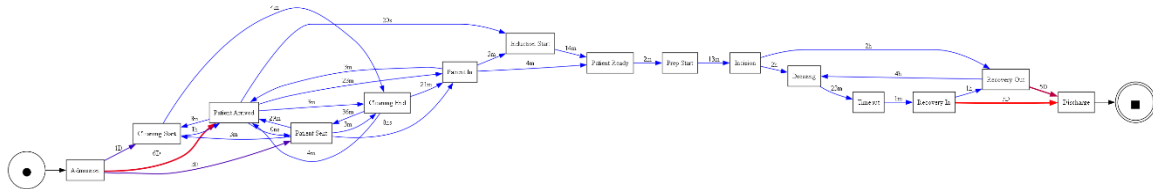


Fig V ENT Surgery (Mean Duration)

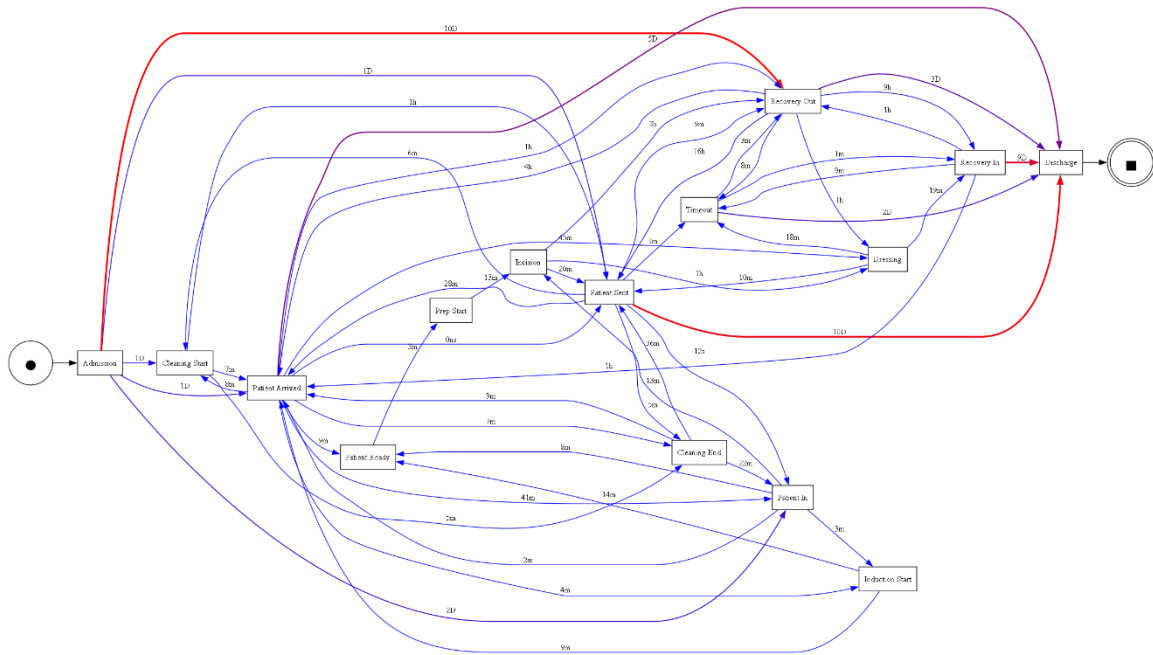


Fig W General Surgery (Mean Duration)

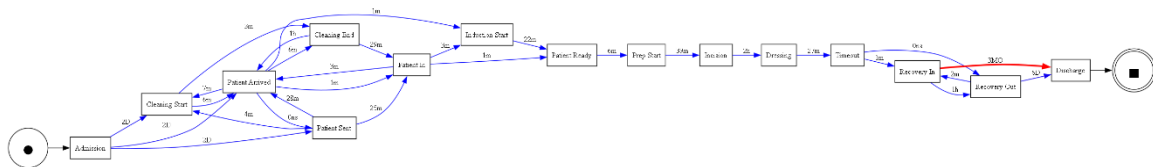


Fig X Neurosurgery (Mean Duration)



Fig Y Ophthalmology (Mean Duration)

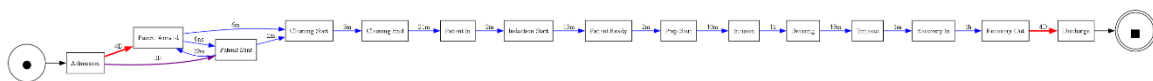
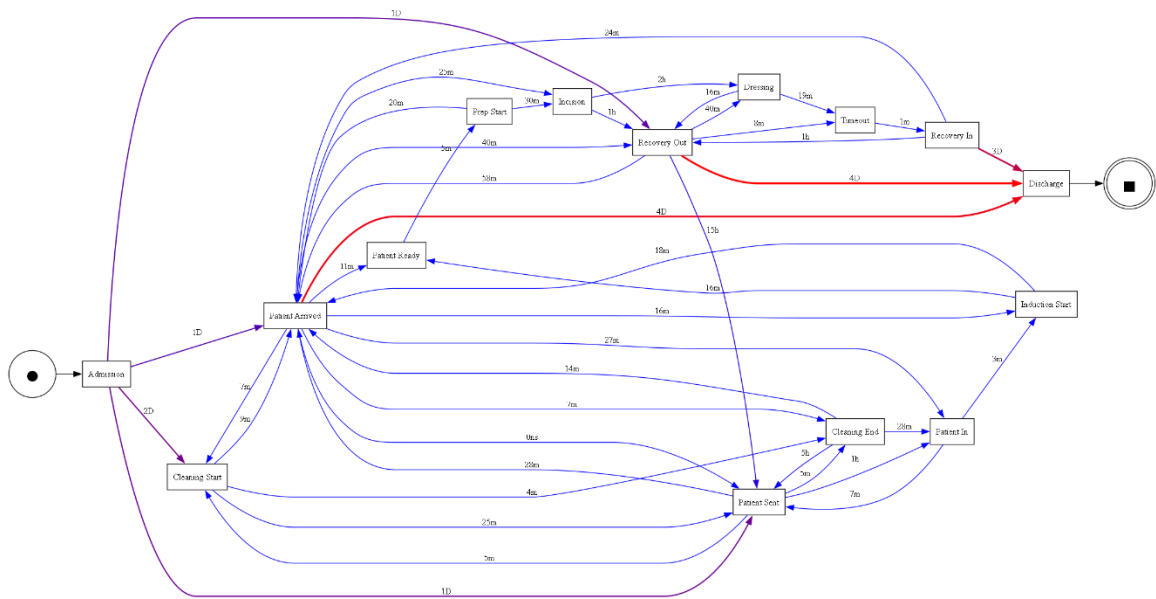
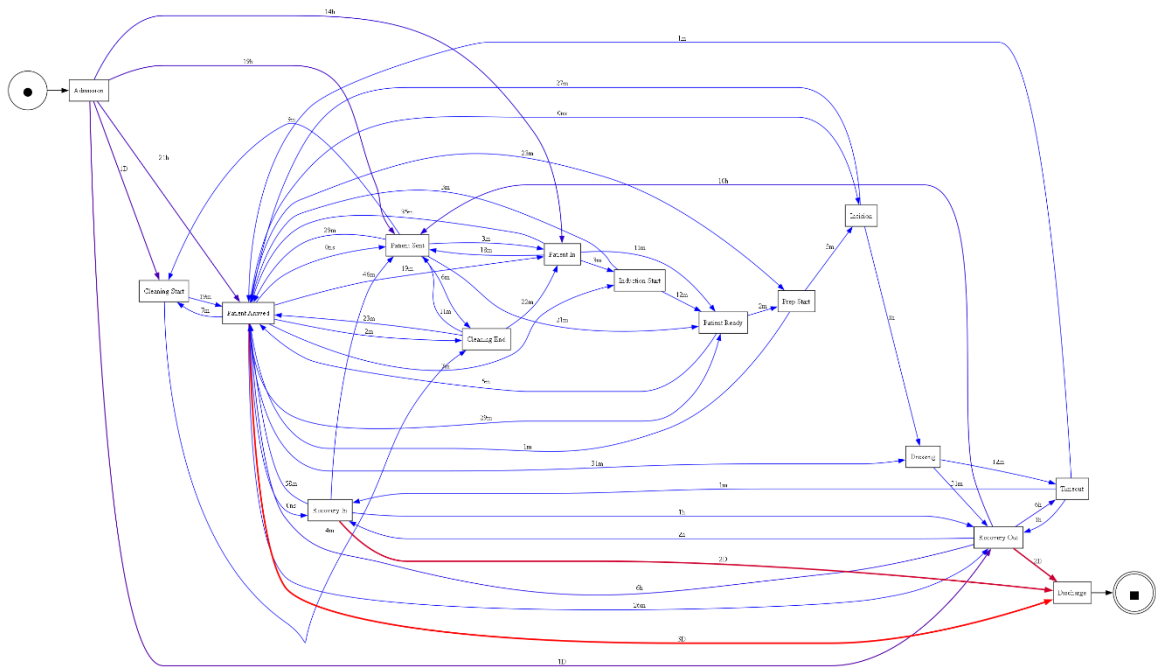
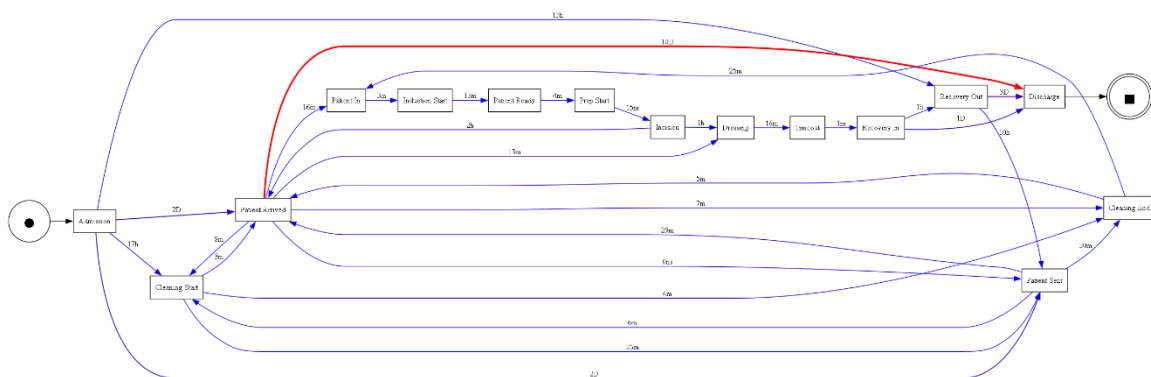
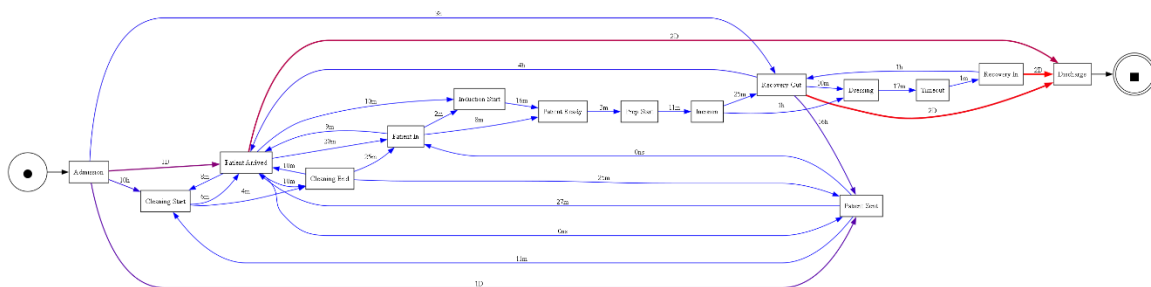
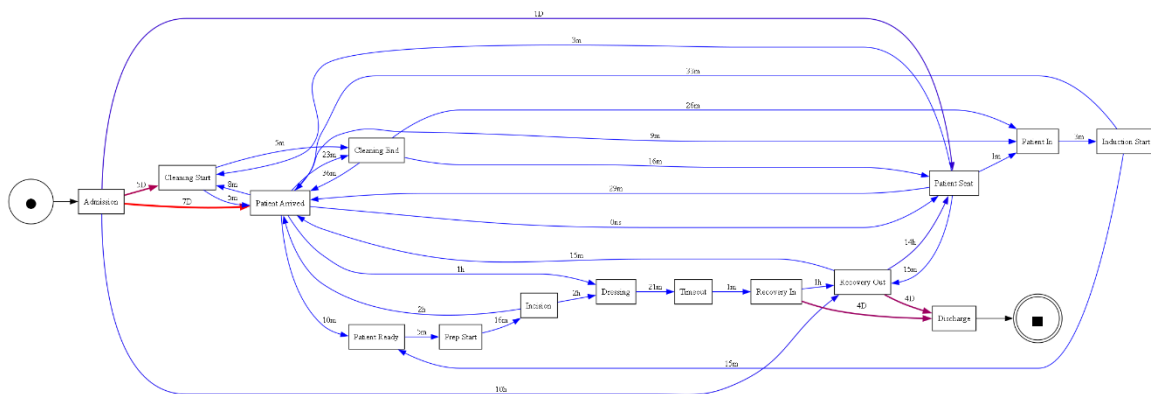
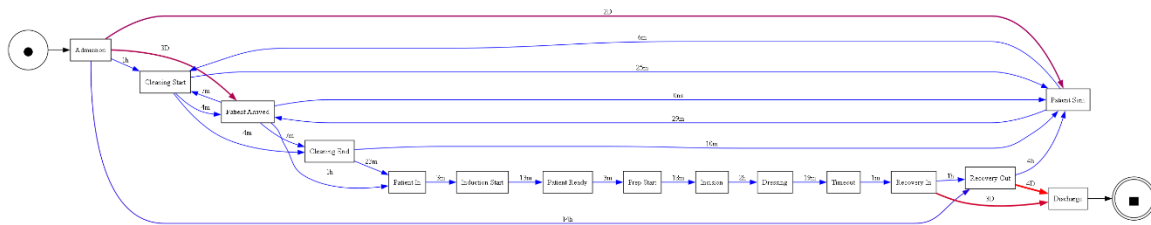


Fig Z Pediatrics Surgery (Mean Duration)





B.1.2 Surgery Location-Level Graphs

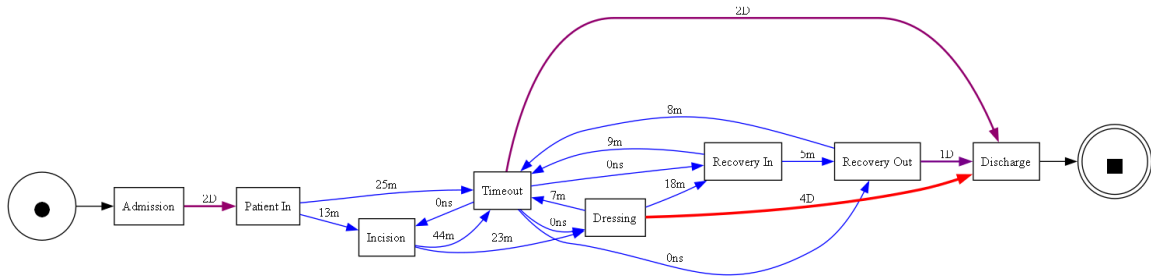


Fig GG Endoscopy (Mean Duration)

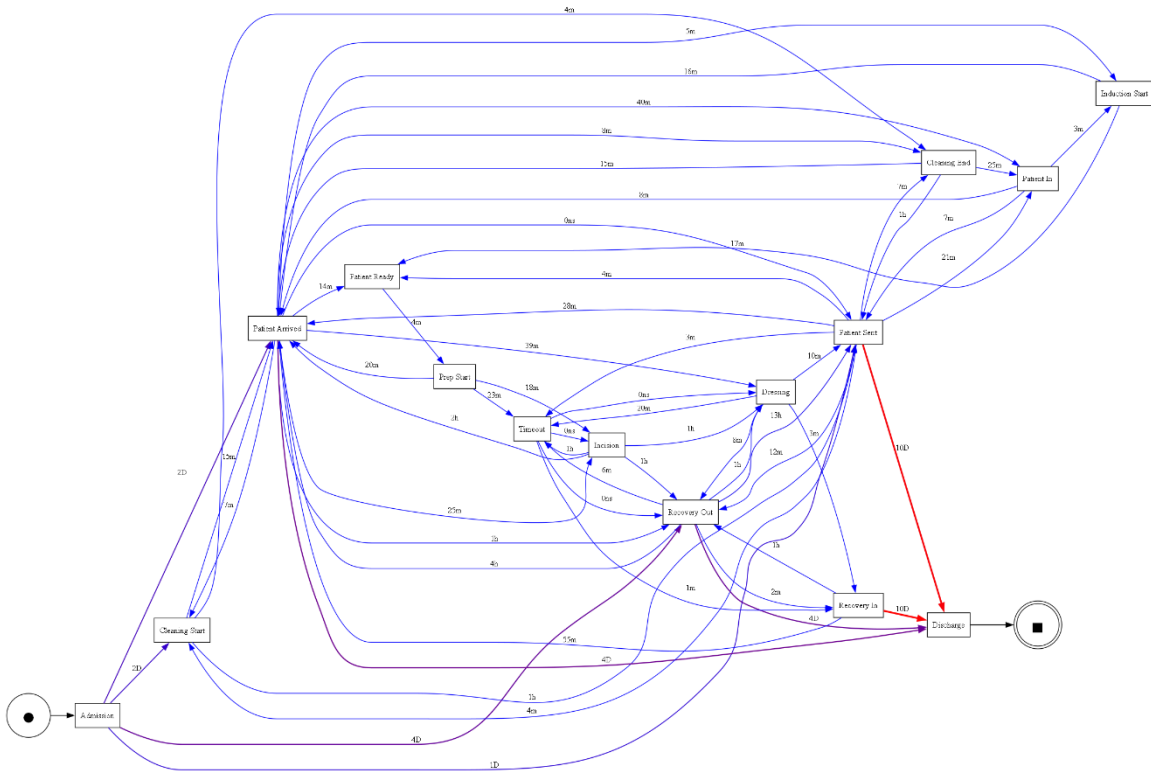


Fig HH Main-OR (Mean Duration)

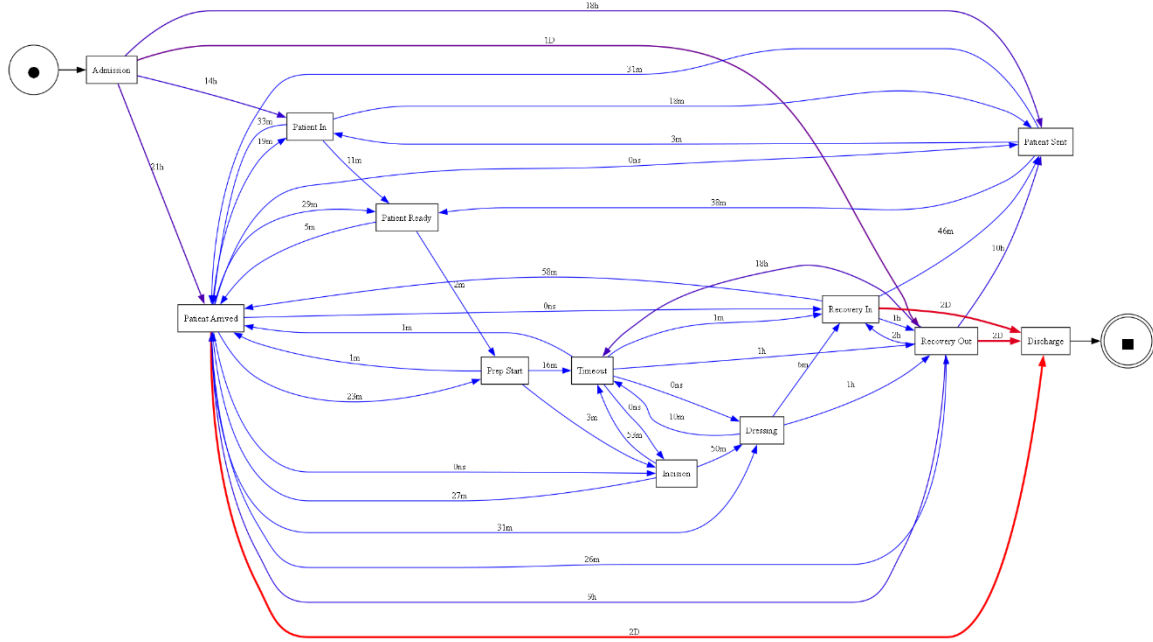


Fig II Obgynae (Mean Duration)

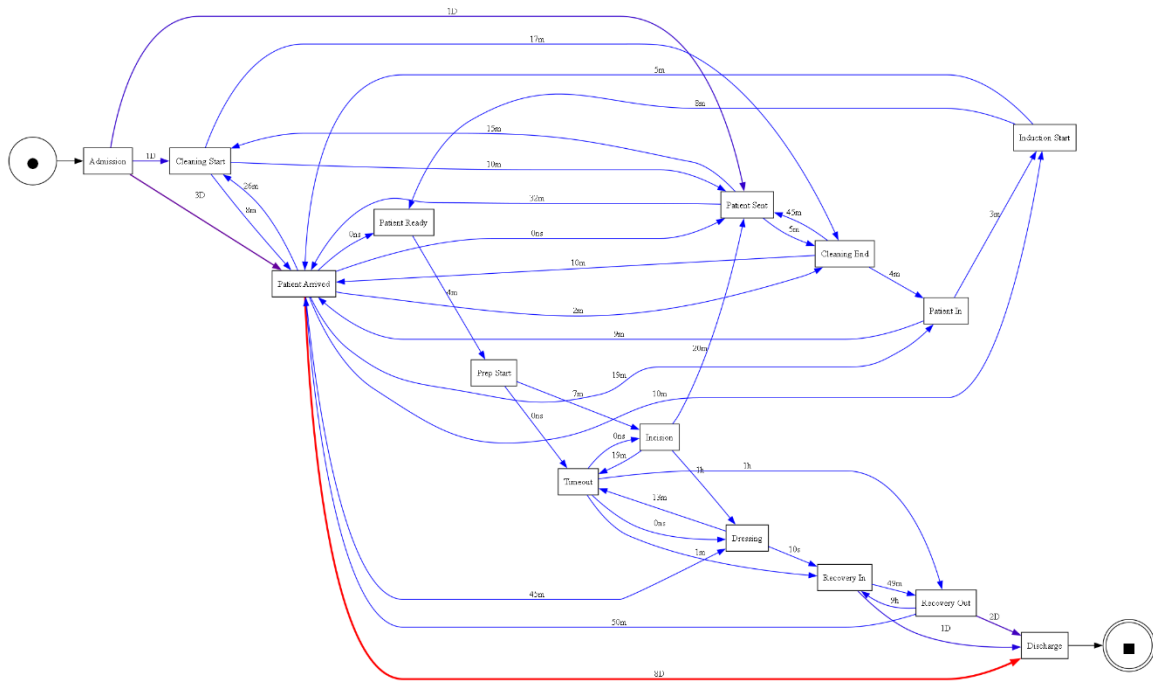


Fig JJ SDC (Same Day Care) (Mean Duration)

B.2 DFG Graphs with Median Duration

B.2.1 Surgery Specialty-Level Graphs

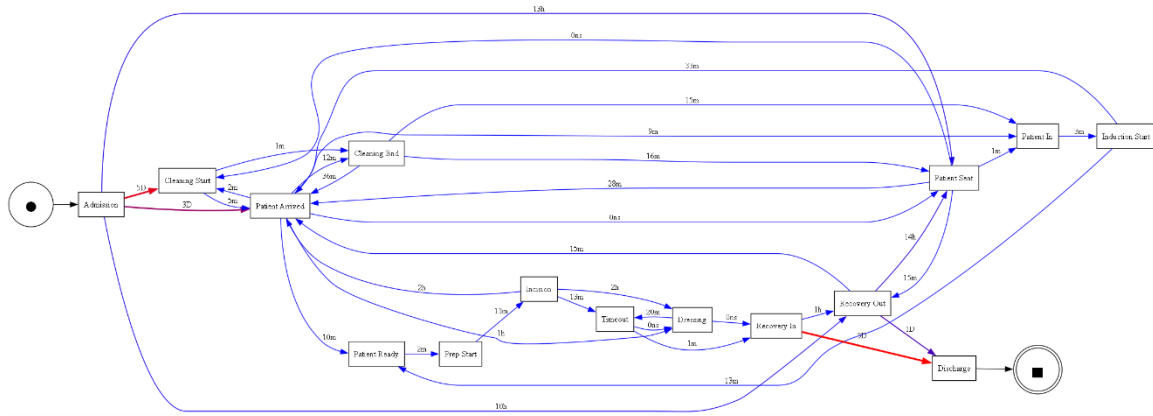


Fig KK Breast Surgery (Median Duration)

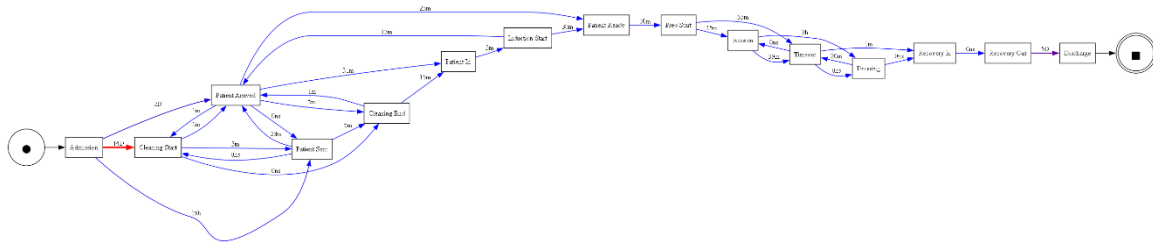


Fig LL Cardiac Surgery (Median Duration)

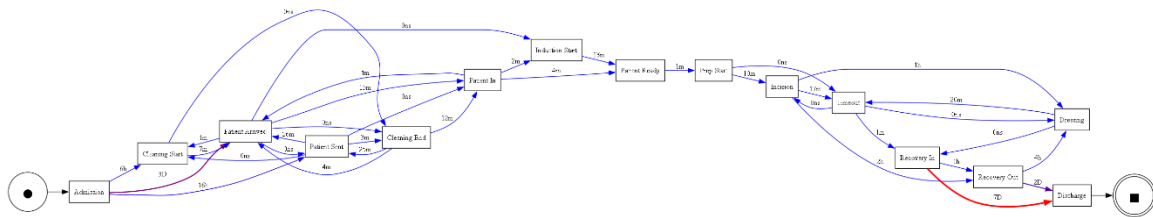


Fig MM ENT Surgery (Median Duration)

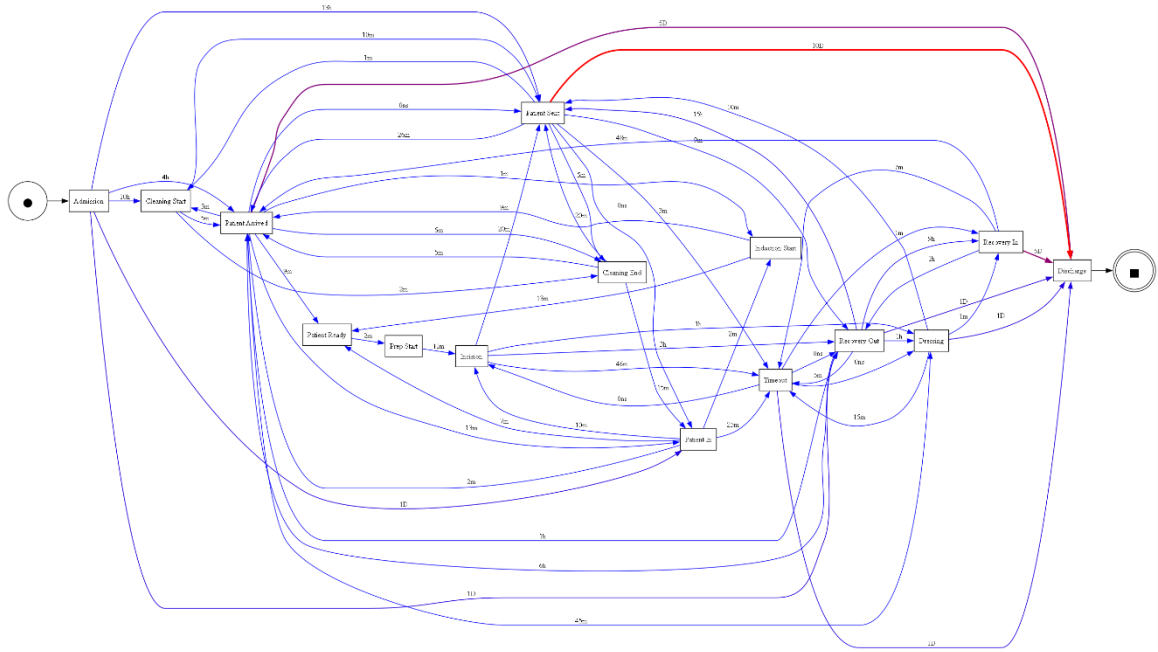


Fig NN General Surgery (Median Duration)

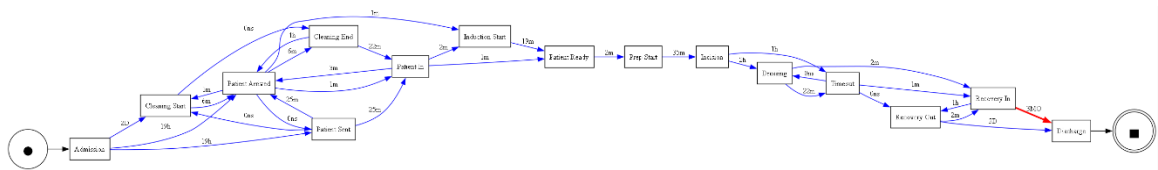


Fig OO Neurosurgery (Median Duration)

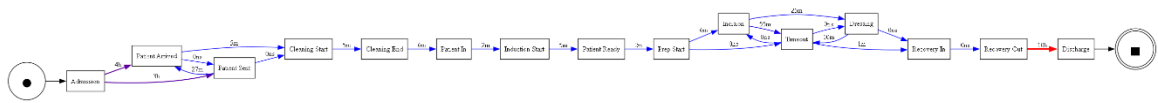


Fig PP Ophthalmology Surgery (Median Duration)

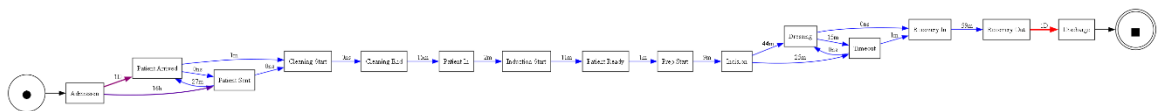


Fig QQ Pediatrics Surgery (Median Duration)

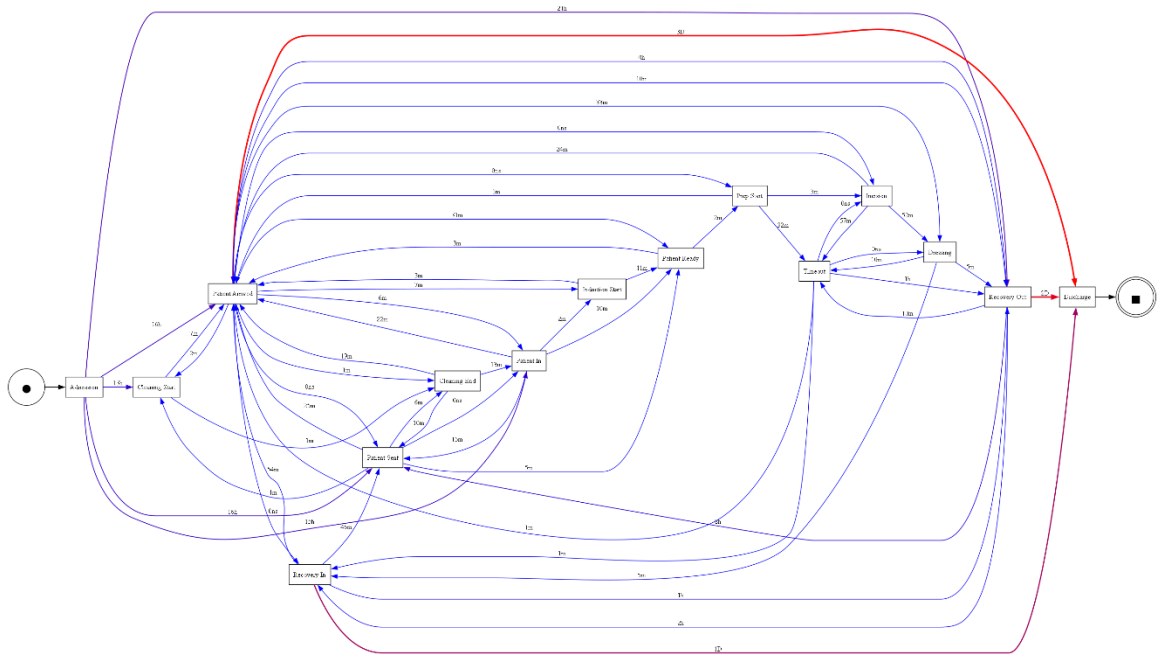


Fig RR Obgynae Surgery (Median Duration)

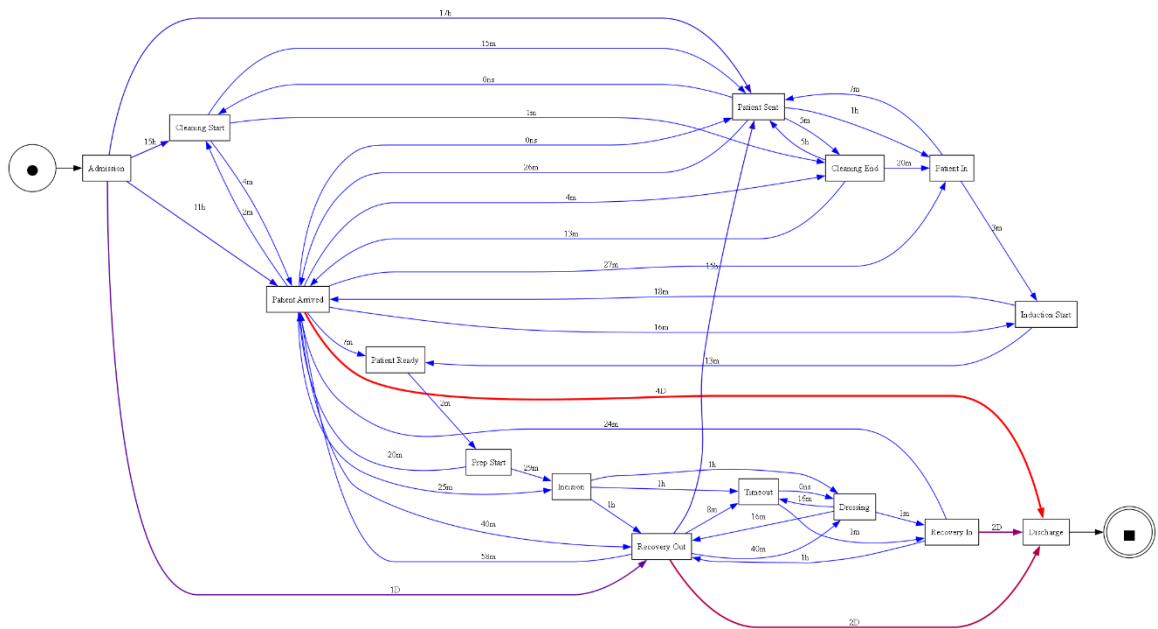
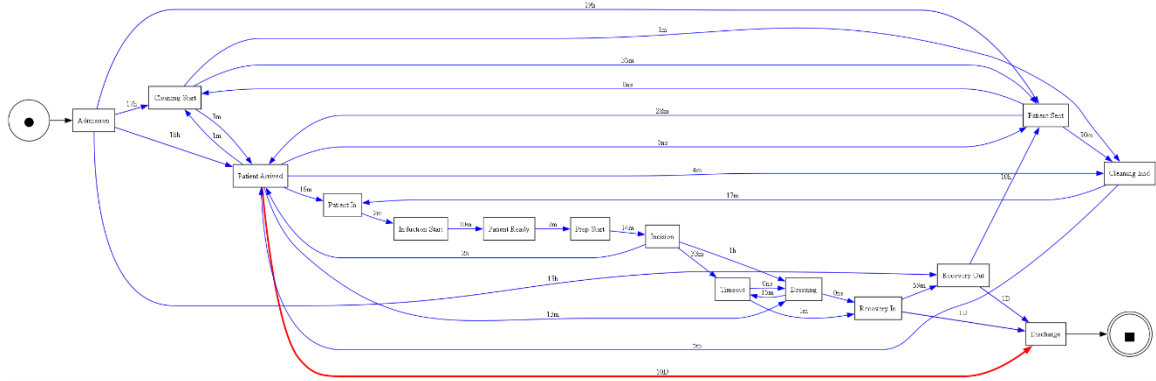
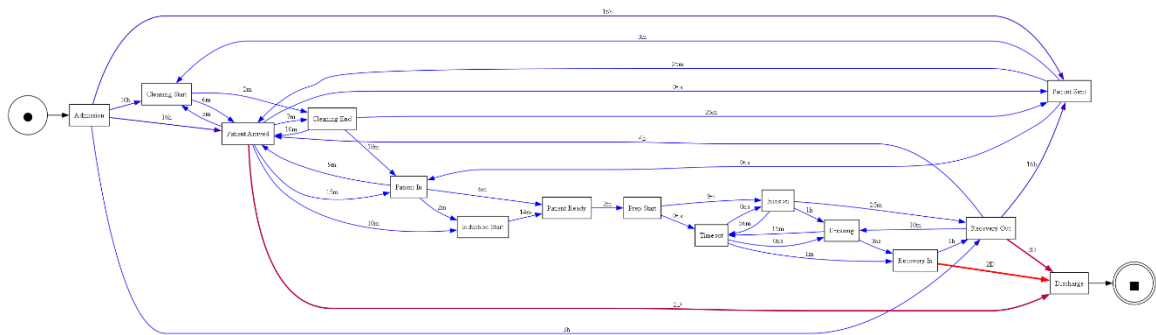
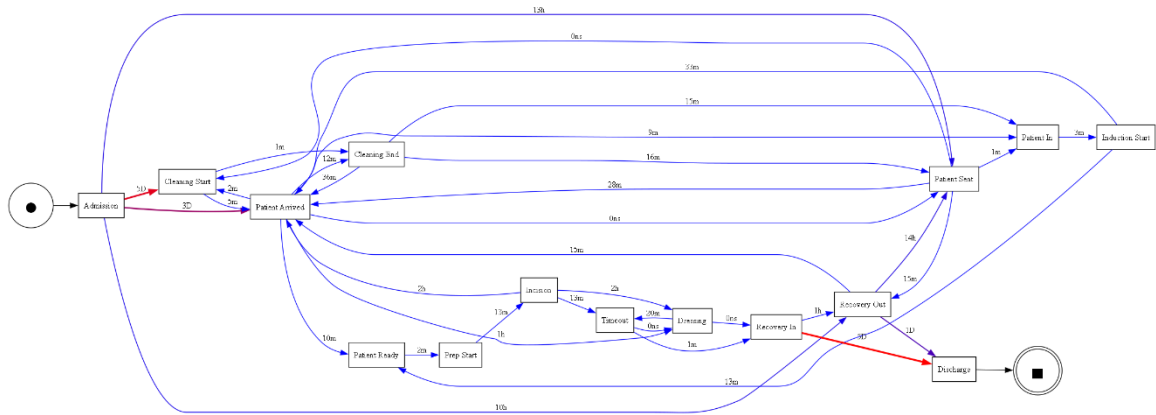
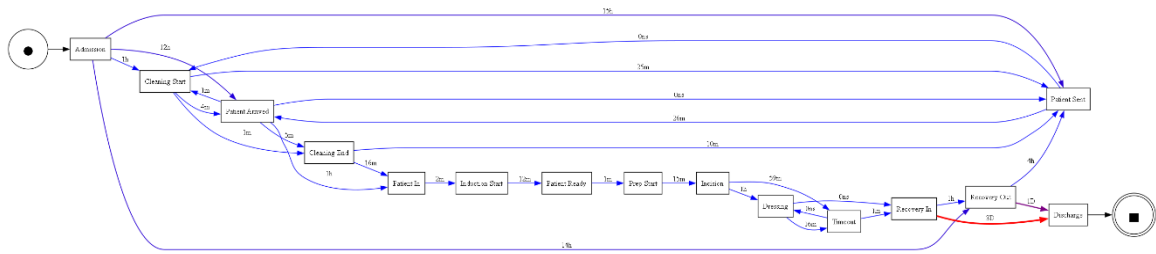


Fig SS Orthopedics Surgery (Median Duration)



B.2.2 Surgery Location-Level Graphs

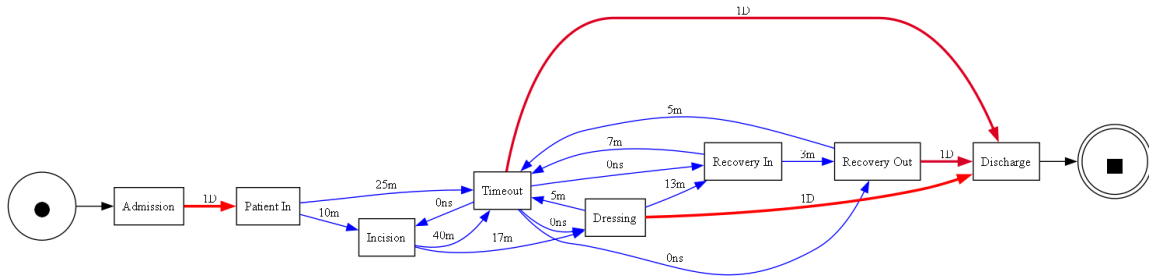


Fig XX Endoscopy (Median Duration)

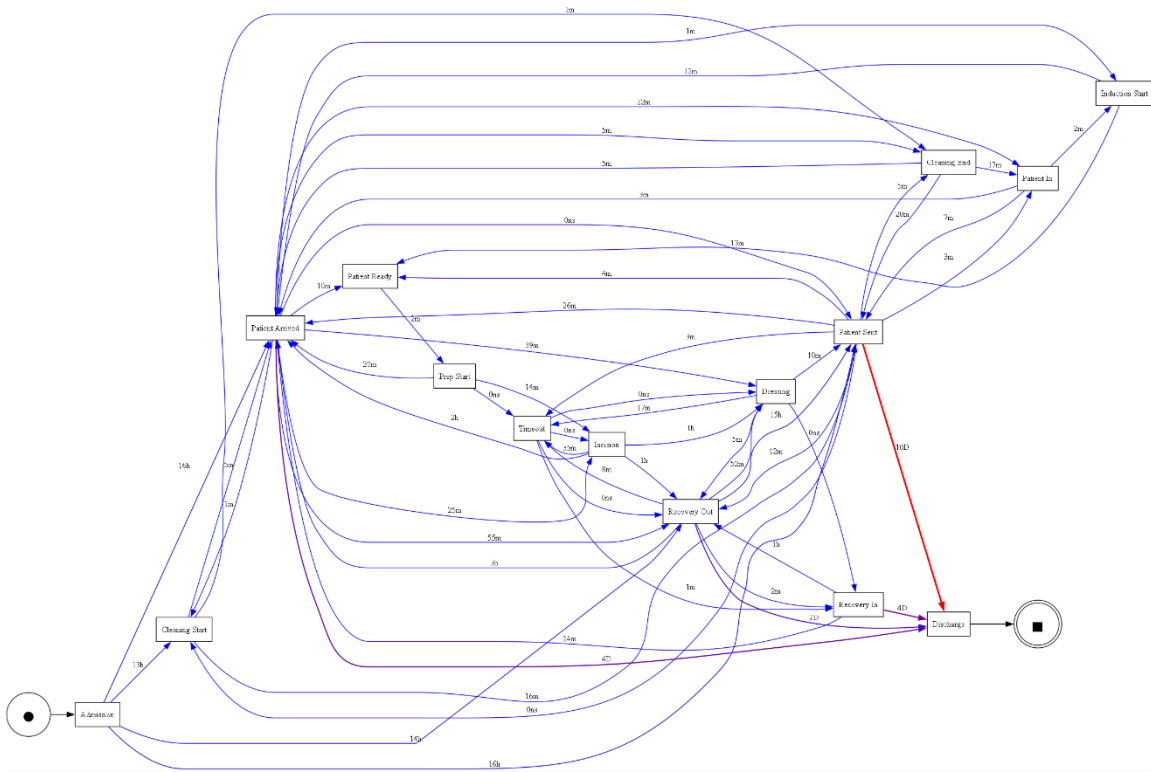


Fig YY Main-OR (Median Duration)

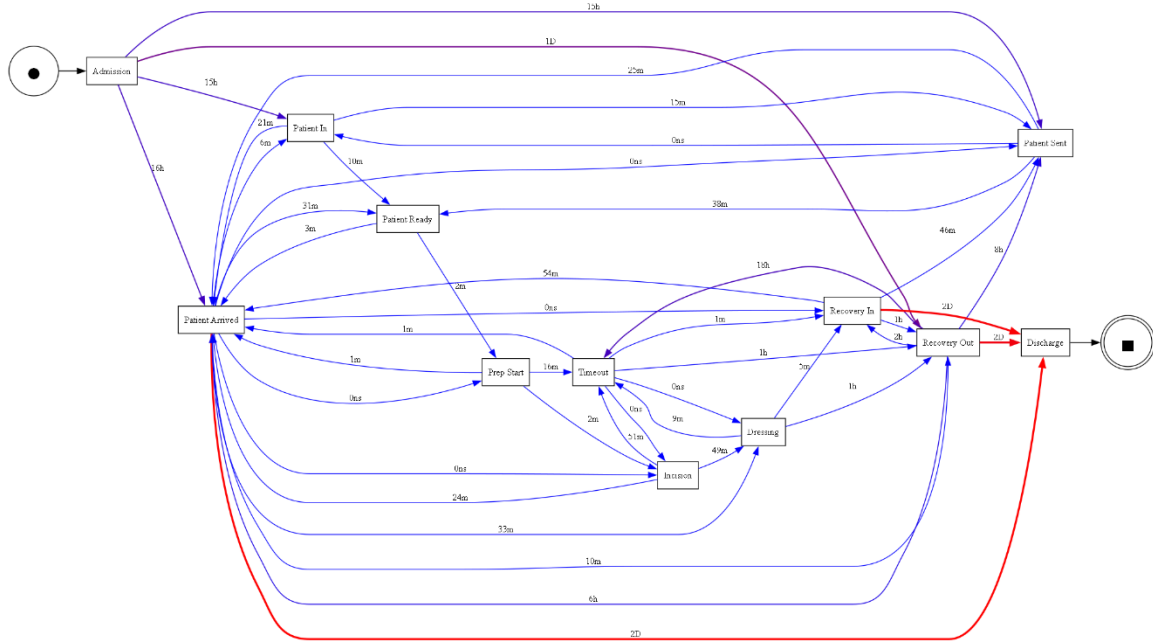


Fig ZZ Obgynae (Median Duration)

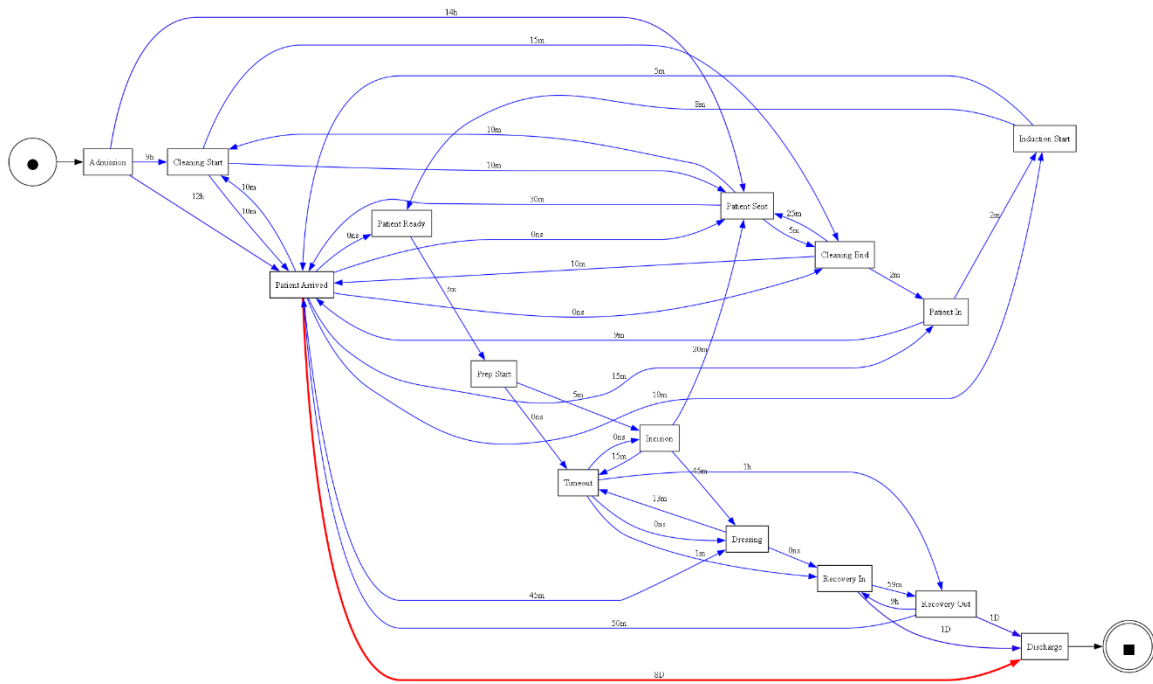


Fig AAA SDC (Median Duration)

B.2.3 Top 5 Highest-Volume Surgeon-Level Graphs

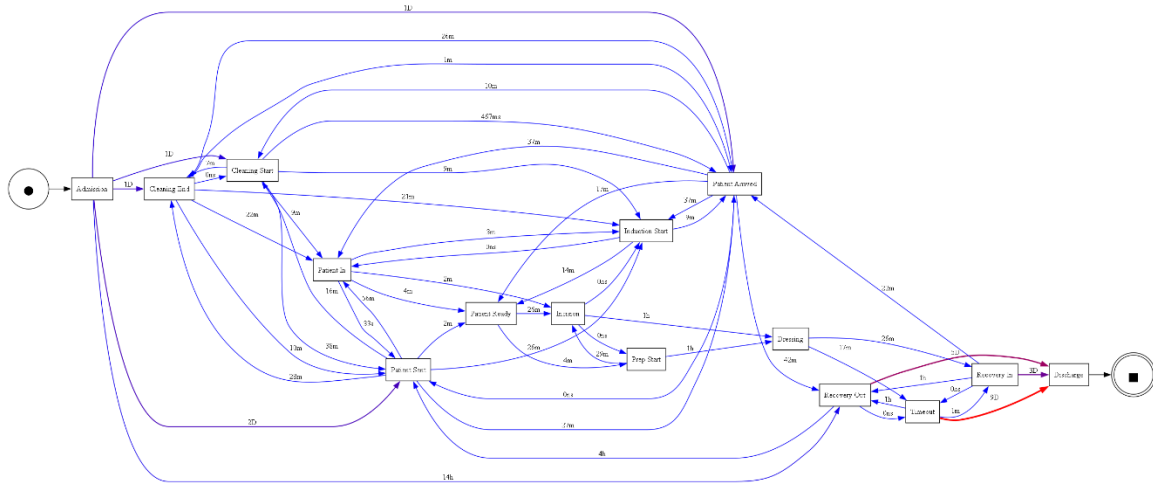


Fig BBB Highest-Volume Surgeon (Median Duration)

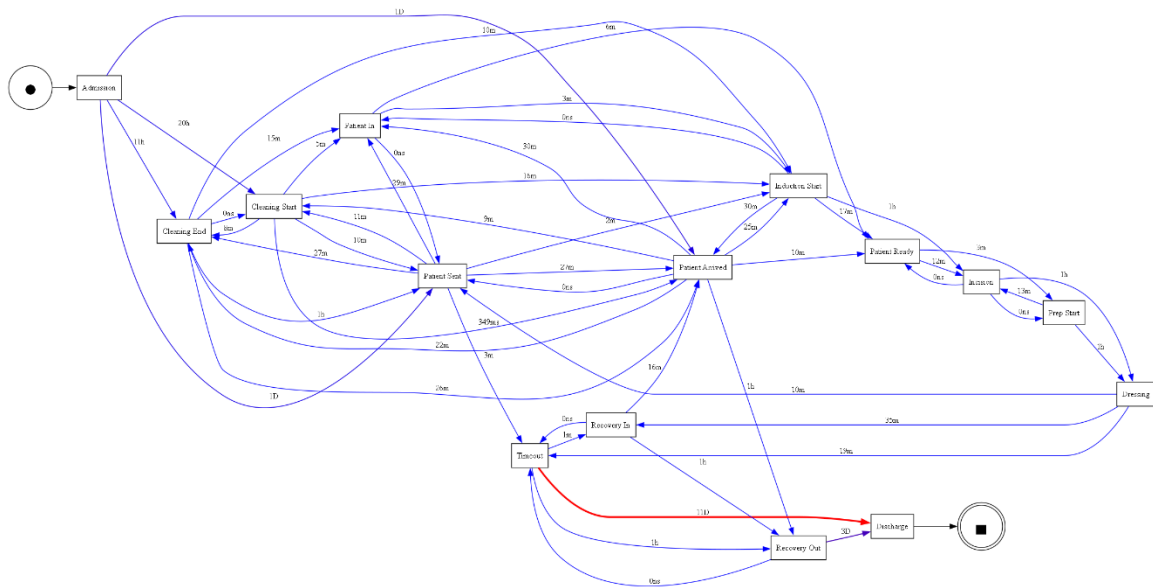


Fig CCC 2nd Highest-Volume Surgeon (Median Duration)

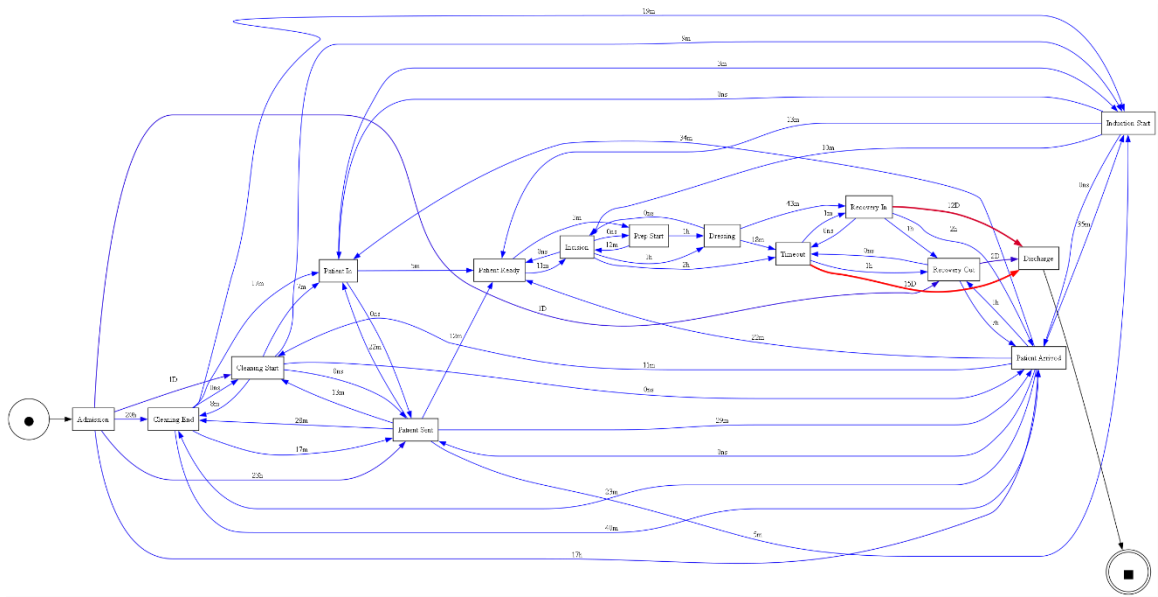


Fig FFF 5th Highest-Volume Surgeon (Median Duration)